

Capital Regulation as a Barrier to Bank Consolidation^{*}

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Abstract

We show that capital regulation has meaningfully depressed bank M&A activity in the U.S. by making leverage-increasing merger financing expensive. Using an instrument for the post-merger capital of target banks, we find that a standard deviation increase in bank capitalization raises the likelihood of banks being purchased within a year by 50%. Using a structural model of banks choosing merger partners subject to capital constraints, we find that weakening of regulatory capital policies around M&A would lead to more mergers, especially between community banks. Total welfare would fall due to consumer welfare losses from lower competition despite fixed cost reductions increasing bank efficiency. Imposing a similar friction on larger, publicly traded banks would have limited a fifth of fixed cost synergies realized in the banking sector since 1992 and protected higher consumer surplus.

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1 Introduction

The United States has an extraordinary number of banks. Until 1994, the number of banks was rationalized by interstate branching regulation: state and federal laws that made it “easier for a New York bank to expand in Kuala Lumpur than Jersey City” (William Jefferson Clinton, 1994).¹ However, today, despite full interstate branching deregulation and thirty years of consolidation, the United States still has 4,000 banks: nearly as many banks as the entire European Union and more banks per capita than any country with a population exceeding ten million people (Figure A.3).

The continued abundance of banks is puzzling. The industry has features that would suggest large economies of scale (Semih Yildirim and Philippatos, 2007; Wheelock and Wilson, 2012; Hughes and Mester, 2013). Banks are network goods where consumers value the expanded availability of ATMs and branches (Saloner and Shepard, 1995; Gowrisankaran and Stavins, 2004; Dick, 2007; Ho and Ishii, 2011), and online banking should have only reduced marginal costs further (Vives, 2019). Furthermore, mergers give the remaining firms market power and allow them to raise markups for consumers.

In this paper, we show that the regulation of bank capital structure constitutes an important barrier to bank mergers. In order to ensure financial stability and to limit the use of the FDIC’s Deposit Insurance Fund, regulators require banks to hold a minimum amount of equity capital against their assets. However, by capping the leverage potential merger combinations can take on, capital regulation *also* caps the ability of banks to finance acquisitions in cash. We call the shadow cost that this friction imposes on potential mergers the *regulatory capital wedge* (RCW). We show correlational evidence of the RCW and verify our results by using lagged cross-sectional differences in the maturity of banks’ Treasury portfolios to construct a Bartik instrument for capitalization. Using this instrument, we find that a standard deviation increase in capitalization raises the one-year probability of a bank being purchased by 50%. The effect of RCW on bank consumers depends on whether the prevented mergers help or harm consumers. Embedding this friction in a structural model of endogenous mergers, we find that the allocative efficiency gains prevented by RCW are outweighed in magnitude by the negative effects of market power in the deposit and loan markets.

We begin by providing causal evidence that consolidation since 1990 has indeed been profitable for banks. We build a population of 5,018 mergers between commercial banks. 80% of the mergers since 1990 were between community banks, a designation for banks holding fewer than \$10B in assets that applies to 90% of currently operating U.S. commercial banks. In order to measure the causal effect that mergers have on the merging parties, we use difference-in-differences to compare the outcomes of the combined bank around the merger date to untreated controls. We demonstrate that realized mergers raised the average profits of the merging parties (the *pro-forma* entity) by 10.4% by reducing non-interest expenses by -4.7%. The effects survive relabeling of merger timing to bank holding company consolidation dates and matching merging pairs just to similar untreated banks. We also show that acquirers seem not to target counties set to move on differential trends:

¹Paraphrasing the remarks of Thomas Labrecque, then CEO of Chase Manhattan Bank.

counties with and without mergers evolve in parallel on a variety of economic prosperity and other proxies for demand for deposits and loans.

Armed with causal evidence that makes the lack of further consolidation seem puzzling, we present and empirically test one explanation – that minimum capital regulation makes financing cash mergers costly. The mechanism is as follows. While mergers can be financed in either stock or in cash, private firms struggle to pay targets with stock because their stock is illiquid (Brau and Fawcett, 2006). As a consequence, firms choose to fund acquisitions with readily available cash rather than offering their target equity in a stock-for-stock merger. However, unlike other firms, banks face strict minimum requirements for the amount of equity capital they must hold to support their assets. Bank regulators – Federal Reserve, OCC, and FDIC – impose the minimum on banks for reasons unrelated to the M&A: increasing bank safety and limiting the damage of bank runs. Cash mergers are difficult to implement without breaching minimum capital requirements, and stock-for-stock mergers remain costly as in other industries, and so potentially profitable cash mergers are untenable. We refer to the combination of these forces as the regulatory capital wedge (RCW).

First, we show the relevance of this mechanism to the banking sector; namely, that many acquisitions in the banking sector would be subject to such a restriction. 80% of banks are privately held, making their stock illiquid. Accordingly, just 24% of acquisitions by private buyers since 1990 were conducted exclusively in stock. Private acquirers with smaller regulatory capital ratios have lower probabilities of occurring. The merger rate for public acquirers is more than twice as large as for privately held acquirers. The consequences are significant for realized mergers. Mergers done between more capitalized banks had lower average returns. We show that profits increase by 17% for the least capitalized quantile of mergers but just 2% for the most capitalized quantile since 1990 using the stacked difference-in-differences design.

Next, we design two additional tests that show a causal link between higher capital and higher propensity to engage in M&A. First, we leverage the 2001 change in merger accounting for the market value of a target bank's securities to show that the market value of equity shifts merger propensity. Banks are regulated on their book value – namely, the quantities of assets listed on their balance sheets, valued at the amount that banks paid for them. However the 2001 FASB phase out of pooling of interest accounting meant that 2002-onward, mergers had to revalue the target at their fair market value, rather than their accounting book value. The “unrealized gains” on a target's securities, namely the difference in their market and book value, consequently shift bank merger propensity while being fairly irrelevant for day-to-day operations. Consistent with this argument, the magnitude of unrealized gains is predictive of when a bank becomes a merger target.

Since unrealized gains could still be endogenous to the quality of banks as merger targets, we construct an instrument to shift unrealized gains independently from the merger synergies of the target. Our Bartik instrument interacts changes in the 5-year Treasury yield with the share of Treasury securities with a remaining maturity of greater than 5 years a bank held two years prior in order to create variation in leverage ratios due to unexpected yield changes rather than bank

skill in managing its securities portfolio. The argument is that while securities portfolios can gain or lose value due to skill, all banks stand on equal footing predicting the future changes in the 5-year Treasury yield, and so shifts in unrealized gains due to the Bartik instrument come from a bank holding more long-maturity securities being unlucky rather than incompetent. With this construction, we can shift bank capital without shifting a bank's appeal as a merger target. We then apply the instrument to shift the post-merger capital of the target firm and find the same results. The instrument suggests that a standard deviation increase in a target's capital ratio raises the 12-month propensity of acquisition by 50%. The first stage is strong in both regimes, but the second stage is significant only after 2001. The differential effect is evidence of RCW: such shifts in "unrealized" capital are relevant only in a time period in which its accounting is mandatory.

To study the consequences of RCW for efficiency and consumer welfare, we develop a model of heterogeneous banks who choose whether to sell to a competitor and raise the value of the combined banks before competing in the deposit and loan markets. Banks are simultaneously auctioned off to one another in a merger market. Each bank bids on every other bank, each of whom has a reservation value of remaining independent. With a realistic simplification of banks to consider each potential merger in isolation and once-and-for-all, the process allocates selling banks to their highest merger surplus buyer. Surplus has two components: profit synergies and one-time costs. To allow banks to choose merger partners in a sophisticated manner, our model posits that banks draw unobserved shocks that affect profit synergies and increase product market profits of the banks as standalone entities. Mergers can also impose one-time costs that reduce surplus including the regulatory capital wedge. We parameterize the regulatory capital wedge as the interaction of a bank's Tier 1 Leverage Ratio and whether it is privately traded. After the merger market clears, banks earn profits from lending at a spread above the risk free rate and offering deposits at a spread below the risk free rate.

Estimation of the structural model proceeds in two parts. We estimate depositor and borrower demand with [Berry \(1994\)](#) and use an assumption of Bertrand Nash conduct to identify bank-time-level quality, marginal costs, and fixed costs. We then use these estimates to estimate merger market parameters via simulated maximum likelihood. The simulation-based estimation accommodates unobserved heterogeneity between banks while using realized merger pairs to identify parameters. Our model estimates suggest that RCW provides a significant deterrent to bank-to-bank mergers. One-time merger costs are 6% of seller assets, and capital constraints add as much as an additional cost of 1.25% of seller assets, declining in the pair's Tier 1 Leverage Ratio post-merger.

Providing capital regulation relief around mergers would create fixed cost synergies but harm consumer welfare enough to reduce total surplus. Removing the one-time capital cost of performing cash mergers as a private firm increases the merger rate by 39% since 1992. In this counterfactual, 1,532 more mergers would have occurred between banks within state between 1992 and 2024, reducing the number of community banks with fewer than \$10B in assets dramatically. While increasing consolidation in this manner would have increased the efficiency of the 2023 sector by \$5B in fixed costs annually (3%), it would have been at the expense of welfare losses for depositors

and borrowers. While branch networks of the consolidated firms expanded to raise the quality of the combined banks, increases in market power outweighed these effects.

Finally, the ability for public firms to transact in stock has been a significant driver of fixed cost synergies realized in the banking sector. Applying the estimates of RCW to publicly traded banks, we find that costs analogous to those faced by private firms would have increased the total fixed costs paid to operate the banking sector by 19% by dissuading public firms from acquisitions. That said, the benefits to allocative efficiency have also come at the detriment of consumers, who would have enjoyed an additional \$37B in surplus annually if such RCW costs had been faced by public banks.

The results suggest that capital regulation can be an important merger deterrent. In industries where transaction-by-transaction review is costly or administratively constrained, a rule-based policy that automatically deters some anti-competitive transactions could complement or lessen the load of antitrust merger review. Our empirical results show that the load would be lessened the most on deals between the smallest firms.

Related literature:

Merger waves. Policymakers and antitrust regulators have raised concerns about consolidation or “roll up” of several industries, especially industries who receive government funding (Cumming, 2026; Kessler, 2026). In healthcare markets, prior work examining hospitals (Dafny, 2009; Gaynor et al., 2013; Dafny et al., 2019), dialysis centers (Wollmann, 2024), and fertility clinics (La Forgia and Bodner, 2025) has shown that consolidation harms consumers. However, in other contexts, consolidation also has cost efficiency and productivity benefits (Jeziorski, 2014; Clark and Samano, 2022; Demirer and Karaduman, 2026). Theoretically, the cost efficiency of consolidation can come from economies of scale or from efficient firms buying inefficient firms, as posited by Jovanovic and Rousseau (2002); Maksimovic and Phillips (2001).

We show that leverage regulation, while not explicitly designed to limit consolidation, makes such a roll-up untenable for U.S. banks. A typical rationale for a lack of rollup is that the Source of Strength Doctrine (noa, 2010) prohibits meaningful entry of private equity into the banking sector. While the doctrine could contribute, we establish that bank-to-bank mergers are also constrained by regulation. The requirement that a buyer of a bank must be well-capitalized means that the debt, the engine of private equity roll-ups in other industries (Kaplan and Stein, 1993; Axelson et al., 2009, 2013; Shivdasani and Wang, 2011), can only be applied sparingly (Johnston-Ross et al., 2025). In this way, such financing frictions are preventing many mergers from even being proposed by competitors to be brought before antitrust regulators.

Merger incentives. We demonstrate that capital regulation is a potent counterweight to mergers. In isolating this friction in banking, we build on past work characterizing bank merger incentives including cost efficiencies (Hannan and Pilloff, 2009; Beccalli and Frantz, 2013; Akkus et al., 2016) and geographic diversification (Aguirregabiria et al., 2016).² Across industries, mergers can

²Though Wheelock and Wilson (2000) finds that inefficient banks are less likely to be acquired.

be considered potentially value enhancing investment, and so financing constraints (Harford, 2005; Almeida et al., 2011; Gorbenko and Malenko, 2018) and market timing (Shleifer and Vishny, 2003; Li et al., 2018) both play a role in inhibiting or encouraging mergers. Managerial entrenchment, implicit in the design of the structural model but not the paper’s focus, can affect both buy-side (Roll, 1986; Morck et al., 1990; Masulis et al., 2007; Harford et al., 2012; Yim, 2013) and sell-side firms (Jenter and Lewellen, 2015; Kotia, 2025).³⁴

Welfare of bank mergers. We contribute estimates of depositor and borrower welfare compared to counterfactual capital regulation policy. Along the way, we also contribute reduced form evidence on mergers continuing to 2024 that shows stronger evidence of cost efficiencies than previously documented. A large body of prior work has shown mixed evidence of the cost efficiency of bank mergers occurring since the Riegle-Neal Act was passed in 1994. Analyzing realized outcomes in previous bank mergers, Rhoades (1998) and Houston et al. (2001) find evidence of meaningful cost savings, whereas Rhoades (1993), Berger (1998), and Focarelli et al. (2002) do not.

We evaluate the welfare of the banking sector as if capital regulation were not a deterrent in the market. Pakes et al. (2024) finds that consumer welfare is lower than prior to mergers in a sample of markets that experienced a single bank merger relative to markets experiencing none. Our focus is welfare with counterfactual capital regulation policy, where we find negative effects for loosening capital regulation and hence increasing mergers.⁵

Capital regulation. This paper shows that capital regulation affects industry structure through M&A. Prior work has investigated the first half of the mechanism, that illiquid stock incentivizes acquirers toward cash mergers (Faccio and Masulis, 2005; Brau and Fawcett, 2006; Barger et al., 2008; Massa and Xu, 2013; Huang et al., 2024),⁶ but we show that the interaction of this financing friction with capital regulation creates a potent merger deterrent. Granja et al. (2017) finds evidence of capitalization playing a role in willingness to buy by documenting that failed banks are purchased by the most well capitalized acquirers. Using an IV affecting not just failing banks, we show this correlation extends to the entire population of banking targets and study its implications on the sector.⁷ Prior work studies its effect on entry and exit (Corbae and D’Erasmus, 2021; Corbae and Levine, 2025; Corbae, 2025), but M&A is far more common in the modern U.S. banking sector than entry and exit. Our empirical design isolates distance from the effect of “microprudential reg-

³⁴A great deal of work relates M&A propensity to forces external to both the buyer and seller, such as policy uncertainty Bonaime et al. (2018).

⁴Many other motivations for acquisition like the option value of exposure to new products and technology or to respond to changes in the cost curve are irrelevant in banking, where deposit-taking and lending are a static, single product homogeneous goods market.

⁵Without explicitly calculating welfare, a large body of work has examined prices and quantities on the loan side Berger et al. (1998); Sapienza (2002); Kahn et al. (2005); Di Patti and Gobbi (2007); Erel (2011); Nguyen (2019); Brunetti et al. (2026) and the deposit side Prager and Hannan (1998); Bord (2018); Focarelli and Panetta (2003); Granja and Paixao (2023); Liebersohn (2024).

⁶The phenomenon also aligns with other theory (Myers and Majluf, 1984; Hanson et al., 2011; Admati et al., 2018) predicting that firms, for new investment, tend to reduce assets (via a cash merger) rather than raise new equity and recapitalize (via a stock-for-stock merger).

⁷Focarelli et al. (2002) notes that equity declines after the merger and hence return on equity rises, but we interpret the drop in equity as revealing that costly equity issuance could dissuade acquisition.

ulation” (Hanson et al., 2011) on M&A while putting aside its effect on day-to-day bank operations, which are plentiful but not the focus of the paper (Berger and Udell, 1994; Gorton and Winton, 1995; Peek and Rosengren, 1997; Gorton and Winton, 2017; Gropp et al., 2019; Fraisse et al., 2020; Carlson et al., 2022; Berrospide and Edge, 2024).⁸

Endogenous bank industry structure. In endogenizing merger choices before studying the horizontal competition of firms like Perez-Saiz (2015); Uetake and Watanabe (2020), building on entry models constructed by Bresnahan and Reiss (1991); Seim (2006); Ciliberto and Tamer (2009); Nocke and Whinston (2013), we allow industry structure to be endogenous to mitigate the biased estimation of parameters related to the horizontal competition of firms.

Our model is novel in documenting the quantitative importance of microprudential regulation in mediating endogenous M&A among the fringe of U.S. community banks. The U.S. banking sector is canonically fragmented due to the long-held political power of unit banks (Calomiris, 2000). However, recent work has developed economic rationales for equilibrium entry and exit, and M&A. Corbae and D’Erasmus (2020) develops a model of dynamic oligopoly in which banks can choose to invest in their deposit franchise in order to grow, in the spirit of canonical models of industry dynamics with investment (Ericson and Pakes, 1995). Gowrisankaran and Holmes (2004); Corbae et al. (2025) extends this framework to accommodate mergers, allowing the dominant bank to acquire a chosen measure of the competitive fringe each period.⁹

Unlike these papers, our model accommodates simultaneous mergers between non-atomistic banks and focuses on a novel financing friction. Due to the vast number of banks simultaneously operating in markets, the sequential method of Jeziorski (2023); Igami and Uetake (2020) is computationally intensive, and so we assume the market clears simultaneously with an assumption of firm myopia to other players. Most similar to our work is Uetake and Watanabe (2020), which models simultaneous bank entry by merger between 1995-2000 where acquirers and targets form stable matchings. Assuming away externalities, we allow buyers to purchase multiple sellers simultaneously and compute downstream consumer welfare from changing merger costs.

2 Data

In order to quantify the reasons for a lack of larger banking sector consolidation, we use standard banking datasets to track bank outcomes on a quarterly basis and to construct the list of bank mergers.

⁸In using an instrument for capital to study the propensity to merge in the short-run to identify effects, we also do not model how capital structure itself affects bank operations, whether leverage increasing risk-shifting, bank deposits disciplining the use of funds, or deposit insurance creating moral hazard (Calomiris, 1990, 1999; Calomiris and Jaremski, 2019; Calomiris and Chen, 2022).

⁹The models in these papers feature a rich set of forces that ultimately shape banks’ choices to enter, exit, and merge, where these forces – costly equity issuance, deposit insurance, manager risk taking, and market power – mediate the flow payoffs to the bank and consequently whether the dominant bank can remain solvent, exit, or acquire.

2.1 Measuring bank outcomes

Our analysis combines information from several data sources, most of which are commonly used in the banking literature. We obtain novel data about bank holding company (BHC) shareholders from digitizing a population of Form Y-6 forms filed by BHCs between 1995-2024.

2.1.1 Bank-level balance sheets and income statements

We collect the universe of bank quarterly balance sheets and income statements (“call reports”) from 1990 Q1-2024 Q4. We use these forms to track regulatory capital ratios, operating costs, and other important balance sheet characteristics we control for throughout our analyses (e.g., insured deposits-to-assets ratios and funding costs). When revenue and cost items are defined as year-to-date, we use the flow value of the item so that we can measure the revenue and cost of the filer quarter-by-quarter. Details of this procedure are provided in Appendix ?? We also use these data to construct our primary source of exogenous variation in the capital-intensity of a potential acquisition: the fair value losses on the securities portfolio that would be booked in the event that a bank is acquired. This variable is constructed by summing the difference between fair value and amortized cost on each bank’s held-to-maturity (HTM) and available-for-sale (AFS) securities portfolios, both of which are booked at cost for the purposes of calculating regulatory capital but are marked to fair value in the event the bank is acquired under the current accounting standards. We obtain annual, branch-level information on deposit volumes spanning 1990 to 2024 from the FDIC’s Summary of Deposits filings.

2.1.2 Bank holding company level balance sheets, income statements, and ownership

We collect the analog to the call reports at the holding company level, the Y-9 series, over the same time period. We use these data primarily to track issuance activity of equity and long-term debt. We assign individual banks that file the call reports to their bank holding companies, if they have one, using the National Information Center relationships file. We do so by following the procedure outlined in the Federal Reserve’s data dictionary.¹⁰

2.2 Constructing a bank merger database

We obtain information on the history of bank organizational relationships and merger transactions from the National Information Center’s (NIC) “relationships” and “transformations” files. When studying financing methods, we use the population of bank mergers maintained by S&P Global Market Intelligence. The database of bank mergers filters out activities more aptly characterized as corporate reorganizations and recognizes mergers at the bank level.

¹⁰See https://www.federalreserve.gov/apps/mdrm/data-dictionary/search/item?keyword=9364&show_short_title=False&show_conf=False&rep_status=All&rep_state=Opened&rep_period=Before&date_start=20190808&date_end=20190808; accessed 07/17/25.

2.2.1 Merger transaction dates and merger deal considerations

We collect data from NIC on the history of ownership relationships between different banks and bank holding companies (“relationships” file) and on mergers and acquisitions (“transformations” file). The data are at the bank charter level, where each charter is assigned an RSSD. That RSSD-level is how we define a bank, but some of the largest “banks,” namely all of the Global Systemically Important Banks (G-SIBs) are colloquially referred to as “banks” while actually acting as a Bank Holding Company owning multiple RSSD’s. Because some entities even contain multiple layers of BHC’s, we refer to the true parent bank holding company as the “high holder.” Our empirical results use the date of the merger as the date reported in NIC that the RSSD of the acquired bank was merged into the acquirer.

We supplement this information on mergers with additional data from S&P Global Market Intelligence (SNL) deal filings. In particular, we retrieve data on announced deal values and consideration mix (cash versus equity) for a subsample of mergers, largely transactions involving at least one publicly-traded company.

2.2.2 Constructing the merger sample

Building an accurate dataset of bank mergers entails several challenges. First, many bank-to-bank mergers are not true changes in control of the sort we are interested in: in the wake of the passage of Riegle-Neal in 1994, many multi-bank holding companies merged subsidiary banks into one charter. This type of activity is more aptly characterized as a corporate reorganization. Second, many mergers are officially between holding companies, with the underlying subsidiary banks merged into a combined charter at some later date; this procedure should be considered a “true” merger, but on paper it resembles a corporate reorganization in the sense that, at the time the two banks are combined, they are already under the same holding company.

Our data construction procedure is intended to minimize the impact of issues of this sort by ensuring had a distinct high holder from its acquirer at least two years prior to the merger date. To implement this algorithm, we begin with our dataset of commercial banks linked to their high holder at every point in the panel from NIC. Next, we merge on the RSSD ID of the commercial bank with the RSSD ID of targets in the NIC transformation file (subsetting to drop bank failures), keeping only matches on RSSD ID where the indicated high-holder relationship is either the current or penultimate link relative to the date of the merger. We store the current and previous high-holder RSSDs from this match, including the windows in which each are active, in our updated dataset. This population has a mix of true mergers and of internal corporate reorganizations. Finally, we merge this file back into the relationship file on the RSSD ID of the *acquiring* bank, again storing the date windows of the current and previous high-holders. We classify a true, inter-company bank-to-bank merger as a merger in the resulting dataset for which the target and acquirer banks have different high holders either at the time of the merger or at any point in the prior two years. The choice of window length is meant to account for the normal length of time that banks take, after

merging holding companies, to consolidate the subsidiary bank charters.

Our resulting sample contains 5,018 mergers between commercial banks from 1990 to 2025. The sample statistics reveal a distinguishing feature of the banking sector: the vast majority of M&A activity is between very small firms. Figure 1a plots the percent of mergers done by acquirer size bin. 80% of mergers are done by banks that have fewer than \$10B in assets. Banks of this size, referred to by regulators as “community banks” comprise 90% of U.S. banks and provided just 20% of bank lending in 2024 (Figure 1b). Table 1 shows descriptive statistics of acquirers versus targets and uninvolved banks. Acquirers have larger return on equity and are more cost efficient than their targets and than banks uninvolved in the merger market. The statistics are suggestive evidence that bank mergers are motivated to realize cost synergies. Table B.2 shows means by five-year cohorts of acquisitions between 1990 and 2024. The trend of efficient acquirers purchasing inefficient targets holds throughout the panel. Figure A.8 shows the distribution of characteristics for acquirers and targets in histograms. Figures A.5 and A.6 show additional statistics about the merger sample. Table B.3 shows a difference in means table for county-year observations that have a presence of merging bank versus county-year observations that have not. Presence is defined as the combined bank, either the acquirer or target, operating at least one branch in the county between 2005-2024. County-year observations that have experienced mergers have lower unemployment rates, lower poverty rates, higher household income, higher populations, more highly educated individuals, fewer retired individuals, and are less rural.

2.3 Measuring economic outcomes

In Section 3, we use economic outcomes of counties affected by bank mergers to evaluate the plausibility of the parallel trends assumption. We use the American Community Survey (ACS), available from 2005-2023 to measure the average unemployment rate, average poverty rate, average household income, and population by U.S. county. We use the publicly available Census Business Dynamics Statistics as our measure of establishment entry, exit, net job creation, job reallocation, and firm-death rates.

3 Profitable consolidation

Using differences-in-differences in order to estimate the causal effect on merging parties, we show that realized mergers since 1990 raised the net income of merging banks by 10.4% by reducing non-operating expenses by -4.7%.

3.1 Identification strategy

In order to estimate the causal effect on merging parties, we create a pro-forma bank for every merger, use banks who do not participate in the merger market as a control group, and assume

parallel trends in outcomes counterfactual to the mergers and controls. With the sample construction and assumptions, we estimate the causal effect of the merger on the pro-forma bank using difference-in-differences.

The pro-forma bank is a single entity, and every outcome variable for the pro-forma bank is built from the acquirer and target in the merger of interest. Let entities involved or uninvolved in mergers be indexed by i . For outcomes measured in dollars, an entity’s outcome $Y_{\mathcal{M}(k,j)t}$ is the sum of the buyer k and seller j outcomes. For instance, prior to the merger, the pro-forma bank’s net income is the sum of the acquirer and target’s net incomes. For a given merger consummating at date g , we include it in our analysis sample if the acquirer has had no other merger within eight quarters before or after g .

The control banks are constructed to create a set of banks untreated by the merger. For our baseline specifications, we use every bank uninvolved in a merger between 1990-2024 as a control group for every merger cohort between 1990-2024. This population is referred to as a “stacked control group” and is useful for achieving statistical power and avoiding ad hoc assumptions for the comparability of the pro-forma bank to any particular untreated banks. For any date t that the control bank files a Call Report, they can act as a control bank at event time s for cohort date $g = t + s$. The granularity of cohort fixed effects that the design allows mitigates bias from heterogeneous treatment effects (Baker et al., 2025).

With this population of pro-forma mergers with cohort dates $g(i)$, we run the following regression for outcome variables $Y_{ig(i)t}$

$$Y_{ig(i)t} = \alpha_{ig(i)} + \tau_{g(i)t} + \sum_{s=-8, \neq -2}^8 \beta_s \cdot \mathbf{1}(t - g(i) = s) \quad (1)$$

Lastly, we make two assumptions: the Stable Unit Treatment Value Assumption (SUTVA), that potential outcomes of the control banks depend only on their own treatment status, not others around them nor the merged group, and Parallel Trends, that the control group of banks would have moved in parallel to the *pro-forma* banks counterfactually without the merger having occurred conditional on controls. If additionally, the pre-merger pro forma bank empirically had no differential trends relative to the control banks, the assumptions posit that the pro-forma bank, as well as control banks, absent their merger, would have continued on their parallel pre-merger trajectories that they were on relative to controls. We show that indeed, treated merger combinations do not have differential trends relative to control banks. Additionally, while the econometric assumptions are fundamentally untestable empirically, in Section 3.4, we show a additional results that suggest they are reasonable.

3.2 Bank profitability

First, realized mergers improve the profitability of the combined firms. In order to document this fact, we estimate Equation (1) for the logarithm in profits of the *pro-forma* entity. The outcome

variable, in the pre-period when both the acquirer and target’s profits are observed, is the logarithm of the sum of the separate quarterly profits. In the post-period, the outcome variable is the logarithm of the cash flows for the acquirer in the Call Reports. So, β_s is the change of the logarithm of the sum of the buyer and seller profits relative to $t = -2$ – two quarters prior to the merger date. Figure 4 shows the resulting estimates of β_s for growth in profits, where profit is defined as

$$\text{Profit}_{it} \equiv \text{Net income}_{it} + \text{Good will impairment losses}_{it} + \text{Amortization expense}_{it}. \quad (2)$$

Figure 4 shows that, in the quarter prior to combination, profit drops precipitously. Otherwise, there is no significant pre-trend in the outcome, consistent with our assumption of parallel trends between the merging and control banks. Then, in the quarter of combination, combined profits increase by 4.3% and grow in the two years following to 10.4%. To demonstrate the importance for accounting profits as well, Figure A.10 shows the same specification run for net income. Merger timing could realistically be earlier than the date reported as the charter retirement date in NIC. We repeat the analysis at the date that bank holding companies were combined – an often simultaneous but sometimes earlier date – and the results are quantitatively similar (Figure C.1).

3.3 Mechanisms: Cost efficiency

The primary channel by which profits rose was by increases in the cost efficiency of merging banks. We present results for several cost categories that together show that the bulk of cost reduction occurred through non-interest expenses, rather than changes in interest-related expenses, such as deposit rates paid on checking accounts.¹¹

Our first outcome is non-interest expense. Bank costs are typically divided into two categories: 1) interest expense, directly related to borrowing (from depositors and other creditors) and lending and 2) non-interest expense, which captures the remainder and encompasses the overhead rent on bank branches, IT, and salaries for loan officers and workers that allow banks to offer their products and services. Figure 5a shows the estimates of β_s for the logarithm of non-interest expense, where we then report the coefficient as the percent growth in the outcome. In the quarter prior to the merger date, non-interest expense grows by 2.5%. After the merger, non-interest expenses decline steadily to a -4.7% decline at the combined firm. The negative estimate implies that the total cost base declined. From this analysis alone, however, it is unclear whether the combined bank became more efficient or simply shrunk after the merger. As a result, we turn our next outcome, non-interest-expense normalized by a lending business’s notion of revenue.

Our second outcome is the level of the “inefficiency ratio,” a common practitioner measure of cost efficiency.¹² Using the Call Reports data, we construct the “inefficiency ratio” of the pro-forma

¹¹In Section 5, we will use a model of depositor and borrower demand to distinguish whether profitability increases stem from marginal and fixed cost reductions, as well as whether mergers are harmful to depositor and borrower welfare.

¹²In truth, the inefficiency ratio as defined here is called the “efficiency ratio” by practitioners, despite the fact that a higher ratio indicates cost inefficiency. To limit confusion, we relabel this quantity. A higher inefficiency ratio means the

j for commercial banks that merge in our sample. The inefficiency ratio is defined as

$$\text{Inefficiency Ratio}_{jt} \equiv \frac{\text{Non-Interest Expenses}_{jt}}{\text{Net Interest Income}_{jt} + \text{Non-Interest Income}_{jt}} \quad (3)$$

The inefficiency ratio, by measuring the bank’s non-interest expenses as a fraction of its income (net of funds laid out or lost in the event of the default of its borrowers), is intended to capture the cost efficiency for the bank without confounding from the heterogeneity of loan portfolios between banks. Banks could have starkly different interest expenses due to, for instance, charge-offs set aside for risky borrowers.

We then estimate Equation (1) for the inefficiency ratio. The resulting event study coefficients are plotted in Figure 5b. We find that bank inefficiency ratios decrease -1.9pp (4%) from their prior level and remain persistently lower for two years after the merger. Breaking the cost items down further in Figure A.11, the decline persists for salaries, fixed expenses, and miscellaneous expenses. The stark effects are consistent with intuition about mergers that cost savings arise from eliminating redundancies in admin and legal.

Importantly, difference-in-difference analysis suggests that profitability increases are not due to increases in revenue. Figure A.12 provides plots of the estimates of β_s from Equation (1) for the growth in net interest income and non-interest income as outcomes. Both are insignificant. We also find suggestive evidence of negative but potentially modest effects on consumers due to the remaining banks having more market power. Figure 6 shows the estimation of Equation (1) on price for loans and deposits and growth in loan and deposit quantities. Market shares decline simultaneously with the merger. Deposit rates decrease by 4bps two years after the merger and loan rates increase immediately but the effect vanishes two years after the merger. These small effects are consistent with prior work (Granja and Paixao, 2023; Brunetti et al., 2026).

3.4 Robustness

If the assumptions of parallel trends and SUTVA do not hold in this setting, the estimate of a causal effect of mergers would be biased. We discuss potential threats to identification and show that empirical evidence consistent with our assumptions.

3.4.1 Parallel trends

In evaluating the plausibility of the assumption, we discuss the ideal experiment. An ideal experiment to estimate the average treatment effect on treated banks would randomly select a pair of separately organized banks to run as a combined pair, then measure the change in $Y_{ig(i)t}$ experienced by that pair. *A priori*, parallel trends in pre and post outcomes could seem economically quite questionable. To become an acquirer, banks may be faring better, facing better financing conditions, or facing higher latent demand for their products that could be visible up prior to the merger and

bank is more cost inefficient.

afterward. Sellers could be increasingly struggling with declining revenue before being willing to be purchased by a larger acquirer or run by management interested in ending their involvement in the firm.

However, many of these arguments weaken greatly given that, in Sections 3.2 and 3.3, the *pro-forma* and control banks outcomes do move in parallel prior to the merger. For parallel trends to be present in the pre-period but dissipate at the merger, the merger timing must be precisely coincidental with or immediately preceding changes in fundamentals that spurred merger activity. For instance, for the merger effect on profits to be *overestimated*, the buyer's acquisitiveness or the seller's willingness to sell must be correlated with standalone growth in $t > 0$, growth that is invisible prior to the merger date $t < 0$. For the merger effect on cost reductions to be overestimated as well, the sellers and buyers would have been due to undergo rapid cost reduction after the theoretical merger date for reasons correlated with a willingness to merge. In other words, the timing of the mergers could be endogenous.¹³

We cannot rule endogenous timing of mergers entirely using pre-period covariates, but we will focus on potential differential trends in geographies. Two plausible channels for a rapid post-period cost reduction if banks had remained standalone are 1) changes to the customer demand that would allow banks to service them cheaper or 2) changes in technology that would allow banks to offer the same services at lower cost. In both, it is difficult to imagine how banks in the same market could have heterogeneous exposures to these forces. So, violations in parallel trends could come largely from variation in where the treated *pro-forma* banks are located versus untreated control banks. To mitigate this concern, Figure 3 shows that consumer and firm-related economic variables that could be associated with latent demand for deposits and loans appear parallel for merger and non-merger counties. In Appendix C.1.2, we take this even farther and run placebo tests on these outcomes, using the merger as a treatment on the outcomes. The argument for a parallel trends violation is that banks possess information not available to the econometrician that the county would perform better after the merger, and so the control used in stacked-difference-in-difference would be a biased counterfactual. Consequently, assuming the bank merger itself did not have a direct effect on such outcomes, the bank merger itself should not be predictive of economic improvements to the county. We find no positive effect of a county's first bank acquisition on economic prosperity. If anything, bank mergers are followed by declines in median household income and a transient spike in firm exits, and both effects are economically small. Hence, it is difficult to imagine that the causal, positive effect of mergers on bank profits could be overestimated because of targeting of mergers to geographic markets that about to fare particularly well.

¹³The phenomena make sense through a Coasian lens. In equilibrium, such changes to the boundaries of the firm must clearly be the consequences of changes in exogenous variables upstream of the expansion and shift as soon as the new state variables are realized that yield.

3.4.2 SUTVA

Second, equilibrium logic threatens the assumption of SUTVA for the control banks. Banks in horizontal competition with the merging parties can be directly affected by the merger. For instance, the merging bank realizing cost synergies could charge lower prices, causing the merged entity to steal business from competitors in its market. On the other hand, a reduction in competition in a market could cause competitor banks to raise prices. While in equilibrium, every bank could be treated by the merger of two unaffiliated banks, we include all untreated banks as controls to avoid discretionary choices about which subgroups of banks are more or less affected. We match each pro-forma entity with a bank of similar size operating in a separate state and find quantitatively similar results to our baseline specification. Formally, we choose a control bank for every pro-forma bank by matching the pro-forma to an observably similar bank on assets, efficiency ratio, and Tier 1 Leverage Ratio 12, 9, and 6 months prior the date of merger. Figure C.4 shows the results for profits.

Lastly, although we will also confirm our stacked control group results are robust against a more selected matched group, we show evidence that using all banks as controls is reasonably consistent with a small SUTVA violation. First, mergers are in fairly competitive markets with many banks. Figure 2 shows a map of Federal Reserve defined banking markets colored by their Herfindahl-Hirschman index. Additional scrutiny is levered on combinations that involve banking markets whose HHI exceeds 1800.¹⁴ 92% of the U.S. population lived in a deposit banking markets with Herfindahl-Hirschman index below 1800 in 2022. Second, mergers are between small banks (as established in Figure 1) so consequently have limited externalities on competitive markets.

4 Capital constrains bank mergers

This section presents evidence of a regulatory capital wedge that reduces banks' returns to mergers, whereby binding capital regulation is incompatible with purchasing potential merger targets in cash. We proceed by explaining the mechanism then describing theoretical predictions that we test in data. We test both direct evidence of the mechanism and additional model predictions to validate our findings.

4.1 Mechanism

Asymmetric information causing frictional equity issuance to outside shareholders is well known as a deterrent to firm investment in new assets. However, in the case of M&A investment (Jovanovic and Rousseau, 2002), firms could theoretically overcome this failure of Modigliani and Miller (1958) by issuing new equity to their *target's* shareholders; financing the acquisition of the target in stock by directly paying the target in stock. Ostensibly, target shareholders could be more informed than other investors. They would see the value in the potential combination and would

¹⁴Cooperman and Walsh (2026) establishes the lack of responsiveness of mergers to such market indices.

accept stock in the new firm as payment for their shares. But, if they do not, asymmetric information persists and attempting to issue equity to the target shareholders would be no better than attempting to issue outside shareholders. That bias falls particularly on buyers whose stock is not publicly traded, where the information gap would be the largest and the stock would be most illiquid currency for a potential target.

Firms could then, typically, overcome this additional friction by financing their investment in ways that increase firm leverage in order to pay their targets in *cash*. Two procedures are available. First, firms could procure new cash with debt. They could issue new debt, use that new cash to purchase the target and create a larger combined firm whose assets were supported by the same equity base plus far more debt. Another attractive solution is using cash it has available to pay the target. In this way, firms cut into their existing asset base to pay the target and create a smaller, but less levered, combined firm.

However, neither solution is readily available in industries when firm leverage is capped. When U.S. banks have already funded as many assets in deposits as they can and bind to their minimum equity capital ratio, all new investment must be equity financed. U.S. banks, who lever up to their minimum equity capital requirements by continually funding new loans with deposits, have little remaining wiggle room for cash- or debt-financing large investments such as the acquisition of a peer. Even if firms choose to lever to purchase their target, since the net effect of their transaction must keep them sufficiently capitalized, such debt- and cash-financed transactions could require re-raising equity to “recapitalize”. We refer to this wedge between mergers one would expect to occur in stock than those that do in reality as the regulatory capital wedge (RCW).

4.2 Testing for extensive margin effects of capital

The most direct test of the RCW is that bank merger pairs with more excess capital are more likely, all else equal, to occur. Several analyses are consistent with this mechanism: merger pairs with more capital in the cross-section are more likely to occur, and a Bartik instrument on target capitalization raises 12-month probability by 50% in 12 months.

4.2.1 Background: Capital regulation

We now make the constraint of capital regulation tangible. Bank equity capital is regulated to exceed several well-defined ratios. We will focus on the Tier 1 Leverage Ratio alone in our baseline specification. The Tier 1 Leverage Ratio must exceed 4% and is defined as

$$r_{it} \equiv \text{Tier 1 Leverage Ratio}_{it} = \frac{\text{Tier 1 Capital}_{it}}{\text{Average Total Consolidated Assets}_{it}} \equiv \frac{K_{it}}{A_{it}} \quad (4)$$

where Tier 1 Capital is reported on the Call Reports by banks but is almost entirely composed of common stock and retained earnings. Regulators (the Federal Reserve, OCC, and FDIC) require the maintenance of equity exceeding these ratios by promising immediate corrective action involving

capping dividend payouts or mandatory capital raises.

Method of payment and the post-merger leverage ratio. The Tier 1 Leverage Ratio disciplines mergers because the two methods of payment leave the pro-forma bank at very different points relative to the 4% floor. Consider a buyer i with Tier 1 capital K_{it} and average total consolidated assets A_{it} acquiring a target j with (K_{jt}, A_{jt}) , and normalize the purchase price to the target's book equity K_{jt} .¹⁵

In a stock-for-stock merger, no cash leaves the pro-forma bank and the seller exchanges her equity claim for the buyer's. Both banks' capital and assets survive the transaction:

$$R_{ijt}^{\text{Stock}} \equiv \frac{K_{it} + K_{jt}}{A_{it} + A_{jt}} \quad (5)$$

where we use uppercase R_{ijt} to denote the *combined* bank's new capital ratio. The stock-financed ratio is simply the asset-weighted average of the two standalone ratios: acquiring a better-capitalized target *raises* the buyer's leverage ratio.

In a cash merger, the buyer pays K_{jt} out of its own liquid assets and the target's equity is extinguished. Only the buyer's capital survives to support the combined asset base, and the combined asset base itself shrinks by the cash paid out:

$$R_{ijt}^{\text{Cash}} = \frac{K_{it}}{A_{it} + A_{jt} - K_{jt}}. \quad (6)$$

Comparing (5) and (6),

$$R_{ijt}^{\text{Stock}} - R_{ijt}^{\text{Cash}} \geq 0, \quad (7)$$

with strict inequality whenever the target has positive equity and the pro-forma bank has any deposits or other debt. A cash merger therefore always leaves the pro-forma bank closer to the regulatory floor than the identical merger financed in stock, and the wedge is increasing in the size of the target and in the leverage of both banks. The comparison to the buyer's *standalone* ratio is equally stark: $R_{ijt}^{\text{Cash}} < r_{it} \equiv K_{it}/A_{it}$ whenever $A_{jt} > K_{jt}$, so a cash acquisition of any levered target mechanically consumes the buyer's capital buffer, while a stock acquisition need not.¹⁶

Equation (6) also caps how much a bank can buy without issuing equity. Writing $r_{jt} \equiv K_{jt}/A_{jt}$,

¹⁵Any premium above book, paid in cash, only tightens the comparison below: goodwill and other intangibles are deducted from Tier 1 capital, so a cash premium reduces the numerator as well as the denominator. Normalizing the price to K_{jt} therefore understates the regulatory cost of cash.

¹⁶Consider a numerical example. A bank with \$100 in assets and \$10 in equity would like to purchase a bank with \$20 in assets and \$2 in equity. The buyer could pay management \$2 in cash for their equity and the combined bank would have \$118 in assets and \$10 in equity. The transaction has increased leverage: it has reduced the Tier 1 Leverage Ratio from 10% to 8.5%. Only a stock-for-stock merger, where the seller willingly changes out her equity for the buyer's, retains a capital ratio of 10%.

the constraint $R_{ijt}^{\text{Cash}} \geq 4\%$ holds only for targets no larger than

$$A_{jt} \leq \frac{K_{it} - 0.04A_{it}}{0.04(1 - r_{jt})}. \quad (8)$$

Acquisition capacity in cash is proportional to the buyer's excess capital $K_{it} - 0.04A_{it}$: at a 4% floor and a target capitalized at 10%, each dollar of excess Tier 1 capital supports roughly \$28 of target assets, and a buyer sitting exactly at the minimum can buy nothing at all in cash.

As a result, the simplest variation in RCW would simply be the distance a bank is from its regulatory equity minimum, where RCW is the largest when the distance is the smallest, $RCW \propto -(R_{ijt}^{\text{Cash}})$.

The risk of using this as a proxy for a regulatory capital wedge is that, in a dynamic sense, a banks could retain earnings and hold high Tier 1 Leverage Ratio *because* its cost of external equity issuance is large. We will report associations between Tier 1 Leverage Ratio and merger choice, implicitly requiring the cost of equity issuance to vary independently. However, we also repeat our analysis using plausibly exogenous variation in Tier 1 Leverage Ratio using an instrument in Section 4.2.3.

4.2.2 Tier 1 Leverage Ratio and merger propensity

First, we present correlational evidence that capital increases merger propensity. To do so, we construct potential buyer-seller pairs as our unit of observation. The potential buyer-seller pairs for 4,000 commercial banks just in 2024 is nearly 16 million ordered pairs. In order to keep the analysis tractable, we limit combinations to pairs headquartered in the same state and where the seller is smaller than the buyer, as we will in the empirical model in Section 5. Constructing observations in this way takes into account how constrained a particular merger pair is. The merger pair can be constrained because of the buyer's standalone leverage ratio, the seller's standalone leverage ratio, or, the relative size of the seller to the buyer.

Consistent with our mechanism, Figure 8 shows that the selection of the pairing is increasing in the capital that the target would have post-merger. Figure 8 separates combinations by the combinations value of R_{ijt}^{Cash} , then whether the combination has a privately held or a publicly traded buyer.

4.2.3 Instrumenting for capital with unrealized gains and the 2001 Pooling of Interest Accounting phase out

Because the endogeneity concern that costly equity issuance could be negatively correlated with distance from the Tier 1 Leverage Ratio, we first demonstrate the relevance of the mechanism in ordinary least squares regression using shocks to capital that would only be booked and hence incorporated into regulatory capital ratios in the event of a merger.

Merger-specific capital losses: unrealized gains. Importantly for the construction of our

instrument, banks are regulated on the *book* value of their ratios. So, in general, the frequent changes in the *market* value of their assets are not used to compute their regulatory capital minima. Barring selling the unmarked assets to another party, the only major reason for bank assets to be marked to market on their Call Reports is in the event of being purchased, through the accounting norm called Purchase Accounting.

Our measure of capital loss booked in the event of the merger uses HTM and AFS securities. Conveniently for identification, capital regulation uses a bank’s book value: even though banks report the fair value of their securities, the securities are booked at cost for regulatory purposes.

Since 2001, mergers have been accounted for on books only using purchase accounting, which requires marking the target firm’s assets to market when purchased rather than absorbing the target at a desired value (“pooling of interest accounting”).¹⁷ As a result, unrealized losses are a headwind (tailwind) for the regulatory capital of target banks, only in the event of an acquisition but not in everyday operations. In the framework presented, the unrealized losses and gains exogenously shifts the state variable, capital (and hence c^{Reg}), in a decision about whether to merge, but it leaves unchanged capital’s role as a state variable mediating the value of the standalone banks ($\Delta\pi_{\mathcal{M}(i,j)}$). This distinction makes it ideal in shifting capital to demonstrate the direct link between capitalization and a likelihood of engaging in M&A.

We calculate a measure of unrealized gains on these securities as a fraction of a bank’s regulatory capital.

$$\text{Unrealized Gains}_{jt} \equiv \frac{\text{Unmarked HTM/AFS assets}_{jt}}{\text{Assets}_{jt}}.$$

Dividing the numerator by total assets allows unrealized gains to measure the relative share of capital that is revealed specifically in a merger.

We then evaluate the importance of unrealized gains and post-merger leverage ratio on the extensive margin using this regression specification, for the panel 1994-2024 when fair-value reporting in the Call Reports began:

$$\mathbf{1}(j \text{ is a seller in a merger in } (t, t + h]) = \beta X_{it} + \tau_{\text{Post-2001}} (\text{Unrealized Gains})_{it} \mathbf{1}(t > 2001)_t + \tau_{\text{Pre-2001}} (\text{Unrealized Gains})_{it} \mathbf{1}(t \leq 2001)_t + \varepsilon_{it} \quad (9)$$

Table 3 shows the results of an OLS regression of being acquired on unrealized gains of a seller, for $h = 4$. As expected, $\tau_{\text{Post-2001}} > 0$. All mergers use purchase accounting, and so every merger marks target asset fair values to market. Further consistent with this being a causal association, the association is statistically insignificant prior to 2001 at the 95% confidence level. In the post-2001 regime, the effect of unrealized gains are also sizable, economically. The standard deviation post-2001 of unrealized gains was 1% of assets, in the full panel banks had a 2.2% chance of

¹⁷Prior to 2001, firms could use pooling of interest accounting, but only for stock-for-stock mergers. Effective in fiscal years ending June 2001, the carve out for pooling of interest accounting was phased out and acquisitions moved to purchase accounting.

being targets. So, a standard deviation increase in unrealized gains shifted the likelihood of a bank becoming a merger target by 18bps (8.5%).

The results are robust to using other covariates that may shift the quality of the bank as a potential merger target, such as being a less efficient or lower return on assets bank with room to improve.) Importantly, the regression coefficient on Tier 1 (Leverage) Ratio is negative: times when banks have more capital are *less* likely to be acquired in the next year. This shows why forthcoming causal identification to shift regulatory capital ratios independent of merger synergies is helpful for this setting.

Constructing an instrument for unrealized gains. Next, it is feasible that unrealized gains still retain the endogeneity problem of distance from the Tier 1 Leverage Ratio. Such unrealized gains could be due to especially strong performance in, for instance, mortgage securities that make the bank a more attractive target.

Consequently, we use differences in the duration of bank’s securities portfolios to generate plausibly exogenous movements the post-merger Tier 1 Leverage Ratio that a pro-forma bank would have. In particular, we instrument for unmarked assets with historical portfolio choices interacted with Treasury yield changes. The two equation model is:

$$\mathbf{1}(Seller)_{it} = \tau(\text{Unrealized Gains})_{it} + \beta X_{it} + \varepsilon_{it} \quad (10)$$

$$(\text{Unrealized Gains})_{it} = \gamma W_{it-8} \times \Delta y_t^{5-year} + \beta X_{it} + u_{it} \quad (11)$$

where W_{it-8} is the share of assets eight quarters prior allocated to long-term securities (defined as debt securities issued by U.S. Treasuries, U.S. agencies, MBS, and similar of a maturity of at least five years).

The exclusion argument is as follows. Banks make portfolio decisions with imperfect information about the future, namely, the future changes in 5-year yields. Banks’ portfolio choices, after controlling for covariates, are mean-independent of whether a bank is ultimately acquired. Importantly, the argument rules out any confounder that affects attractiveness as a merger target today that either also causes securities portfolio choice in the past or is directly caused by the historic portfolio choice. One concern is that the choices are sufficiently persistent such that the lagged portfolio is sufficiently confounded with latent factors determining the merger synergies of the target. For instance, if poorly managed banks could be especially strong merger targets, and poor management is persistent such that the poor management in choosing a long-term securities portfolio before an increase in yields, then we would *underestimate* the effect of capital on mergers because this would push banks with unrealized losses (less capital) to become merger targets.

Table 4 shows the results. The strong negative relationship in the first stage has an intuitive asset pricing rationale. The instrument constructs a prediction of the market value of the securities portfolio with the heuristic that long maturity securities are much more sensitive to bond yield changes than short maturity securities. Then, the sign of the yield curve change determines whether

those securities gain or lose value. An increase in the five-year yields means that the long-maturity securities lose significant value. So, one would expect a negative first stage and one that scales with how many long-maturity securities the bank possess. As expected, unrealized gains has meaningful residual variation outside of the Bartik instrument: only *some* of unrealized gains came from asset pricing identities imposed by external yield curve changes. Those changes are the changes that shift capital but are exogenous to the quality of the bank. The first stage F statistic is large; 2411 for the instrument without covariates. As expected, the significant second stage appears only in the post-2001 population. Table B.4 repeats the regression separately for the two subpopulations. Consistent with unrealized gains being a proxy for capital and with RCW being a salient cost of mergers, unrealized gains are only a significant predictor post-2001. If one were concerned that unrealized gains were positively correlated with other unobserved attributes of a bank, then the strength of the effect coming only when unrealized gains had to be incorporated into post-merger regulatory capital ratios cuts against those concerns.

Constructing an instrument for post-merger Tier 1 Leverage Ratio Now showing the strength of the instrument, we return to creating exogenous variation in a constructed proxy of regulatory cost, the post-merger Tier 1 Leverage Ratio to validate Prediction 1.

The true post-merger ratio involves the marking of the entire asset portfolio to market. The HTM and AFS securities are the only components of the bank's portfolio for which the market value is available (and reported as fair value). As a result, we construct an approximation for the post merger ratio incorporating HTM and AFS into Equation (5) as follows:

$$\tilde{r}_{jt} \equiv \frac{K_{jt} + \text{Unmarked HTM/AFS assets}_{jt}}{A_{jt} + \text{Unmarked HTM/AFS assets}_{jt}} \quad (12)$$

Hence, target with unrealized gains would bring a larger leverage ratio post-merger than their pre-merger regulatory ratio constructed using book values. In contrast, targets with unrealized losses would bring a smaller leverage ratio.

The two equation model instrumenting for $\tilde{r}_{jt}$ is

$$\mathbf{1}(\text{Seller})_{it} = \tau \tilde{r}_{it} + \beta X_{it} + \varepsilon_{it} \quad (13)$$

$$\tilde{r}_{it} = \gamma W_{it-8} \times \Delta y_t^{5\text{-year}} + \beta X_{it} + u_{it} \quad (14)$$

Table 5 shows the estimates for Equation (13). The first stage is weaker than instrumenting for unrealized gains, understandably, due to the larger number of confounders discussed above that determine equilibrium Tier 1 Leverage Ratio choice, but the F statistic of 525 is still large. Despite the additional noise, The covariates strengthen the effect such that a 1 pp increase in post-merger Tier 1 Leverage Ratio corresponds to a 19.3bp increase in probability of acquisition. However, given a standard deviation of leverage ratio of 5.63pp, the economic magnitude of the estimate grows. The estimates imply that a standard deviation increase in $\tilde{r}_{jt}$ raises the 1-year likelihood

of being acquired by 50% The effect is stable when adding covariates intended to proxy for the attractiveness of the bank as a merger target.

4.3 Additional predictions

We have shown directly that higher capital ratio merging pairs have a higher propensity to merge. The association holds in the cross-section and causally using an instrumental variable for capital ratio. The presence of RCW yields two additional predictions.

4.3.1 Prediction 1: Prevalence of cash mergers

We show that within realized mergers, the use of cash as payment is larger when a proxy for c_i^{Ext} are higher. The easiest proxy is whether one's stock is publicly or privately traded. The liquid method of payment for acquisitions theory of going public (Brau and Fawcett, 2006) makes the claim that public and privates face this difference in c_i^{Ext} . That prior work is consistent with reality.

To ascertain the transaction medium of bank mergers, we turn to the reported mergers available in S&P SNL. This bank merger dataset is not exhaustive in its coverage of bank mergers, as it relies on public announcements. Availability of RSSD numbers that could match the dataset to the Call Reports is also imperfect. As a result, the analysis is limited to 6,757 bank mergers occurring between 1991 and 2024, but not only commercial bank mergers defined in the same way as in Section 3.

Figure 7 shows the breakdown of mergers by financing medium and shows that while 62% of deals by private buyers are done exclusively in cash, only 25% of deals done by publicly traded buyers are. The difference is highly statistically significant. We can show this in a deal-level regression on percent of the transaction total transacted in cash. The regression we run, for deals i closed in years $t(i)$,

$$(\text{Percent of deal consideration in cash})_{it(i)} = \tau_t + \beta 1(\text{Private buyer})_{it(i)} + \varepsilon_{it} \quad (15)$$

Table 2 shows the relative frequency of cash by estimating the regression. Being a private buyer is associated with funding more than 30pp more of the deal value in cash. The gap between public and private buyers is robust to time fixed effects, the size of the buyer (measured in number of branches), and the size of the target.

4.3.2 Prediction 2: Profitability of constrained mergers

While the tradeoff between c^{Reg} and c^{Ext} has been well-established using cross-sectional variation above, the economic importance of the c^{Reg} mediating regime has not yet been established. In this section, we validate Prediction 3 in the data again using Tier 1 Leverage Ratio as our proxy for c_i^{Reg} . As expected, mergers more constrained by c^{Reg} unlock far more value for merging banks than those

unconstrained. In doing so, we show that the capital constraint on bank mergers has meaningful economic effects, which we will have a framework to quantify formally in Section 5.

Figure 9 reports the coefficients β_s from Equation (1) for two-years after merger for bank mergers in our sample, interacted with the quantile of pro-forma Tier 1 ratio two years prior to combination. We find that profits increased for pro-forma banks with the lowest quantile Tier 1 leverage ratio by 17%. Increases were smaller as the combined bank was more capitalized and as low as 2% for combinations that were the most capitalized prior to the transaction. The gradient in merger synergies between the lowest capitalized banks and highest capitalized banks is consistent with our mechanism. While the figure provides an intuitive proxy for c^{Reg} on the x-axis, a positive association between merger synergies between banks with c^{Reg} could contribute to the trend and justifies the Bartik instrument design we presented above.

Together, the evidence is consistent with these regulations being a meaningful deterrent to high value mergers by low capitalized banks. However, the evidence presented does not provide machinery for aggregating up the effect of changing such regulatory costs. To quantify the aggregate effect and account for heterogeneity between banks and heterogeneous choices, we present a structural model of endogenous mergers subject to such frictions. The model will deliver estimates for merger synergies created heterogeneously by different pairs and subject to capital regulation that can match moments like those presented in Figure 9.

5 Empirical model

In order to simulate the effects of counterfactual policy interventions, we develop and structurally estimate a model of the bank merger market. Vertically- and horizontally-differentiated banks compete à la Bertrand-Nash in both the deposit and loan markets, incurring heterogeneous fixed and variable operating costs. Each period, prior to product market competition, there is a “merger market” in which banks choose whether to sell their bank, buy another bank in their market, or continue operating in their current capacity. Business combinations may produce profit synergies, but combinations also incur non-pecuniary integration costs. Importantly, but complicating the model significantly, we account for selection on levels of merging versus nonmerging banks. In doing so, we estimate parameters of the model while limiting the bias that could arise from

5.1 Setup

Time is discrete and the horizon is infinite. At the beginning of time t in market k , there exist $\mathcal{J}_{kt}$ banks. The *ex-ante* state vector (i.e., prior to any merger activity) is given by:

$$\Omega_{kt} \equiv (A_{kt}, B_{kt}, K_{kt}, \xi_{kt}^D, \xi_{kt}^L, c_{kt}^D, c_{kt}^L, \kappa_{kt})$$

where A_{kt} is the vector of bank assets in market k at the beginning of year t , B_{kt} is the branch network of the banks, and K_{kt} is the vector of Tier 1 capital of every bank. We suppress market

subscripts k for ease of exposition.

Bank fundamentals. We define the bank flow profit function with several parameters in order to accommodate banks targeting cost synergies, market power, or both in their choice of merger partner. Banks indexed by j offer two vertically- and horizontally-differentiated products, loans and deposits, indexed by L and D . Banks earn profits from lending at a spread above the risk-free rate, $\rho_{jt}^L \equiv r_{jt}^L - R_t^F$, and borrowing at a spread below the risk-free rate, $\rho_{jt}^D \equiv R_t^F - r_{jt}^D$. Vertical differences stem from non-price product quality (ξ_{jt}^D, ξ_{jt}^L) and number of physical branch locations B_{jt} . As a result, they face demand for their products, market shares $(s_{jt}^L(\cdot), s_{jt}^D(\cdot))$, that are functions of their prices, quality differences, branches, and the characteristics of their competitors. Banks incur heterogeneous marginal costs in servicing their deposits and loans (c_{jt}^D, c_{jt}^L) , as well as heterogeneous fixed costs of operating, κ_{jt} . Operating profits are thus given by:

$$\pi_j(\Omega_t) \equiv M_t^L s^L(\rho_{jt}^L, \xi_{jt}^L, B_{jt}, \Omega_t)(\rho_{jt}^L - c_{jt}^L) + M_t^D s^D(\rho_{jt}^D, \xi_{jt}^D, B_{jt}, \Omega_t)(\rho_{jt}^D - c_{jt}^D) - \kappa_{jt} \quad (16)$$

where M denotes market size.

Merger technology. With this definition of profit, mergers can produce profit synergies by realizing synergies in bank fundamentals. Namely, the merger can increase quality, decrease marginal costs, or decrease fixed costs. We will assume only fixed cost synergies are created in our baseline results. So, if a bank i purchases bank j , an event we denote by $\mathcal{M}(i, j)$, their fundamentals evolve as:

$$\kappa_{\mathcal{M}(i,j),t} = \kappa_{it} + \kappa_{jt} + A_{jkt} [\delta^\kappa + a^\kappa \mathcal{O}(B_{it}, B_{jt}) - \zeta_{ijt}] \quad (17)$$

$$c_{\mathcal{M}(i,j),t}^d = \frac{A_{it}c_{it}^d + A_{jt}c_{jt}^d}{A_{it} + A_{jt}} \quad (18)$$

The fixed cost expression includes one pair-specific unobservable, ζ_{ijkt} . Each pair of banks i, j in every market k at period t has a pair-specific merger synergy as a fraction of their combined assets. ζ_{ijkt} follows a Type-1 Extreme Value distribution

$$\zeta_{ijt} \sim \text{T1EV}(0, \sigma_\zeta) \quad (19)$$

Finally, quality $\xi_{\mathcal{M}(i,j)}^d$ is the solution to

$$s^d\left(\frac{s_{it}^d \rho_{it}^d + s_{jt}^d \rho_{jt}^d}{s_{it}^d + s_{jt}^d}, \xi_{\mathcal{M}(i,j),t}^d, B_{\mathcal{M}(i,j),t}, \Omega_t\right) = s_{it}^d + s_{jt}^d \quad (20)$$

where branches $B_{\mathcal{M}(i,j),t} = B_{it} + B_{jt}$. In this way, holding prices fixed, consumers do not have higher demand for the combined firm, marginal costs are not lower, but fixed costs can decrease via a deterministic amount δ^κ , branch overlap, and a pair-specific synergy between the acquirer and target.

Selection. To correct for potential selection on levels between merged and unmerged banks, we account for the standalone ψ of merged banks to be unobserved. Bank fundamentals $\psi \equiv (\xi^D, \xi^L, c^D, c^L, \frac{\kappa}{A})$ are parameterized as functions of their lagged values and mean-zero, unobservable shocks that are i.i.d. across banks, markets, and time. So, for $d \in \{D, L\}$,

$$\xi_{jkt}^d = \eta_{\xi_d} \xi_{jt-1}^d + \tilde{\xi}_{jt}^d \quad (21)$$

$$c_{jkt}^d = \eta_{c_d} c_{jt-1}^d + \tilde{c}_{jt}^d \quad (22)$$

$$\frac{\kappa_{jkt}}{A_{jkt}} = \eta_{\kappa} \frac{\kappa_{jt-1}}{A_{jkt-1}} + \frac{\tilde{\kappa}_{jt}}{A_{jkt}} \quad (23)$$

The unobserved shocks are jointly normally distributed, where Σ encodes possible correlations between the unobserved shocks determining profits,

$$\tilde{\psi}_{jt} \equiv (\tilde{\xi}_{jt}^D, \tilde{\xi}_{jt}^L, \tilde{c}_{jt}^D, \tilde{c}_{jt}^L, \frac{\tilde{\kappa}_{jt}}{A_{jt}}) \sim \mathcal{N}(\mathbf{0}, \Sigma). \quad (24)$$

Timing. The timing of the game for a period t is as follows.

0. Bank fundamental ψ_t and merger-payoff shocks ζ_t are drawn from their joint distribution.
1. Banks are simultaneously consider *merger participation decisions* with allocation run by a central planner. All objects are updated to reflect any merger-induced changes.
2. Remaining banks post loan and deposit rates.
3. Depositors and borrowers choose banks, and both markets clear. Time progresses to period $t + 1$.

The remainder of this section describes the model step by step.

5.2 Step 0: Information

All banks, as well as the econometrician, observe the realizations of (A_{kt}, B_{kt}, K_{kt}) . All banks, but not the econometrician, observe ψ_t the shocks to fundamentals.

5.3 Step 1: Merger formation

The expected surplus to merging is change in the buyer i and the seller j 's sum of the discounted present value of utility the induced by a merger in period t . Denoting the present values of utility by U_j :

$$\mathbb{E}[U_{\mathcal{M}(i,j)} - U_i - U_j \mid \Omega_t] = \underbrace{\mathbb{E}\left[\sum_{s=0}^{\infty} \beta^s \Delta \pi_{\mathcal{M}(i,j)} \mid \Omega_t\right]}_{\text{Merger surplus}} - \underbrace{\mathcal{C}^B(\mathcal{O}(B_{it}, B_{jt}), \frac{K_{it}}{A_{it} + A_{jt} - K_{jt}}) - \mathcal{C}^S(A_{jt})}_{\text{Merger frictions}}. \quad (25)$$

Merger surplus. $\Delta\pi(\cdot)$ captures the difference in profits under the newly-combined bank relative to the sum of utilities of the two stand-alone banks in a counterfactual in which they do not merge. The precise function is given by Equations (17)-(18) through Equation (16). The expectation is taken with respect to the bank’s information set, which, in the model, is the full market’s Ω_t .

Merger frictions. Equation (25) makes explicit that merger surplus will be decreasing in merger frictions: the seller’s reserve value and the one-time costs of the merger. The one-time costs of the merger will be the focus of RCW, presented in Section 4.2.3.

The seller’s reserve value is derived from the present value of the utility of a bank’s owner. This quantity is the sum of the bank’s expected future profits and non-pecuniary benefits of operating a standalone bank, $\mathcal{C}^S(\cdot)$.

$$\mathbb{E}[U_j \mid \Omega_t] \equiv \mathbb{E}\left[\sum_{s=0}^{\infty} \beta^s \pi_{js} \mid \Omega_t\right] + \mathcal{C}^S(A_{jt}). \quad (26)$$

where β parametrizes the rate of time preference. One possible source for $\mathcal{C}^S$ is the benefits of keeping the bank “in the family.” Because $\mathcal{C}_{jt}^S$ increases the banker’s utility of operating their bank as a standalone entity, we refer to it as a “merger friction”.¹⁸

The one-time costs of the merger hold the bank’s financing friction. The buyer side frictions are a function $\mathcal{C}^B$ where $\mathcal{C}^B$ takes arguments for branch overlap and the Tier 1 Leverage ratio of the combined firm, if the buyer were to pay cash for the seller.

Choice of buyer. Each bank can sell to a bank that is larger, in assets, within its market. Banks sit out of the merger market by “selling” to themselves, realizing no merger synergies but paying no merger frictions. This means that for a market of size $\mathcal{J}_t$ in a given year, $\frac{\mathcal{J}_t(\mathcal{J}_t-1)}{2}$ potential mergers could occur. We now describe how this choice occurs.

Both to capture industry practices regarding the projection of merger returns better and to greatly simplify computational burden, we adopt a reduced-form approach to characterizing merger formation. We assume that the allocation mechanism is centralized and leverages pairwise “allocation merger surplus” quantities computed *ex ante*. In particular, the allocation merger surplus for a given pair is the merger surplus as if no other mergers occur in the pair’s market and as if no mergers will occur in the future. We define the surplus quantity for a given pair i, j as $\Phi(\mathcal{M}(i, j), \Omega_t)$.¹⁹

¹⁸This is not separately identified from a one-time cost of doing mergers, where purchasing a larger target incurs larger one-time costs of hiring supporting services to assess and implement the integration. Whether this cost that scales with seller assets is related to the buyer or seller side is not separately identified Agarwal, Li, and Somaini (Agarwal et al.) and so we will not interpret it. This is another advantage of our model setup that does not require observing prices in order to estimate parameters.

¹⁹In general, expected surplus and allocation merger surplus need not be equivalent. The realization of a merger between i, j could affect the surplus of another potential merger between ℓ, m , $\mathbb{E}_{-\mathcal{M}(\ell, m)}[\frac{\Delta u(\Omega_t)}{1-\beta}]$. For instance, one consequence of such an externality is *free-riding*: that mergers by other pairs in your market that realize sufficiently limited efficiency gains encourage firms remaining independent because the smaller number of firms increases market power and hence increases the value of remaining as a standalone firm (Stigler, 1968; Gowrisankaran and Holmes, 2004). Such free-riding incentives shrink with the number of firms in the market. Given the competitiveness of the

Given the data generating process specified, the sum of future profits can also be simplified into a deterministic, discounted sum.

$$\Phi(\mathcal{M}(i, j), \Omega_t) \equiv \begin{cases} 0 & i = j \\ \frac{\Delta\pi_{\mathcal{M}(i, j)}(\Omega_t)}{1-\beta} - \mathcal{C}^B(\mathcal{O}(B_{it}, B_{jt}), \frac{K_{it}}{A_{it}+A_{jt}-K_{jt}}) - \mathcal{C}^S(A_{jt}). & i \neq j \end{cases} \quad (27)$$

Every bank submits the identity of the bank (self-inclusive) to which selling has the highest pairwise “allocation merger surplus”. The choice set of a seller j of buyers is the set of banks in the market (self-inclusive) who have at least as many assets as seller j , which we symbolize with $\mathcal{B}_{jt} \subset \mathcal{J}_t$. Formally, for a hypothetical merger between i and j , $\mathcal{M}(i, j)$, the choice of identity of buyer i that seller j submits is:

$$i = \arg \max_{\ell \in \mathcal{B}_{jt} \cup \{j\}} \Phi(\mathcal{M}(\ell, j), \Omega_t) \quad (28)$$

Denote the resulting choices of every bank j over identities of buyers i with a $\mathcal{J}_t \times \mathcal{J}_t$ matrix $\mathbf{P}_t$, where for bank j with a maximum surplus buyer i , $P_{i,j,t} = 1$ and $P_{\ell,j,t} = 0$ for all other banks in the market.

Despite our simplifying assumption that allocation uses myopic surplus, Equation (27) is still a complicated, nonlinear function of all banks’ characteristics, primitives, and equilibrium strategies of acquisition and sale. For example, the computation of profit synergies $\Delta\pi$ requires performing a merger simulation of the surplus created for a seller selling to each potential buyer – computing post-merger quality, marginal cost, and fixed cost for every partner and solving the fixed point problem of Bertrand-Nash pricing given competitor prices for every partner. In assuming a mechanism that matches on highest surplus, we can use allocation to identify parameters without taking a stance on equilibrium prices, which are often unobserved.²⁰

5.4 Step 2: Price-setting in deposit and lending markets

After the merger market has cleared, variables for banks that merged are updated to their combined values. Then, banks set prices to maximize profits. For deposits and loans, this price is the spread – the amount above the risk free rate that banks are willing to lend at, and the amount below the risk free rate that banks are willing to offer depositors. Banks set prices in the deposit and loan markets ($d \in \{D, L\}$) following the standard Bertrand-Nash pricing equation:

$$\rho_{jt}^d = \arg \max_r \left\{ (r - c_{jt}) M_t^d s_{jt}^d(r, B_{jt}, \Omega_t) \right\} \quad (29)$$

banking sector and the importance of cost efficiencies in merging established, allocation merger surplus excludes this force.

²⁰This centralized allocation can also be micro-founded by separate ascending auctions, separate second price auctions, or a centralized planner, as we discuss here. In the auction setup, banks must send bids for each other bank separately, without taking into account how their valuation of two banks may be larger (or smaller) than twice their valuation for holding each bank separately. We discuss the observational equivalence of these microfoundations in Appendix E.2.

where s_{jt}^D denotes the bank's deposit market share.

5.5 Step 3: Depositor and borrowers select banks

We will now make explicit that demand for deposits and loans (s_{jt}^D, s_{jt}^L) come from the utility maximizing decisions of consumers. Deposit and lending markets are structured identically; the exposition here, laid out in terms of the deposit market, extends without alteration to borrowers in the lending market.

Households demand deposit accounts and can choose one bank j from the remaining set $\mathcal{J}_t$ available in market k to deposit their funds. Alternatively, as an outside option, households can elect to place their funds in an investment vehicle paying the market return (e.g., a money market fund). Depositors derive utility from promised deposit rates and other bank services. The indirect utility derived by depositor i from bank j in market k at time t is given by:

$$u_{ijt}^d = \alpha^d \rho_{jt} + \beta^d \log(B_{jt}) + \xi_{jt}^d + \varepsilon_{ijt}^d. \quad (30)$$

The utility of the outside option is normalized to 0. Without loss of generality and for ease of exposition, we write the utility of a depositor in terms of deposit spreads (ρ_{jt}) rather than deposit rates. The spread then reflects the “price” that depositors pay for banking services. The parameter $\alpha < 0$ measures the disutility depositors experience from paying higher spreads (i.e., prices).

Depositors also derive utility from banking services not captured by the price, captured by $\beta \log(B_{jt}) + \xi_{jt} + \varepsilon_{ijkt}$. We use the logarithm of number of branches as the product characteristic. The term ξ_{jt} appears in the merger market: it reflects additional unobserved quality differences at the bank-by-market-by-time level. For example, Bank of America may offer superior services relative to Citibank in a specific market, for instance due its online user interface. The term ε_{ijkt} captures horizontal differentiation in the marketplace. In order to avoid burdening the merger market estimation with too many parameters, the demand model is stylized to allow for unobserved quality and to match the granularity of the state-level merger market.

All depositors select the bank that maximizes their indirect utility. Following the demand estimation literature, we assume that the depositor-bank-market-time specific utility shocks (ε_{ijkt}) are distributed as Type-1 Extreme Value (Berry, 1994; Berry et al., 1995). This depositor optimization implies that the market share of bank j in market k at time t (i.e., the probability a depositor selects bank j in market k at time t) is given by:

$$s^d(\rho_{jt}^L, \xi_{jt}^L, B_{jt}, \Omega_t) = \frac{\exp(\alpha^d \rho_{jt} + \beta^d \log(B_{jt}) + \xi_{jt})}{1 + \sum_{\ell \in \mathcal{J}_t} \exp(\alpha^d \rho_{\ell t} + \beta^d \log(B_{\ell t}) + \xi_{\ell t})} \quad (31)$$

5.6 Solution concept

The allocation of banks to merger partners and subsequent prices and allocations of deposits and loans to consumers proceeds as follows.

Step 1: Merger formation. The central planner uses the following algorithm to assign merger pairs:

1. Each bank is matched to their highest allocation surplus buyer
2. If the match leads to a chain where a buyer is also a seller, the final buying bank purchases all selling banks.

Step 2: Price-setting. Remaining banks compete horizontally within market Bertrand-Nash and accordingly set profit-maximizing interest rates for deposits and loans.

Step 3: Depositors and borrowers select banks. Depositors and borrowers choose banks to offer deposit accounts and loans by maximizing utility.

This allocation is unique. The statement is formalized in Theorem 1 with a proof in Appendix E.2.

Theorem 1. *With bank merger choices Equation (28), price setting Equation (29), and consumer choice Equation (30) with primitives Ω_t , there is a unique matching (allocation) of sellers to buyers M , surviving banks $\mathcal{J}'_{kt}$, and set of prices ρ^D, ρ^L .*

In Appendix E.2, we present a simultaneous auction microfoundation. We also prove that the uniqueness from Equation (28) can be recovered from the auction microfoundation of the merger market.

5.7 Discussion of model set-up and assumptions

Our approach offers two clear advantages for empirical work. First, this assumption has the attractive quality that in the two-sided market case “competitive assignment game”, this allocation is efficient (Shapley and Shubik, 1972; Chiappori et al., 2012).

Second, this modeling choice is parsimonious. Rather than choosing an extensive form game whose details we do not observe and may vary across merger negotiations, we impose that seller choices are efficient given a design of merger surplus that aligns with industry practice. Limiting how externalities can affect merger choices greatly simplifies the model, but through modeling merger synergies, a bank merger still generates equilibrium spillovers on other banks through product market competition with new ξ^D, c^D, κ .

Lastly, complementary to the difference-in-difference evidence in Section ??, our model explicitly accounts for some econometric endogeneity of mergers. There can be *selection on gains*, or “essential heterogeneity,” in the spirit of Roy (1951): banks are more likely to pursue mergers that are more profitable. Difference-in-differences identifies the average effect on the treated, and extrapolating this estimate to become the average treatment effect for *any* bank pair would be to assume away selection on gains. To capture selection on gains, we allow merger synergies between

prospective bank pairings to differ on both observables as well as an unobservable shock that is additively separable in the profit function and independent of all other shocks.²¹ In this way, if more mergers occurred, the fixed cost synergies of the marginal merger would decline, and since mergers created less consumer choice, the detriment of the marginal merger consumers would increase. We turn now to estimation of the model.

6 Estimation

Estimation first uses the approach of [Berry \(1994\)](#) to estimate consumer demand and parameters of bank profit functions. Then, the central econometric challenge is the joint endogeneity of merger decisions with unobserved shocks to bank fundamentals (selection on levels) and to pair-specific synergies (selection on gains). We address both by constructing the joint likelihood of merger outcomes and post-merger primitives under the model’s distributional assumptions, and integrating over the unobservables that drive selection. In this way, banks select merger partners in a way akin to product choice with logit demand and random coefficients. That “first stage” selection of merger partners can then be corrected for in the “second stage” in a manner similar to a [Heckman \(1979\)](#) selection correction.

6.1 Step 3: Depositors and borrowers select banks

The selection of depositors and borrowers into banks identifies their demand parameters, namely the price elasticities, their demand for branches, and the latent quality of each firm. As before, the deposit and loan markets proceed analogously. In order to account for the endogeneity of price and the quality of a given bank’s deposits or loans, we instrument ρ_{jkt}^d for $d = \{D, L\}$ with operating cost intensity, defined as 1-year lagged fixed expenses as a share of bank assets. The two equation model becomes:

$$\log s_{jkt}^d = \alpha^d \rho_{jkt}^d + \beta^d \log(B_{jkt}) + \eta_{kt}^{d,2} + \varepsilon_{jkt}^d \quad (32)$$

$$\rho_{jkt}^d = \gamma^d \text{Operating cost intensity}_{jkt} + \eta_{kt}^{d,1} + \varepsilon_{jkt}^d \quad (33)$$

for market and time fixed effects $\eta_{kt}^{d,1}, \eta_{kt}^{d,2}$. The assumption is that this rolling average is excluded from latent demand and so can act as an instrument for ρ and can identify α^d . Inverting the multinomial-logit demand system at observed shares and prices from Equation (31) yields the unobserved quality residuals for each (j, k, t) ,

$$\xi_{jkt}^d = \log s_{jkt}^d - \alpha^d \rho_{jkt}^d - \beta^d \log(B_{jkt}) - \eta_{kt}^{d,2}, \quad d \in \{D, L\}. \quad (34)$$

²¹It is possible to allow for more general forms of selection on gains, though at greatly-increased computational cost in estimation – we discuss the contours of such an approach in the estimation section.

6.2 Step 2: Price-setting in deposit and lending markets

Given (α^d, ξ_{jkt}^d) , the Bertrand-Nash first-order condition in Equation (29) gives variable costs:

$$c_{jkt}^d = \rho_{jkt}^d - \frac{1}{\alpha^d(1 - s_{jkt}^d)}. \quad (35)$$

The net fixed cost is the residual between observed total profits and the sum of model-implied variable profits:

$$\kappa_{jkt} = M_{kt}^D s_{jkt}^D (\rho_{jkt}^D - c_{jkt}^D) + M_{kt}^L s_{jkt}^L (\rho_{jkt}^L - c_{jkt}^L) - \pi_{jkt}^{\text{obs}}. \quad (36)$$

The parameters from Steps 2 and 3 are then used as data in Step 1.

6.3 Step 1: Merger market

Under the model's distributional assumptions, the joint density of observed primitives and merger assignments factors into an outcome-stage and a selection-stage contribution:

$$\log \mathcal{L}(\theta) = \sum_{k,t} \left[\log L_{kt}^{\text{out}}(\theta) + \log L_{kt}^{\text{sel}}(\theta) \right]. \quad (37)$$

The selection stage uses data on selection of sellers into potential buyers or in staying alone. The outcome stage uses data of the level of parameters $y_{ikt} \equiv (\xi_{ikt}^D, \xi_{ikt}^L, c_{ikt}^D, c_{ikt}^L, \kappa_{ikt})$ given merger selection choices. For non-merger banks, the recovered AR innovations would be unconditionally distributed as $\mathcal{N}(\mathbf{0}, \Sigma)$ – but observing that bank j *did not* merge is itself a selection event, so the conditional distribution of the shocks given the realized assignment in (k, t) is mean-shifted. We absorb this shift with a non-merger Mills correction $\hat{\mu}_{jkt}$, analogous to the classical inverse-Mills correction (Heckman, 1979). For merger pairs, we observe only the post-merger combination $\psi_{\mathcal{M}(i,j),kt}$, not the standalone (y_{ikt}, y_{jkt}) ; the residual after subtracting the deterministic prediction and a control-function shift $\hat{\lambda}_{\mathcal{M}(i,j),kt}$ that absorbs the conditional mean of the chosen pair's T1EV synergy shock is treated as Gaussian.

Define the draw r for one bank j in market k at time t as $\tilde{y}_{kt}^{(r)} \equiv (\tilde{\xi}_{kt}^{d,(r)}, \tilde{c}_{kt}^{d,(r)}, \frac{\tilde{\kappa}_{kt}^{(r)}}{\mathbf{A}_{kt}})$, or $\tilde{\mathbf{y}}_{kt}^{(r)}$ for the whole market vector. For each (k, t) and each Monte Carlo replication $r = 1, \dots, R$, we draw a vector $\tilde{y}_{kt}^{(r)}$ from the appropriate conditional distribution (Equation (24)).²² The non-merger and merger Mills shifts are both constructed as importance-weighted averages over Monte Carlo draws of the latent shocks, with weights given by the per-draw selection likelihood.

²²The base shocks $\varepsilon_{kt}^{(r)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ are drawn *once* at the start of estimation and held fixed across optimizer iterations; θ -dependent affine transformations convert these standardized draws into conditional draws at any candidate θ . This holds the simulated likelihood smooth as a function of θ , which materially speeds optimization.

6.3.1 Likelihood function for merger choices

We start by deriving the selection term of the likelihood.

Counterfactual profits. Given $\Omega_{kt}^{(r)}$, we construct the profit synergies that drive selection. The baseline (no-merger) per-bank profit at draw r is

$$\pi_{j,kt}^{(r)} = M_{kt}^D s_{jkt}^{D,(r)} (\rho_{jkt}^D - c_{jkt}^{D,(r)}) + M_{kt}^L s_{jkt}^{L,(r)} (\rho_{jkt}^L - c_{jkt}^{L,(r)}) - \kappa_{jkt}^{(r)}, \quad (38)$$

where shares $s_{jkt}^{d,(r)}$ are computed from the logit demand system at the drawn $\xi^{d,(r)}$, holding all banks' prices fixed at their observed values. For each candidate pair (i, j) , the counterfactual profit of the merged entity is

$$\pi_{\mathcal{M}(i,j),kt}^{(r)} = M_{kt}^D s_{\mathcal{M}(i,j),kt}^{D,(r)} (\rho_{ikt}^D - c_{\mathcal{M}(i,j),kt}^{D,(r)}) + M_{kt}^L s_{\mathcal{M}(i,j),kt}^{L,(r)} (\rho_{ikt}^L - c_{\mathcal{M}(i,j),kt}^{L,(r)}) - \kappa_{\mathcal{M}(i,j),kt}^{(r)}, \quad (39)$$

where the merged-entity primitives are constructed as a function of δ via (17)–(18).²³

Merger frictions. We parameterize merger frictions for estimation. We impose that $\mathcal{C}^S$ scales with seller assets,

$$\mathcal{C}_{jkt}^S \equiv c_S A_{jkt}. \quad (40)$$

Then, because branch overlap $B_{\mathcal{M}(i,j)kt}$ appears in our merger synergy equations, the outcome equation analog, (Heckman, 1979), we allow it to enter the first stage. We parameterize RCW to scale with seller assets and to apply differentially to privately held, rather than publicly traded buyer banks, consistent with Prediction 1 and empirical merger propensities. We define the indicator for privately held as $D_{ikt} = \{0, 1\}$.²⁴ We allow the bank's distance from the Tier 1 Leverage ratio is

²³Holding non-focal banks' prices fixed at observed values is the key simplification that makes the per-pair counterfactual tractable. The merged entity's logit denominator updates in closed form: $D_{ij,kt}^{(r)} = D_{kt,\text{baseline}}^{(r)} + \exp(V_{ij,kt}^{\mathcal{M},(r)}) - \exp(V_{ikt}^{(r)}) - \exp(V_{jkt}^{(r)})$, where V denotes the indirect utility index. This avoids re-solving the equilibrium for every (R, J, J) triple.

²⁴Public banks, by virtue of $\mathcal{C}^S$, still face one-time merger costs, but since banks are either private or public, they also must act as the reference category and are subsumed into the fixed cost estimated for $\mathcal{C}^S$.

estimated flexibly in a piecewise manner as the following:

$$\begin{aligned}
C_{jkt}^B \equiv & c_4 D_{ikt} \mathbf{1}\left(\frac{K_{ikt}}{A_{ikt} + A_{jkt} - K_{jkt}} \leq 4\%\right) A_{jkt} + c_5 D_{ikt} \mathbf{1}\left(\frac{K_{ikt}}{A_{ikt} + A_{jkt} - K_{jkt}} \in [4, 5)\%\right) A_{jkt} \\
& + c_6 D_{ikt} \mathbf{1}\left(\frac{K_{ikt}}{A_{ikt} + A_{jkt} - K_{jkt}} \in [5, 6)\%\right) A_{jkt} + c_7 D_{ikt} \mathbf{1}\left(\frac{K_{ikt}}{A_{ikt} + A_{jkt} - K_{jkt}} \in [6, 7)\%\right) A_{jkt} \\
& + c_8 D_{ikt} \mathbf{1}\left(\frac{K_{ikt}}{A_{ikt} + A_{jkt} - K_{jkt}} \in [7, 8)\%\right) A_{jkt} + c_{10} D_{ikt} \mathbf{1}\left(\frac{K_{ikt}}{A_{ikt} + A_{jkt} - K_{jkt}} \in [8, 10)\%\right) A_{jkt} \\
& + c_{12} D_{ikt} \mathbf{1}\left(\frac{K_{ikt}}{A_{ikt} + A_{jkt} - K_{jkt}} \in [10, 12)\%\right) A_{jkt} + c_{max} D_{ikt} \mathbf{1}\left(\frac{K_{ikt}}{A_{ikt} + A_{jkt} - K_{jkt}} \geq 12\%\right) A_{jkt} \\
& + c_{Branch} \frac{B_{\mathcal{M}(i,j)kt}}{B_{ikt}} A_{jkt}
\end{aligned} \tag{41}$$

Selection probabilities. Using the merger-surplus expression (27), the deterministic surplus of pair (i, j) at draw r is

$$\bar{\Phi}_{ij,kt}^{(r)} \equiv \frac{\Delta \bar{\pi}_{ij,kt}^{(r)}}{1 - \beta} - c_{i,kt}^{B,(r)} - c_{j,kt}^{S,(r)}, \tag{42}$$

The pairwise deterministic profit gain is $\Delta \bar{\pi}_{ij,kt}^{(r)} \equiv \pi_{ij,kt}^{\mathcal{M},(r)} - \pi_{i,kt}^{(r)} - \pi_{j,kt}^{(r)} - \zeta_{ijkt}$. Then define the parameters of the synergy function $\theta \equiv (c_S, c_4, c_5, c_6, c_8, c_7, c_8, c_{10}, c_{12}, c_{max}, c_{Branch}, \delta^\kappa)$.

Applying the distributional assumption for ζ_{ijt} from Equation (19) and (27), the multinomial-logit probability that after Step 1 (before chain resolution), buyer i is matched to any given seller j is:

$$\Pr(P_{ij,kt} = 1 \mid \tilde{\mathbf{y}}_{kt}^{(r)}; \theta) = \frac{\exp(\bar{\Phi}_{ij,kt}^{(r)} / s_\zeta)}{1 + \sum_{\ell \in \mathcal{B}_{jkt}} \exp(\bar{\Phi}_{\ell j,kt}^{(r)} / s_\zeta)}, \quad s_\zeta \equiv \frac{\sigma_\zeta A_j}{1 - \beta}. \tag{43}$$

Accordingly, the probability that seller j remains standalone is the probability of choosing the outside option:

$$\Pr(P_{jj,kt} = 1 \mid \tilde{\mathbf{y}}_{kt}^{(r)}; \theta) = 1 - \sum_{\ell \in \mathcal{B}_{jkt}} \Pr(P_{\ell j,kt} = 1 \mid \tilde{\mathbf{y}}_{kt}^{(r)}; \theta) = \frac{1}{1 + \sum_{\ell \in \mathcal{B}_{jkt}} \exp(\bar{\Phi}_{\ell j,kt}^{(r)} / s_\zeta)}. \tag{44}$$

Like estimation of matching in trading networks, (Fox, 2017) the selection likelihood is the likelihood of every realized sale occurring (whether to a another buyer or two oneself). By our assumption myopic surplus, decisions are made independently across sellers. So, the per-draw conditional likelihood of the entire ownership matrix before chain resolution is therefore

$$\Pr(\mathbf{P}_{kt} \mid \tilde{\mathbf{y}}_{kt}^{(r)}; \theta) = \prod_{(i,j): P_{ij,kt}=1} \Pr(P_{ij,kt} = 1 \mid \tilde{\mathbf{y}}_{kt}^{(r)}; \theta) \prod_{j: P_{jj,kt}=1} \Pr(P_{jj,kt} = 1 \mid \tilde{\mathbf{y}}_{kt}^{(r)}; \theta). \tag{45}$$

Note that the potential for chains to arise between buyers and sellers, and hence a lack of observability of $\mathbf{P}_{kt}$, creates a partial identification problem. Define the ownership matrix as $\mathbf{M}_{kt}$. Each element of the ownership matrix captures whether bank i chooses to sell to itself ($M_{ii} = 1, M_{i\ell} = 0, \forall \ell \neq i$) or to another bank ℓ ($M_{\ell j} = 1$). The model specifies that the ownership

data reflect the merger market after resolving chains within a market and year, $\mathbf{M}_{kt} = P^{\mathcal{J}_{kt}-1}$. However, because more than one matrix P could create $\mathbf{M}_{kt}$, multiple sets of parameter values are observationally equivalent. One salient preimage P of $\mathbf{M}_{kt}$ is the matrix where no chains needed to be resolved: $P = P^{\mathcal{J}_{kt}-1}$. In this case, the initial choice of sellers to match to their highest surplus buyer, where the first part of Step 1 choice created an allocation without chains. For estimation, we estimate parameters using this root. In other words, we will assume that no chains needed to be resolved in every market and year and so $\mathbf{M}_{kt}$ can substitute for P_{kt} .

$$L_{kt}^{\text{sel},(r)}(\theta) = \prod_{(i,j): M_{ij,kt}=1} \Pr(M_{ij,kt} = 1 \mid \tilde{\mathbf{y}}_{kt}^{(r)}; \theta) \prod_{j: M_{jj,kt}=1} \Pr(M_{jj,kt} = 1 \mid \tilde{\mathbf{y}}_{kt}^{(r)}; \theta). \quad (46)$$

Integrating over draws r ,

$$\hat{L}_{kt}^{\text{sel}}(\theta) = \frac{1}{R} \sum_{r=1}^R L_{kt}^{\text{sel},(r)}(\theta) \quad (47)$$

6.3.2 Likelihood function for outcomes

In this section, we correct for selection on levels between merging and nonmerging banks using the outcome likelihood. The key difference in the outcome likelihood is that is based on the configuration of sellers matched to buyers that actually occurred. So, notably, if banks A and B were sold simultaneously to the same buyer C, the seller selection likelihood used the sellers A and B's myopic choices of the highest surplus buyer C to rationalize realized merger configuration. But, once A and B's selections are made, the outcome of the new bank ABC combines ζ_{ACt} , ζ_{BCt} , ψ_{At} , ψ_{Bt} , ψ_{Ct} .

The per-draw conditional likelihoods $L_{kt}^{\text{sel},(r)}(\theta)$ defined in (46) induce normalized importance weights

$$w_{kt}^{(r)}(\theta) \equiv \frac{L_{kt}^{\text{sel},(r)}(\theta)}{\sum_{s=1}^R L_{kt}^{\text{sel},(s)}(\theta)}. \quad (48)$$

For any function of the latents conditional on the realized assignment:

$$\hat{\mathbb{E}}[g(\tilde{\mathbf{y}}_{kt}) \mid \mathbf{M}_{kt}] = \sum_{r=1}^R w_{kt}^{(r)} g(\tilde{\mathbf{y}}_{kt}^{(r)}).$$

Hence, the model-implied conditional expectations of the latent fundamentals are

$$\hat{\mathbb{E}}[\tilde{\kappa}_{jkt} \mid \mathbf{M}_{kt}] = \sum_r w_{kt}^{(r)} \tilde{\kappa}_{jkt}^{(r)}, \quad (49)$$

The gap between the unconditional means (zero) and the posteriors in (49) quantifies the magnitude of selection on levels in the estimated equilibrium, separately for fundamentals and for frictions.

Control function for ζ on mergers. A standard property of the T1EV distribution gives the conditional mean of the chosen pair's synergy shock in closed form:

$$\mathbb{E}[\zeta_{ij,kt} \mid M_{ij,kt} = 1, \mathbf{y}_{kt}] = \sigma_\zeta (\gamma_{EM} - \ln \Pr(M_{ij,kt} = 1 \mid \mathbf{y}_{kt}; \theta)), \quad (50)$$

where γ_{EM} is the Euler-Mascheroni constant. Integrating over the simulated posterior of the latents,

$$\hat{\lambda}_{ij,kt} \equiv \sum_{r=1}^R w_{kt}^{(r)} (\gamma_{EM} - \ln \Pr(M_{ij,kt} = 1 \mid \mathbf{y}_{kt}^{(r)}; \theta)), \quad (51)$$

with $w_{kt}^{(r)}$ the importance weights from (48).

Merged firm selection correction. For each bank j who sold to itself in the merger market or newly merged banks $\mathcal{M}(i, j)$ in (k, t) , the post-merger combination outcome $\psi_{\mathcal{M}(i,j),kt}$ are equal to

$$\psi_{\mathcal{M}(i,j),kt} = \tilde{\psi}_{ikt} + (\boldsymbol{\delta}_s \odot \tilde{\psi}_{jkt}) + (\boldsymbol{\eta} \odot \psi_{i,k,t-1}) + \boldsymbol{\delta}_s \odot (\boldsymbol{\eta} \odot \psi_{j,k,t-1}) + \zeta_{ij,kt} A_{jkt} e_\kappa \quad (52)$$

where $\odot$ denotes elementwise multiplication and e_κ is the unit vector selecting the κ coordinate. Naively, $\tilde{\psi}_{jkt} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ unconditionally. However, for any given outcome for a merger pair, the residual must be corrected for the potential covariance of that residual with a high draw of ζ . In particular, the shocks to the merged firm, $\tilde{\psi}_{ij,kt}^{\mathcal{M}}$, can be represented as:

$$\tilde{\psi}_{\mathcal{M}(i,j),kt} \equiv \psi_{\mathcal{M}(i,j),kt} - \odot(\boldsymbol{\eta} \odot \psi_{i,k,t-1}) - \boldsymbol{\delta}_s \odot (\boldsymbol{\eta} \odot \psi_{j,k,t-1}) - \sigma_\zeta \hat{\lambda}_{ij,kt} e_\kappa. \quad (53)$$

After the control-function shift, the ζ component of this residual is a centered Gumbel with scale σ_ζ , with variance $\pi^2 \sigma_\zeta^2 / 6$.

The second assumption we make for tractability of estimation is that the control function correction affects mean, but not variance. As a result of this assumption, we can represent the covariance of the full residual as

$$\mathbf{C}_{\mathcal{M}} = (1 + \boldsymbol{\delta}_s \boldsymbol{\delta}_s^\top) \odot \boldsymbol{\Sigma} + \frac{\pi^2 \sigma_\zeta^2}{6} e_\kappa e_\kappa^\top. \quad (54)$$

We treat (53) as Gaussian for the likelihood evaluation. This is the sole distributional approximation in the outcome stage: the centered Gumbel on the κ row is right-skewed rather than symmetric, but its mean (zero) and variance ($\pi^2 \sigma_\zeta^2 / 6$) are matched exactly.

Unmerged firm selection correction. Non-mergers are themselves selected events: for each non-merger bank j in (k, t) , the realized assignment matrix $\mathbf{M}_{kt}$ encodes the fact that j *could have* merged but did not. The conditional distribution of $\tilde{\psi}_{jkt}$ given the realized selection therefore carries a non-zero mean shift, analogous to the shift on ζ for mergers. Define the non-merger Mills

shift as the importance-weighted posterior mean of the bank's drawn innovation:

$$\hat{\mu}_{jkt} \equiv \sum_{r=1}^R w_{kt}^{(r)} \tilde{y}_{jkt}^{(r)}, \quad (55)$$

We subtract $\hat{\mu}_{jkt}$ from $\tilde{y}_{jkt}$ before evaluating the Gaussian density.²⁵

The per-period outcome log-likelihood is

$$\log L_{kt}^{\text{out}}(\theta) = \sum_{j \in \mathcal{N}_{kt}^{\text{nm}}} \log \phi(\tilde{y}_{jkt} - \hat{\mu}_{jkt}; \mathbf{0}, \Sigma) + \sum_{(i,j) \in \mathcal{M}_{kt}} \log \phi(\tilde{\psi}_{\mathcal{M}(i,j),kt}; \mathbf{0}, C_{\mathcal{M}}), \quad (56)$$

where $\phi(\cdot; \mathbf{0}, V)$ is the multivariate normal density, $\mathcal{N}_{kt}^{\text{nm}}$ is the set of non-merger banks in (k, t) , and $\mathcal{M}_{kt}$ is the set of realized merger pairs. Both sums apply Mills shifts $-\hat{\mu}_{jkt}$ for non-mergers, $\sigma_{\zeta} \hat{\lambda}_{ij,kt} e_{\kappa}$ for mergers – to absorb the conditional mean of the relevant selection shock before evaluating the Gaussian density.

With the terms of the likelihood defined, we then use simulated maximum likelihood for draws $R = 100$ to recover the parameters. Initial values for η_{ψ} (Equation 21) and similar were set from estimating an AR(1) process in the data. Initial values for merger frictions come from estimating logit demand on choices with $\Delta\pi$ replaced by a constant. Initial values for merger synergies (δ_{κ}) were estimated from realized pre and post merger differences in fixed costs.

7 Model estimates

This section describes the demand parameters and consequential changes in marginal costs, fixed costs, and quality of merging banks and depositor and borrower welfare from bank mergers. Using those estimated parameters in our model of endogenous mergers, we verify significant pair-specific motivations for mergers and find meaningful costs of merging on both the seller and buyer side.

7.1 Stage 1: Deposit and loan demand estimates

Here we describe the demand estimates for bank-specific deposits and loans. The demand estimation is standard, intended to proceed similarly to prior work (Egan et al., 2022; Koont, 2023; Wang et al., 2022).

Consumers value branches and are elastic to deposit rates and loan rates. Estimates from Equation (32) and of the Berry (1994) model are shown in Table 6. The instrument “operating cost intensity” is significant at the 1% level. Columns 1 and 2 show the first stage of the 2SLS demand system. The dependent variables are the endogenous deposit and loan rates. The instrument, by shifting cost of operating the bank, corresponds to a lower interest rate paid on deposits and a

²⁵Omitting $\hat{\mu}_{jkt}$ is equivalent to treating non-merger residuals as unconditionally distributed, which biases $\text{diag}(\Sigma)$ and propagates into compensating biases in δ_s . The merger sum already corrects for selection through $\hat{\lambda}_{ij,kt}$ on the κ row of (53); the non-merger sum requires the analogous correction on the full innovation vector.

higher loan rate charged. The F statistics for both exceed 10. Columns 3 and 4 show the OLS regressions. As is typically found in demand regressions, without instruments for prices, demand seems fairly price inelastic. However, Columns 5 and 6 show the second stage of the 2SLS system. Using the operating cost intensity as the instrument for deposit rate and loan rate, banks command higher market shares when they offer higher deposit rates and lower loan rates.

Differences in the median level of the profit parameters between acquirers and borrowers is significant. Table 7 shows the difference in means table for acquirers, targets, and uninvolved banks between 1990-2024. As was reported with accounting data in Table 1, acquirers are more cost efficient than merger targets. Compared to targets, acquirers have a 6bp advantage in the year prior to merger in deposit marginal costs and a 52bp advantage in loan marginal costs. Fixed costs are larger in dollar terms for acquirers, but this difference reflects their scale. When normalizing by assets, acquirers have significantly smaller fixed costs than targets. Acquirers have higher unobserved quality in providing deposits, but lower unobserved loan quality, insignificant at the 90% confidence level.

7.2 Stage 2: Merger market

Estimates of the merger market parameter suggest that regulatory capital constraints for private firms create large one-time merger costs. The model estimates suggest mergers create value through fixed cost synergies.

Table 8 presents estimates of the structural model parameters. First, incidental to quantifying merger synergies and merger frictions with selection, we estimate parameters related to the data-generating process of fixed costs, κ . Row 1 reports the persistence of the AR(1) process for κ/A_i . From this persistence we also recover, row 2, the variance of shocks to fixed cost κ/A . Variance in these shocks mediate selection on levels: the merged pair could have received particularly negative fixed cost shocks, meaning the banks poised to become more cost efficient merged and fixed cost synergies would be over-estimated. Rows 3 and 4 list the parameters δ . The values for δ are almost equal to 1. Merged entities, without pair-specific synergies have essentially the weighted average of pre-merger fixed costs, or, in dollar terms, the sum of the fixed costs of the separate entities. Next in row 5, counterintuitively, $c_{Branches}$ is slightly positive, corresponding to higher branch overlap actually predicting fewer fixed cost reductions. Instead, branches contribute to the propensity for particular buyers and sellers to match, as going from 0% to full branch overlap of the buyer's branches increases the surplus of the a potential merger combination by \$4.5M. Fixed cost synergies are loaded into pair-specific shocks ζ . The variance of ζ_{ij}/A_j is on the order of the profitability changes seen in Section 3. The standard deviation of ζ_{ij}/A_j is 0.07% of seller assets annually in profits, which is roughly 10% of an average change in a typical bank's annual return on assets. The parameter implies that fixed cost synergies from bank mergers, should they arise are pair-specific.

With merger synergies mediated primarily by pair-specific shocks, we turn to costs that affect merger selection but are excluded from the real merger outcomes used in estimation. Most

important for policy, mergers involving pro-forma banks close to the Tier 1 Leverage Ratio are significantly more expensive. Figure 10 plots the coefficients for the piecewise function of Tier 1 Leverage ratio. Consistent with theory, the mergers closest to the threshold have the highest cost. The regulatory capital wedge is 1% of seller assets for mergers that would breach capital ratios. Since banks hold significant buffer from their capital ratios, just 87 mergers estimate the $< 4\%$ bucket (see count in Figure 8). Then, merges that would place the combined firm just above the 4% minimum have a shadow cost of 1.5% of seller assets. For reference, this figure is equivalent to eight years of profits from the average target, or, for a merger driven entirely by pair-specific synergies, this statistic would be equivalent to more than 20x the capitalized value of the pair-specific profit synergies expected from the merger or a 20 year payback period.²⁶ As expected, moving up toward larger capital ratios, estimates fall, consistent with theory that the quantity $\min\{c^{Reg}, c^{Ext}\}$ declines with Tier 1 Leverage Ratio.

The results suggest that temporary capital relief around mergers or permanent adjustments would have large effects on the aggregate merger market. We now use the model estimates to quantify the effect of alternative policy.

8 Counterfactual capital regulation

We have established that the realized mergers between U.S. banks between 1990 suggest a large RCW from reduced form and estimates of the structural model. Using counterfactual simulations of banking sector mergers and competition from the model parameters, we now document how adjustments to capital regulation would shape industry structure. Temporary relief from capital regulation around mergers would increase consolidation drastically among the smallest banks because capital regulation makes acquisitions by privately traded banks costly. Such relief would increase banking sector efficiency while reducing consumer welfare. Imposing this friction on the largest, publicly traded banks would have constrained fewer, larger bank mergers, and preventing these would have squandered 19% of the fixed cost efficiencies realized in the banking sector created since 1992, in favor of consumer welfare.

8.1 Temporary capital regulation relief

The first counterfactual tests a policy that allows banks to breach their Tier 1 Leverage ratio temporarily. Temporary relief from capital regulation around mergers would increase consolidation drastically among the smallest banks because capital regulation makes acquisitions by privately

²⁶Holding fixed σ_ζ , the coefficients in merger surplus are identified through their independent variation from one another and are identified only as a multiple of σ_ζ . σ_ζ is then pinned down by the variance of choices relative to the deterministic merger surplus Φ as well as its role in correcting the value of post-merger κ . Baseline estimation makes profits stylized to fixed costs alone, and a more flexible first stage merger selection will increase the variability of profits and increase σ_ζ and estimated capital costs. As a result, reporting capital costs as a fraction of σ_ζ is a reasonable normalization.

traded banks costly. Such relief would increase allocative efficiency by reducing fixed costs through more mergers, but decrease consumer welfare.

We implement this by setting the coefficients on post-merger Tier 1 Leverage Ratio to zero, or assuming there is no regulatory capital wedge, then measuring how the 1992-2023 world would have progressed. To isolate just the change in industry structure of the regulatory capital wedge, we first repeatedly simulate counterfactual banking sectors by state and year from our parameter values to use as a baseline. Appendix ?? describes the procedure. The simulations begin with the 1992 population of U.S. commercial banks, then plays the merger market and horizontal competition given the opponents available every year, then uses the newly consolidated sector for the next year. We do not use an outside option in the computation of optimal rates set in the deposit and loan markets, so the estimates of consumer welfare harm are likely high upper bounds. The setup allows We then compare these simulations with simulations from the same parameter values, but RCW parameters set to zero. The results below show the comparison of 45 simulations. Appendix F shows how the re-simulated data fit against empirical merger data.

Removing capital constraints changes the number and also type of mergers that happen than in the data. Figure 11 shows the mean number of mergers occurring annually in the baseline versus counterfactual. The policy makes merging less costly and, as expected, increases the rate of consolidation drastically in the smallest banks, the vast majority of which are privately held. Setting the RCW to zero does almost nothing to the regional and large national banks, as shown in the last two panels.

We can then assess the welfare of these counterfactual industry structures in the 2023 market. Figure 12 shows the resulting total welfare, broken down by producer surplus via fixed cost synergies, increases in variable profits, and increases (decreases) in depositor and borrower welfare. The increase in market power reduces consumer surplus.²⁷ Loans and deposits contribute comparably. Furthermore, because banks are horizontally differentiated, changes in which banks merge has affects on aggregate variable profits. For instance, by increasing the consolidation of the community banking sector, the variable profits of the sector actually decline slightly because the remaining banks have more formidable competition.

8.2 Imposing costs on public banks

The results of Section 8.1 show that the effects on the banking sector of RCW are concentrated in the small bank fringe. We next turn to quantifying how this friction, which affects the majority of acquirers but a minority of the total banking sector market share, would affect the acquisition activity of the largest banks. In this way, we provide a quantification for how much access to liquid stock has shaped the ability for the banking sector to consolidate frictionlessly. Figure 11 shows that imposing the private bank costs on public firms would reduce the number of mergers by suppressing consolidation of the largest banks. Analogously to *removing* RCW in the last section,

²⁷The model does not allow for marginal cost synergies that, if present, would create additional surplus to be divided between consumers and banks.

adding it for the firms doing the largest deals since 1992 reduces the merger rate by 30%. In Figure 12, we quantify the welfare effects of this policy. First, the public acquirer mergers have played a substantial role in creating allocative efficiency in the banking sector, as predicted when Riegle-Neal was originally passed. With equivalent costs imposed on public firms, sector fixed costs would be \$28B larger, a 19% increase from the simulated 2023 baseline. That said, with these costs, depositors and borrowers would have enjoyed \$37B in additional surplus.

9 Conclusion

Despite thousands of mergers between commercial banks since 1990, the United States still has 4,000 operating commercial banks. This paper demonstrates that capital regulation, by creating a shadow cost of leverage, makes financing profitable M&A challenging consequently dissuades consolidation. We present a conceptual framework with predictions that we verify empirically. The incidence of this regulatory capital wedge is greatest for firms who cannot easily finance mergers by using their own stock as payment, making this wedge largest for smaller bank acquirers. In the first section of the paper, we first prove out the premise that, without accounting for additional one-time costs and frictions associated with mergers, that business combinations are indeed profitable for the merging parties. Next, we show empirical evidence that costly equity issuance paired with capital regulation creates this wedge and dissuades consolidation by less capitalized bank combinations. In the final part of the paper, we evaluate how a regulator could improve the efficiency of the banking sector by adjusting the shadow cost of capital regulation around M&A. We estimate that \$5.6B fewer fixed costs would be paid by the banking sector annually today if affected acquirers were allowed to breach capital ratios in order to purchase another bank, but such benefits would be swamped in aggregate by reductions in consumer welfare.

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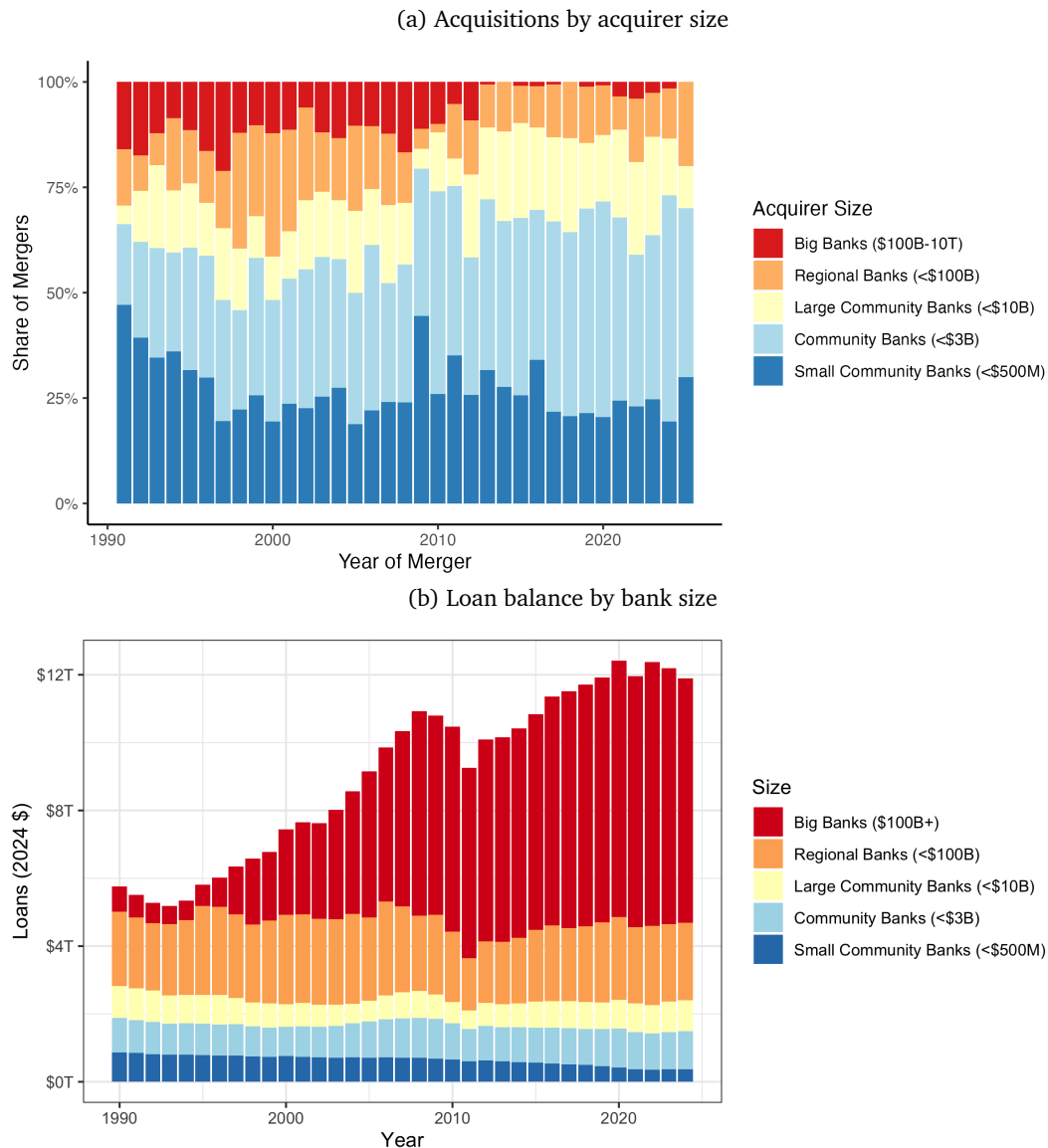
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Figures

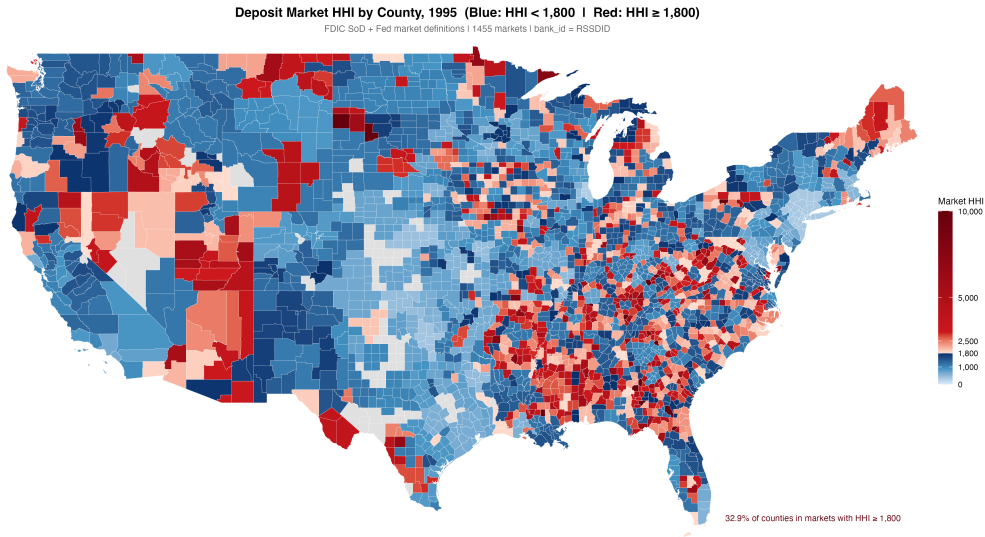
Figure 1: Community banks in the merger market



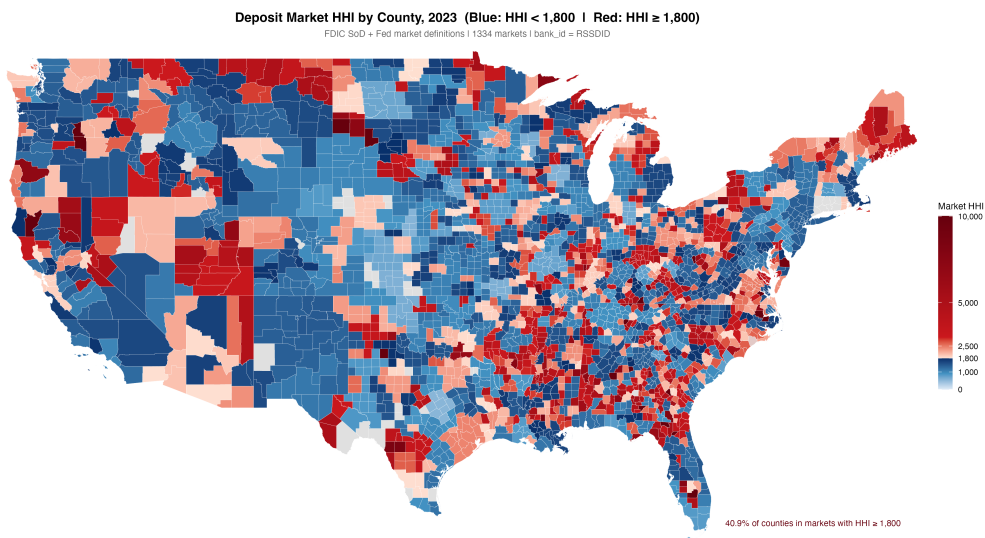
Note: Panel (a) shows the share of mergers done annually by the size of the acquiring bank. The majority of acquirers are community banks, banks have less than \$10B in assets (in real terms, 2023 USD) in the quarter before the acquisition. To put in perspective the small size of the typical merger transaction, Panel (b) shows the dollars of loans on the balance sheet of banks filing Call Reports, summed by size of the bank. Only 20% of loans in 2024 were given by community banks (banks holding <\$10B in assets, are shown in blue and yellow).

Figure 2: Competitiveness of U.S. banking sector

(a) 1995

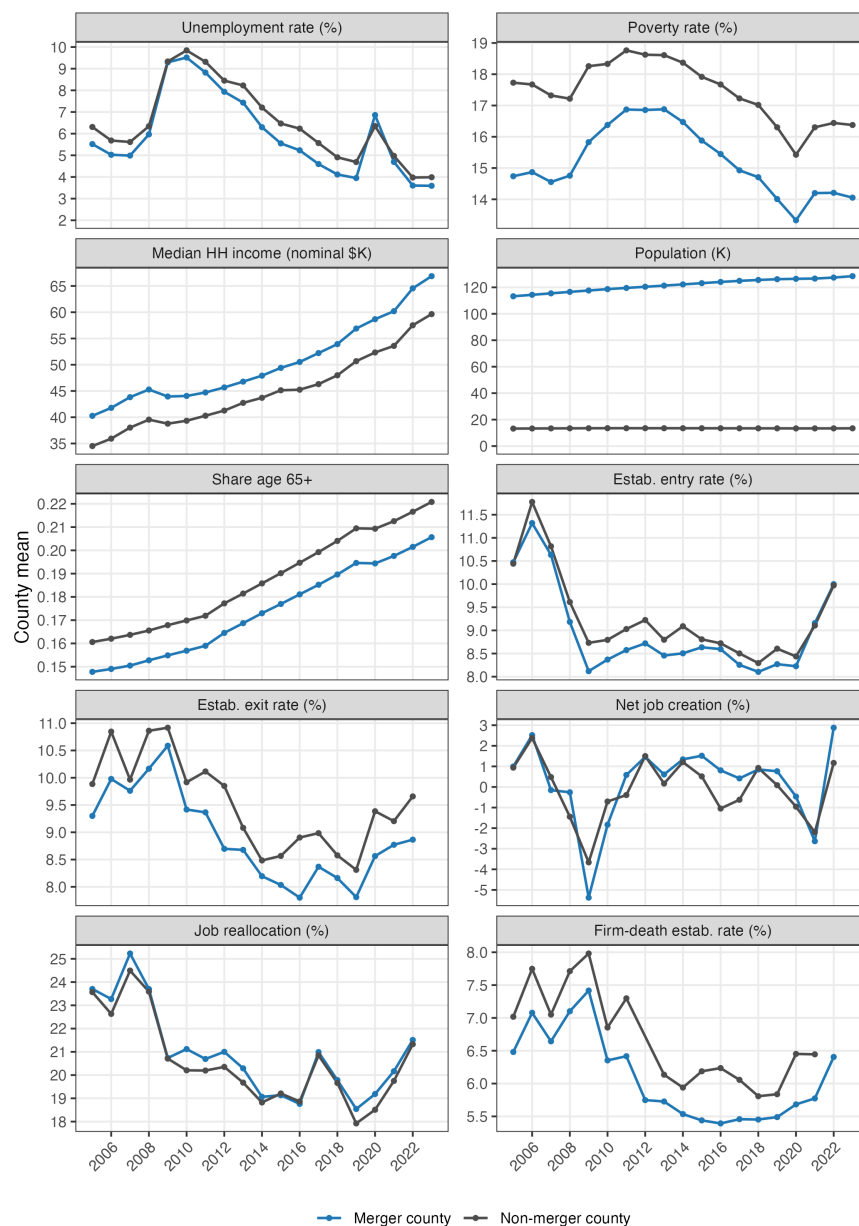


(b) 2023



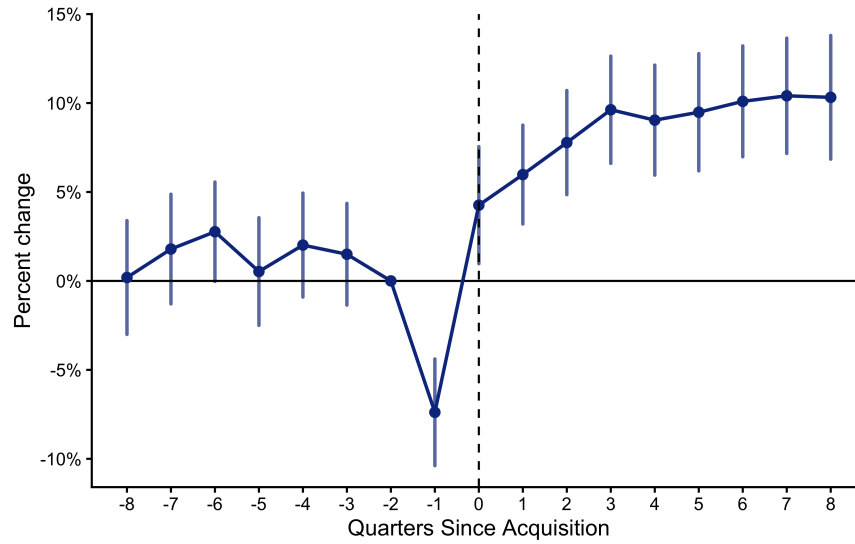
Note: Panels (a) and (b) show a heatmap of Federal Reserve banking market maps, colored by their deposit market HHI. HHI's are computed with Call Reports filing institutions and do not include credit unions. 1995 market maps are assumed to be equivalent to 2009 maps provided by CASSIDI. 92% of the U.S. population lived in a competitive banking market as classified by the DOJ as moderately concentrated ($HHI < 1800$) in 2022, using population estimates from the ACS.

Figure 3: Economic outcomes of merger versus non merger counties



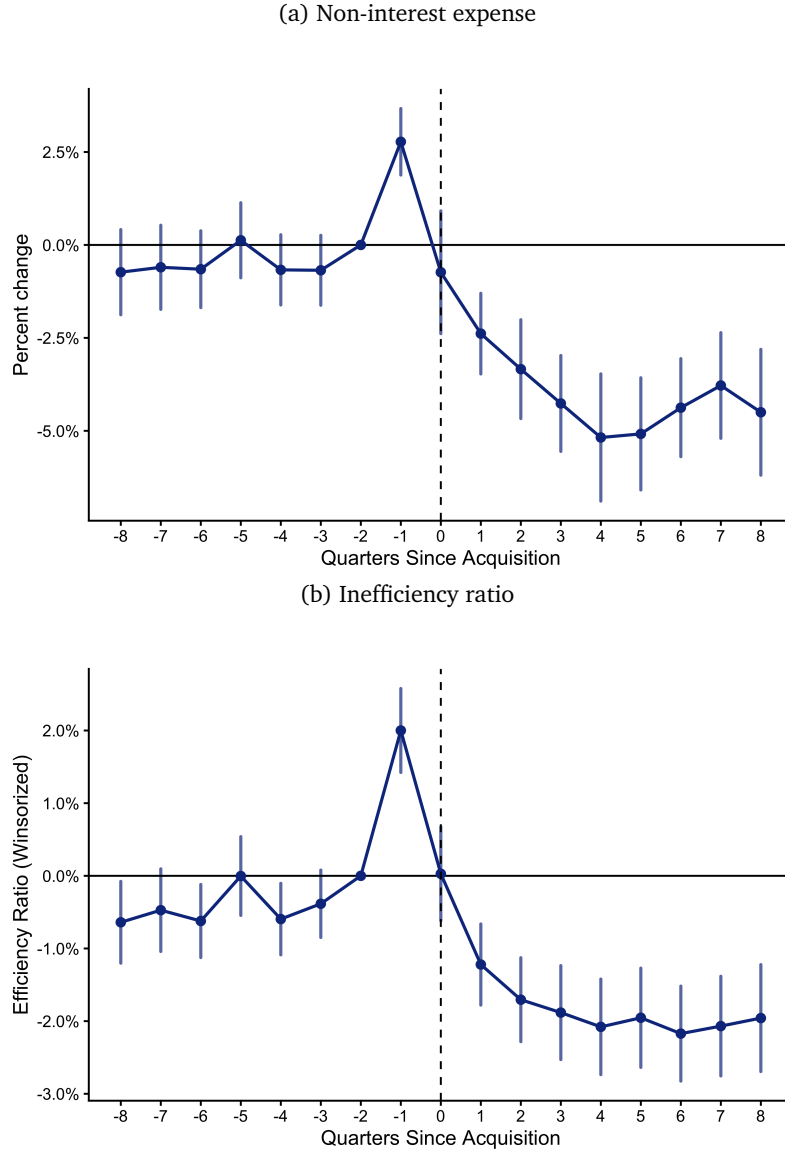
Note: This figure plots unweighted cross-county means by year for counties ever exposed to a bank merger (blue; a bank with a branch in the county was acquired in some year, $N = 2,569$) versus counties never exposed (red, $N = 640$), pooling the two years before and after each county's own reference period. Panels show, from top-left: unemployment rate, poverty rate, nominal median household income, population, the population share aged 65+, and Census Business Dynamics Statistics (BDS) establishment entry, exit, net job creation, job reallocation, and firm-death rates (all in percent). Income and population figures are reported in nominal (not inflation-adjusted) dollars/counts. Sample restricted to 2005–2023.

Figure 4: Profit synergies for merged banks



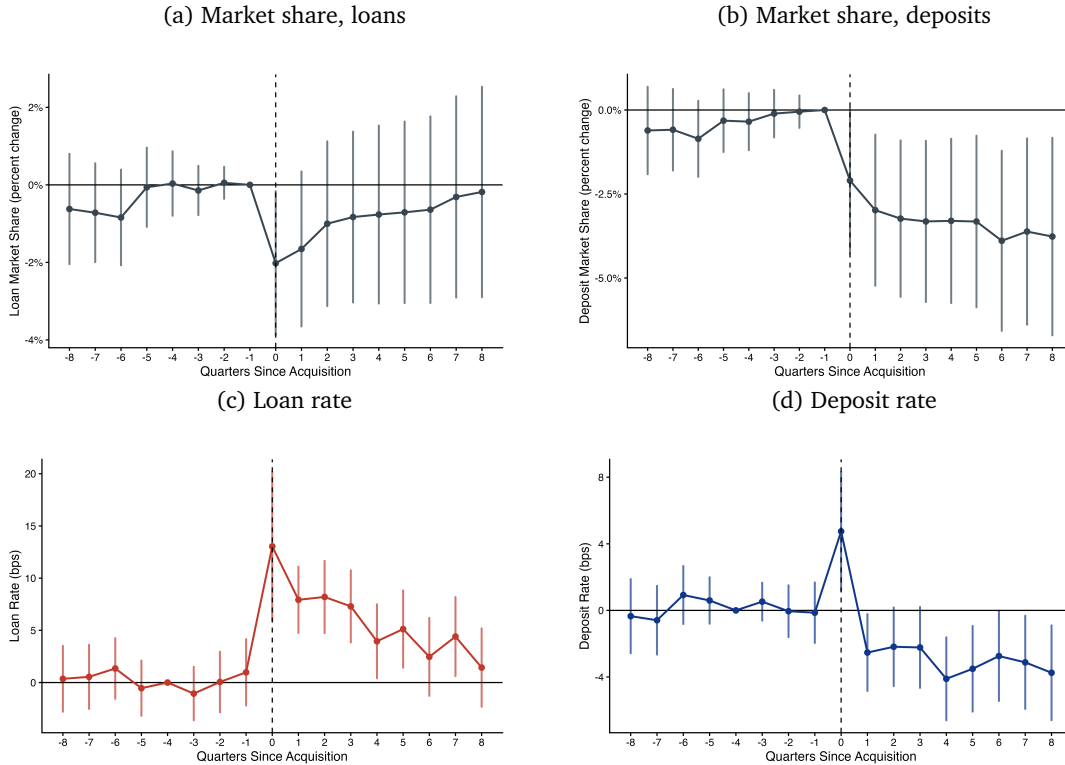
Note: This figure shows the estimates of β_s from Equation (1) on the constructed population of bank mergers compared with a stacked population of banks that were never acquirers nor targets between 1990-2024. The outcome is the logarithm of profits of the pro-forma bank, then the y-axis transforms log points to percent points. Each merged bank observation is a *pro-forma* bank that consists of the financial statements of the acquirer and target in the pre-period combined as if operating as one entity, then the financial statements of the acquirer beginning in $t = 0$. The estimates imply that profit of the *pro-forma* (combined) bank increased by 10.4% two years after the merger. Standard errors are clustered at the bank level. Vertical bars show 95% confidence intervals.

Figure 5: Cost synergies for merged banks



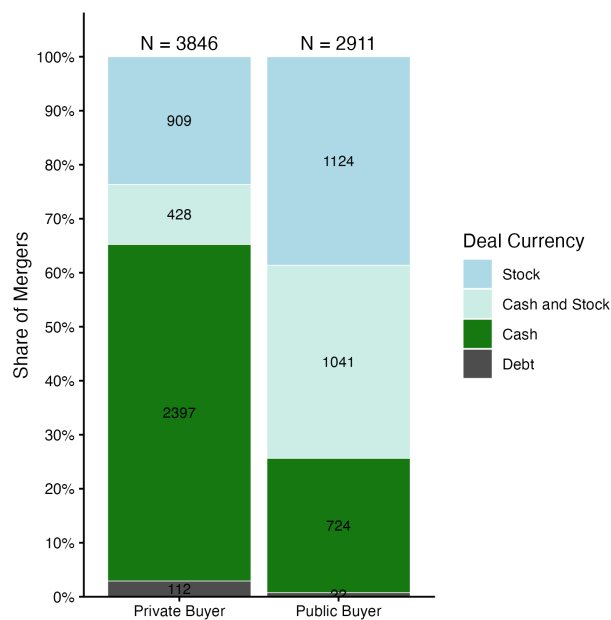
Note: This figure shows an increase in the cost efficiency of merging banks by plotting β_s from Equation (1), where the inefficiency ratio is defined in Equation (3). In the period before the acquisition, bank inefficiency ratios spike, likely reflecting one-time merger related costs. After the merger, bank inefficiency ratios fall by -1.9pp two years after the merger. Standard errors are clustered at the bank level. Vertical bars show 95% confidence intervals.

Figure 6: Prices and quantities



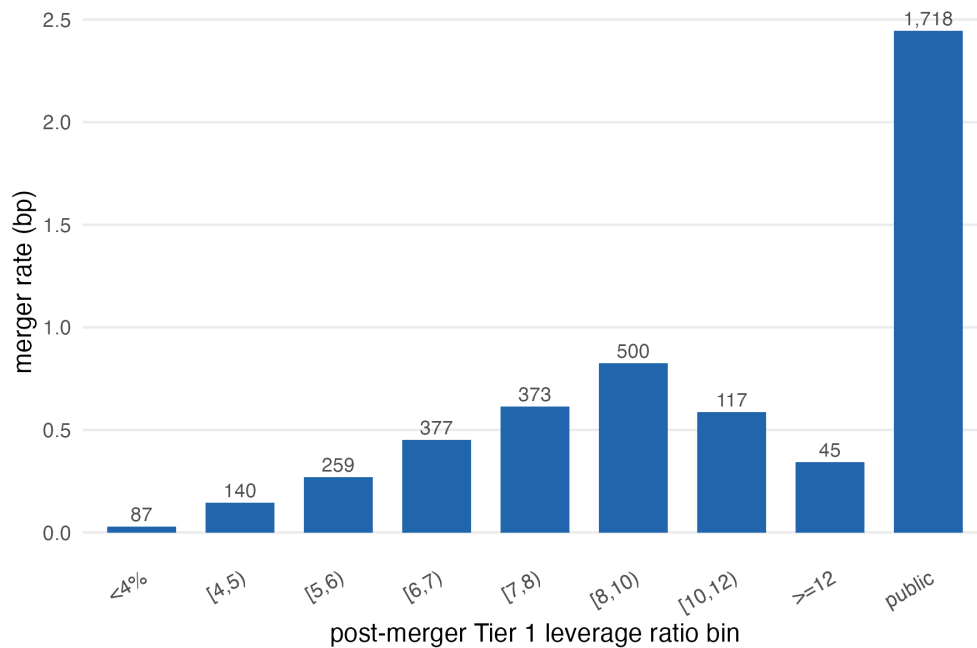
Note: This figure shows the estimates of β_s from Equation (1) on the constructed population of bank mergers compared with a stacked population of banks that were never acquirers nor targets between 1990-2024. Panel (a) plots the estimates of β_s for $s \in [-8, 8]$ for growth in dollars of deposits. Panel (b) plots the estimates of β_s for growth in dollars of loans. The graph shows a visible pre-trend prior to the merger where merging banks had higher growth in loans than control banks. Panel (c) plots the estimates for deposit rates, defined as deposit income. Deposit and loan rates are winsorized at the 1% and 99% level. Standard errors are clustered at the bank level. Vertical bars show 95% confidence intervals.

Figure 7: Prediction 1: Transaction medium, SNL bank merger data



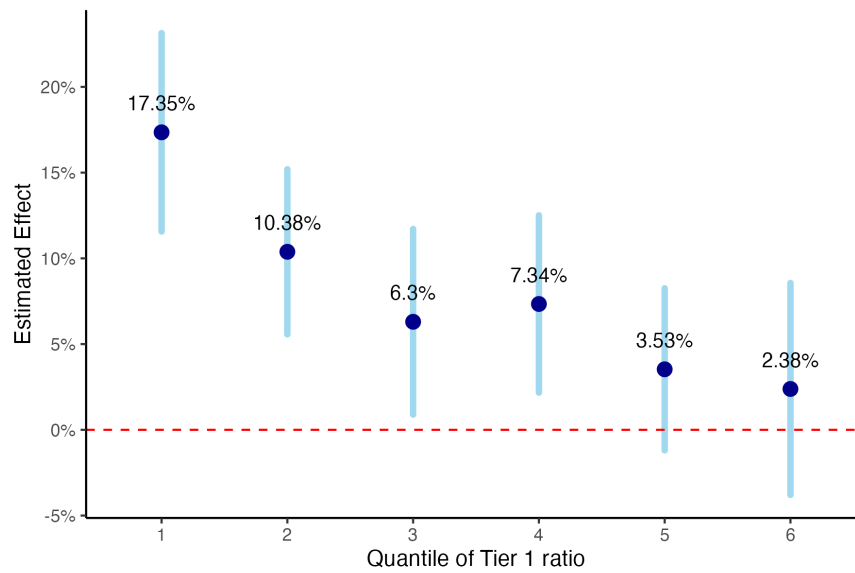
Note: This graph shows the breakdown of transaction medium for bank-to-bank acquisitions reported in the S&P SNL data for bank mergers announced between 1991-2024. 62% of private buyer acquisitions use cash alone to pay for the acquiring bank, compared to just 25% of public buyer acquisitions. 11% of private acquisitions use a mix of cash and stock, and 24% use stock alone. In contrast, stock is part of the deal value for 74% of public buyer transactions.

Figure 8: Merger propensity by Tier 1 Leverage Ratio



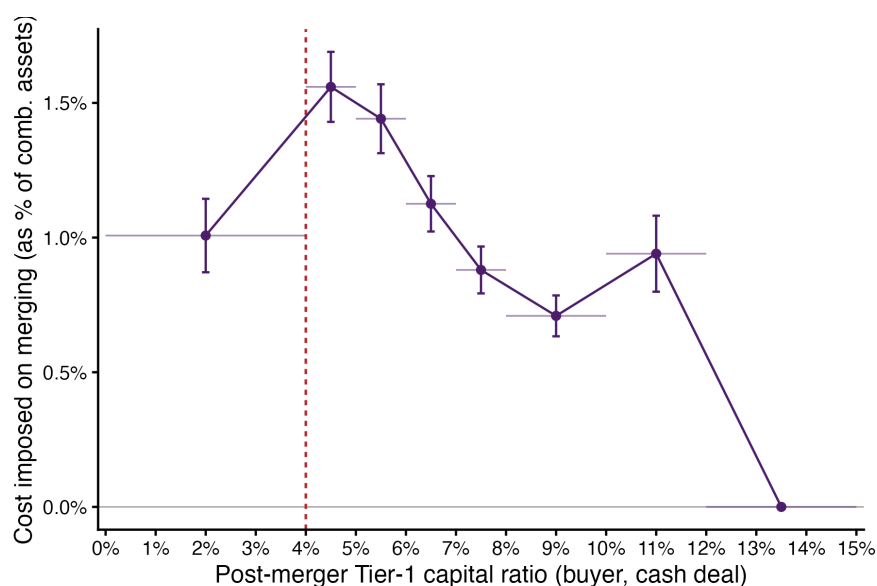
Note: This graph shows the merger rate for potential pairs of banks, cut by the post-merger leverage ratio. The leverage ratio is the Tier 1 Leverage Ratio, computed using reported Tier 1 Capital and Tier 1 Assets in Call Reports. Post-merger Tier 1 Leverage ratio is computed assuming that the buyer buys the target in cash, reducing the pro-forma company's assets by the value of the target's equity. The equation is given in Equation (6). Each observation is a potential merger combination, where potential merger combinations are within-state mergers between 1990-2024 where the seller is smaller than the buyer bank in the year prior to the merger.

Figure 9: Net income gains by Tier 1 Leverage Ratio



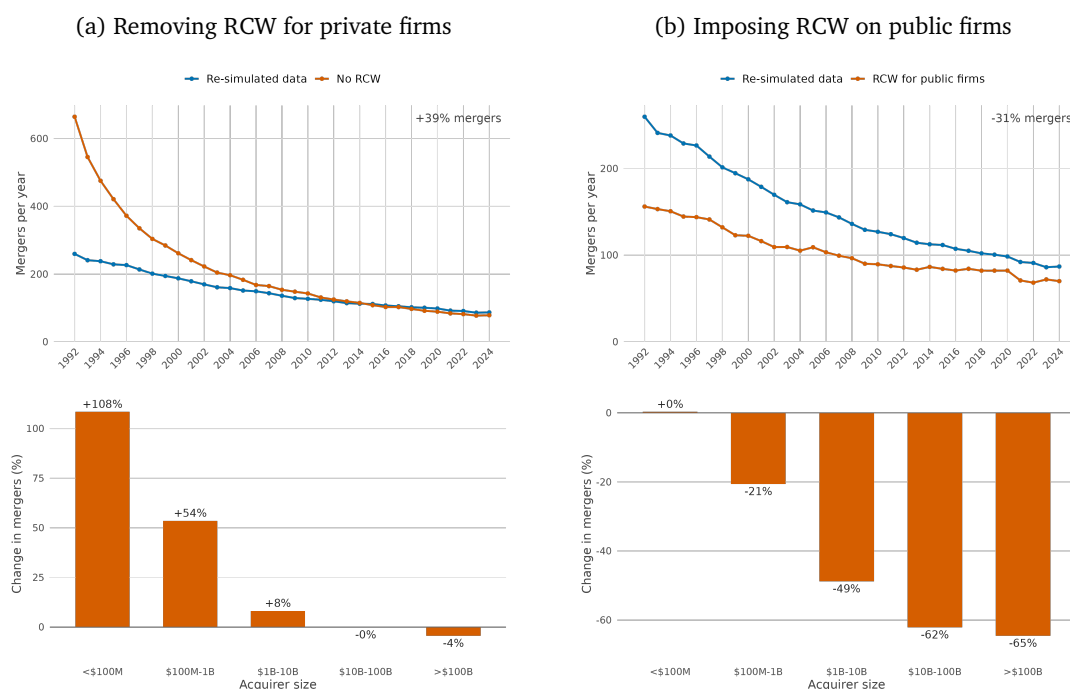
Note: This figure reports the coefficients β_s for net income from Equation (1) for two-years after merger for bank mergers in our sample, interacted with the quantile of pro-forma Tier 1 ratio two years prior to combination. The regression implies that net income increased for pro-forma banks with the lowest quantile Tier 1 ratio by 17%. Increases were smaller as the combined bank was more capitalized and as low as 2% for combinations that were the most capitalized prior to the transaction. Standard errors are clustered at the bank level. Vertical bars show 95% confidence intervals.

Figure 10: Regulatory capital wedge, as percent of seller assets



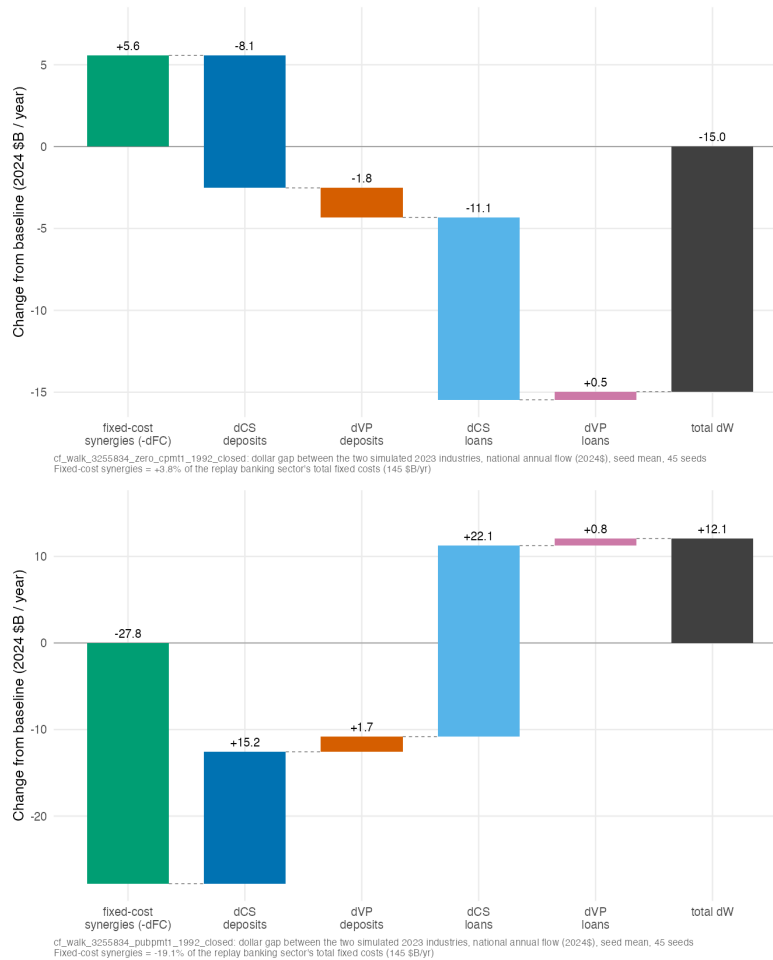
Costs increase approaching the 4% minimum Tier 1 Leverage Ratio. At its peak, combinations that would have a 4-5% post-merger Tier 1 Leverage Ratio face a shadow cost of 1.5% of seller assets. For reference, this figure is equivalent to eight years of profits from the average target, or, for a merger driven entirely by pair-specific synergies, this statistic would be equivalent to more than 20x the capitalized value of the pair-specific profit synergies expected from the merger or a 20 year payback period.

Figure 11: Counterfactual bank mergers with changes to RCW



Note: Panel 11a shows the effect on the merger rate if RCW were set to zero. Without a wedge from capital regulation making mergers costly, private firms would have consolidated more, especially through the 1990's. The number of mergers with very small acquirers, with assets <\$100M would have doubled, but there would be no effect on regional and large banks with assets exceeding \$10B. In contrast, in Panel 11b, applying the RCW estimates to publicly traded banks would have done the inverse. The merger rate would have been lower, with depressed M&A activity for publicly traded banks, most of whom have at least \$100B assets.

Figure 12: Counterfactual welfare with changes to RCW



Note: Panel a) shows the estimated effect of temporary capital regulation relief for around mergers. Total welfare would decrease because cost efficiencies would increase but to the detriment of depositor and borrower welfare. The additional consolidation would actually hurt variable profits in the deposit and loan markets slightly. Panel b) shows the converse, adding an equivalent RCW for publicly traded banks, who are much larger. In this case, welfare would be larger. Preventing additional mergers would lose significant fixed cost synergies, but the welfare benefits of the efficiencies are smaller than the benefit consumers would face from having less bank consolidation.

Tables

Table 1: Commercial bank merger sample

Variable (pp)	Acquirer	Target	Uninvolved	Acq. - Tar.	Tar. - Unin.	Acq. - Unin.
Inefficiency Ratio	63.8	74.2	73.3	-10.4***	0.9	-9.5***
ROA: Net income / Assets	0.29	0.20	0.96	0.09***	-0.77	-0.67
ROA (adj): Economic profits / Assets	0.30	0.20	0.97	0.09***	-0.76	-0.67
ROE: Net income / Equity	3.01	1.64	2.27	1.37***	-0.63*	0.74**
ROE (adj): Economic profits / Equity	3.14	1.71	2.33	1.43***	-0.62*	0.81***
Tier 1 Leverage Ratio	9.2	10.0	11.5	-0.8***	-1.4***	-2.2***
Loans / Assets	64.3	60.5	59.2	3.9***	1.3***	5.1***
Deposits / Assets	81.1	84.5	83.4	-3.4***	1.1***	-2.3***
Observations	4,204	4,927	1,144,752			

⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Note: This left panel of this table shows the means of profit and balance sheet variables for commercial bank acquirers, commercial bank targets, and all other uninvolved banks in the quarter prior to the merger. The right panel shows the difference in means between acquirers and targets, targets and other banks uninvolved in a merger transaction, and acquirers and uninvolved banks.

Table 2: Use of cash in merger transactions, by listing status of buyer

	(1)	(2)	(3)	(4)	(5)
Private Buyer	0.305*** (0.014)	0.368*** (0.014)	0.340*** (0.015)	0.326*** (0.015)	0.326*** (0.015)
Private Target			0.169*** (0.020)	0.163*** (0.020)	0.158*** (0.020)
Buyer Branches (std.)				-0.045*** (0.007)	-0.038*** (0.007)
Target Branches (std.)					-0.018* (0.007)
Num.Obs.	4495	4495	4495	4495	4495
R2	0.095	0.185	0.198	0.206	0.207
FE: ANNOUNCE_YEAR		X	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: This table shows the result of estimating Equation (15) with increasing covariates using S&P SNL data between 1991-2024 who list the breakdown of the deal value by currency (stock, cash, or debt). We limit the population of commercial bank mergers to those with a reported number of branches for the target and buyer and those who report a breakdown of the deal value. The coefficient on private buyer is interpreted as the additional share of the acquisition funded in cash, as opposed to stock or debt, that is associated with being a privately held (rather than publicly traded) buyer. With an increasing number of fixed effects controlling for the size of the acquirer, the target, and the year of announcement, the relationship between being a publicly traded acquirer and using more cash to finance the merger is stable. In the rightmost specification, being a privately held buyer is associated with funding 32.6pp more of the acquisition's value in cash.

Table 3: Merger propensity versus unrealized gains

	(1)	(2)
Unrealized Gains (pp) x Pre-2001	0.296 (11.128)	-2.620 (11.132)
Unrealized Gains (pp) x Post-2001	17.446*** (4.145)	16.998*** (4.149)
Return on Assets (pp)		-2.854** (0.918)
Efficiency Ratio (pp)		0.014 (0.011)
Tier 1 Ratio (pp)		-4.627*** (0.763)
Securities / Assets (pp)		-1.994*** (0.439)
Bank × (Post 2001) FE	X	X
quarter-year FE	X	X
Observations	788,491	786,822
R ²	0.244	0.244

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: This table shows the estimates from Equation (9), with standard errors clustered at the bank level. The results show that, with bank and time fixed effects, a 1 pp increase in unrealized gains corresponds to a 17 bp increase in the propensity to become a merger target in the next 12 months. With a 2.2% merger rate, a standard deviation increase in unrealized gains shifted the likelihood of a bank becoming a merger target by 18bps (8.5%).

Table 4: Merger propensity versus unrealized gains, using Bartik instrument

	Unrealized Gains (pp)		1(Seller) (bps)			
	First stage		OLS		2SLS	
	(1)	(2)	(3)	(4)	(5)	(6)
$w_{i,t-8} \times \Delta y_t^{5-yr}$	-2.079*** (0.042)	-2.081*** (0.042)				
Return on Assets (pp)		0.003+ (0.002)		-4.443 (2.824)		-4.568 (2.889)
Efficiency Ratio (pp)		-0.000 (0.000)		0.038 (0.032)		0.039 (0.032)
Tier 1 Ratio (pp)		0.006*** (0.001)		-2.855*** (0.632)		-3.034*** (0.640)
Securities / Assets (pp)		-0.003*** (0.000)		-2.613*** (0.211)		-2.519*** (0.212)
Unrealized Gains (pp)			-7.554* (3.722)	-9.152* (3.695)	16.793* (8.464)	20.360* (8.507)
First-stage F					2411.1	2474.1
Size $\times$ quarter-year FE	X	X	X	X	X	X
Observations	768,964	768,964	768,964	768,964	768,964	768,964
R ²	0.564	0.568	0.005	0.006	0.005	0.006

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: This table shows the first stage and second stage of instrumenting for unrealized gains, with standard errors clustered at the bank level. Columns 1 and 2 shows the estimates of Equation (10), with and without covariates. Column 3 and 4 show the OLS regressions, similar to Table 3 but without bank fixed effects. Without bank fixed effects, unrealized gains are not positively associated with probability of sale within 12 months. Columns 5 and 6 show the second stage, Equation (11). Using the instrument for unrealized gains built from yield curve changes and long-maturity securities, a one standard deviation increase in unrealized gains increases the probability of being sold by 17.8bps. With covariates controlling for the bank's attractiveness as a merger target, this rises to 21.0bps (9.7%).

Table 5: Merger propensity versus Tier 1 Leverage Ratio, using Bartik instrument

	$\tilde{r}_{jt}$ (pp)		1(Seller) (bps)			
	First stage		OLS		2SLS	
	(1)	(2)	(3)	(4)	(5)	(6)
$w_{i,t-8} \times \Delta y_t^{5-yr}$	-2.293*** (0.100)	-2.156*** (0.104)				
Return on Assets (pp)		0.211* (0.096)		-4.450 (2.829)		-9.176+ (5.336)
Efficiency Ratio (pp)		-0.000 (0.000)		0.038 (0.032)		0.043 (0.035)
Securities / Assets (pp)		0.025*** (0.005)		-2.591*** (0.210)		-3.145*** (0.312)
$\tilde{r}_{jt}$ (pp)			-4.029*** (0.588)	-2.991*** (0.627)	15.229* (7.697)	19.320* (8.265)
First-stage F					524.5	429.4
Size $\times$ quarter-year FE	X	X	X	X	X	X
Observations	768,963	768,963	768,963	768,963	768,963	768,963
R ²	0.061	0.089	0.005	0.006	0.005	0.006

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: This table shows the first stage and second stage of instrumenting for unrealized gains, with standard errors clustered at the bank level. Columns 1 and 2 shows the estimates of Equation (13), with and without covariates. Column 3 and 4 show the OLS regressions. Columns 5 and 6 show the second stage, Equation (14). Using the instrument for Tier 1 Leverage Ratio built from yield curve changes and long-maturity securities, a 1 pp increase in unrealized gains increases the probability of being sold by 15.2bps. With covariates controlling for the bank's attractiveness as a merger target, this rises to 19.3bps. The increase with covariates is a 109bp rise in the 1-year likelihood of acquisition, or 50% from baseline.

Table 6: Deposit and loan demand

	Rate (pp)		Log market share			
	First stage		OLS		2SLS	
	Deposits (1)	Loans (2)	Deposits (3)	Loans (4)	Deposits (5)	Loans (6)
Fixed expenses _{t-1} / Assets _{t-1} (pp)	-0.598*** (0.021)	0.603*** (0.042)				
Deposit rate (α_d ; pp)			0.111*** (0.011)		2.194*** (0.106)	
Loan rate (α_l ; pp)				-0.157*** (0.012)		-1.886*** (0.144)
Log(# branches)	-0.068*** (0.003)	-0.191*** (0.008)	0.989*** (0.008)	0.994*** (0.008)	1.154*** (0.013)	0.683*** (0.031)
First-stage F (Kleibergen–Paap)	791.3	209.6				
State $\times$ year FE	X	X	X	X	X	X
Observations	250,298	249,660	250,298	249,660	250,298	249,660
R ²	0.858	0.757	0.758	0.754	0.778	0.759

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: [Berry \(1994\)](#) logit demand estimated by 2SLS. Columns 1 and 2 show the first stage of the 2SLS demand system. Dependent variable is the (endogenous) interest rate. The interest rate (deposit rate in column 1, loan rate in column 2) is instrumented with operating-cost intensity (1-year lagged fixed expenses / assets). Kleibergen–Paap is the cluster-robust first-stage F . Columns 3 and 4 show the OLS regressions. As is typically found in demand regressions, without instruments for prices, demand seems fairly price inelastic. However, Columns 5 and 6 show the second stage of the 2SLS system. Using the operating cost intensity as the instrument for deposit rate and loan rate, banks command higher market shares when they offer higher deposit rates and lower loan rates. Annual data, where each observations is the Q4 Call Reports filing. Standard errors clustered by bank.

Table 7: Difference in means, acquirers and targets

Variable	Acquirer	Target	Uninvolved	Acq. - Tar.	Tar. - Unin.	Acq. - Unin.
ξ^D (deposit unobserved quality)	0.265	0.101	-0.002	0.164***	0.102***	0.267***
ξ^L (loan unobserved quality)	0.077	0.127	-0.001	-0.050	0.128***	0.078**
mc^D (deposit marginal cost, pp)	0.408	0.480	0.279	-0.071*	0.200***	0.129***
mc^L (loan marginal cost, pp)	3.386	3.900	3.834	-0.513***	0.066**	-0.448***
κ (fixed cost, \$)	136700.4	18548.8	19419.3	118151.6***	-870.5	117281.1***
κ / Assets	0.0047	0.0061	0.0117	-0.0014***	-0.0056	-0.0070
Observations	4,076	4,711	992,405			

⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Note: This table shows the difference means for acquirers, targets, and uninvolved banks in the merger sample for the estimated profit parameters in Q4 prior to the year of merger. ξ^D, ξ^L are non-price, unobserved quality residuals for every bank estimated from the [Berry \(1994\)](#) demand system for deposits and loans, respectively. Marginal costs c^D, c^L , and fixed cost κ are backed out from observed prices using the assumption that banks compete Bertrand-Nash.

Table 8: Merger market model estimates

Category	Parameter	Estimate	Units	Estimate (\$/year)
Profit parameters	ρ^κ	0.7041	-	
	$\sigma_{\tilde{\kappa}}$	0.0079	-	-
Merger synergies	δ_κ^B	1.00	-	-
	δ_κ^S	0.9977	-	-
	$c_{Branches}$	0.0004	-	-
	σ_ζ	0.07%	Seller assets	
Merger selection	$C^S(a_1)$	6.25%	Seller assets	-
	Private x $1_{<4\%}(a_4)$	1.01%	Seller assets	-
	Private x $1_{4-5\%}(a_5)$	1.56%	Seller assets	-
	Private x $1_{5-6\%}(a_6)$	1.44%	Seller assets	-
	Private x $1_{6-7\%}(a_7)$	1.13%	Seller assets	-
	Private x $1_{7-8\%}(a_8)$	0.88%	Seller assets	-
	Private x $1_{8-10\%}(a_{10})$	0.71%	Seller assets	-
	Private x $1_{10-12\%}(a_{12})$	0.94%	Seller assets	-
	Private x $1_{12+\%}(a_{max})$	-	Seller assets	-
	$c_{Branches}^{Sel}$	4.5M	\$	-

Note: This table shows the estimates for the merger friction and merger synergy parameters. One-time merger costs related to being a private buyer for a combination close to the Tier 1 Leverage ratio minimum are larger than costs for more capitalized combinations. Fixed cost κ is fairly persistent over time, with $\rho^\kappa = 0.7$. δ_κ^B and δ_κ^S are both almost 1, suggesting that the merger synergies that are realized come entirely through the pair-specific synergy shock ζ that performs a Heckman selection correction in outcomes for κ .

A Additional figures

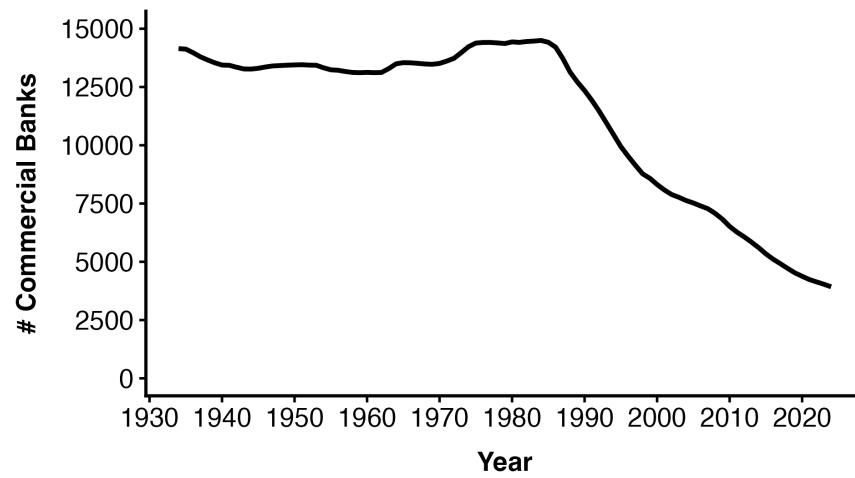
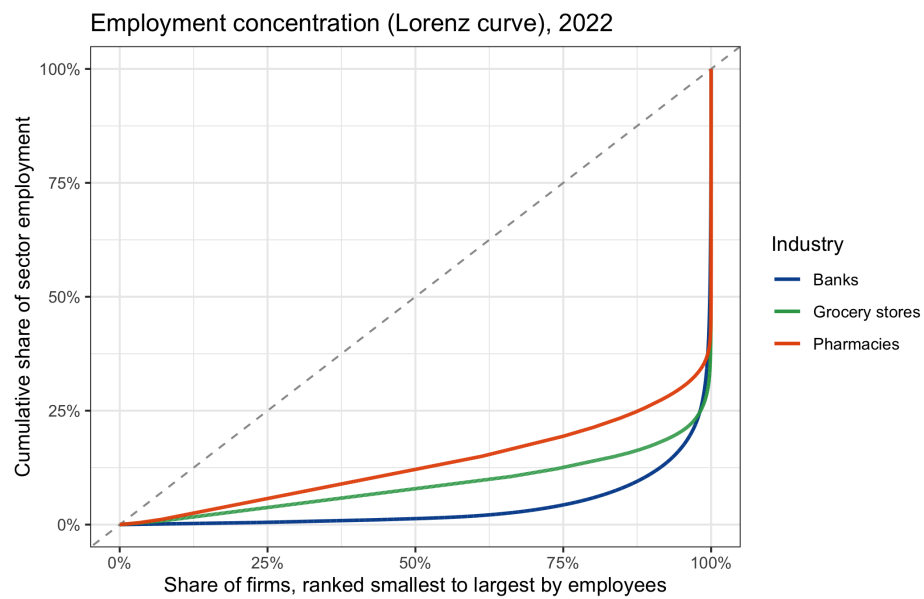


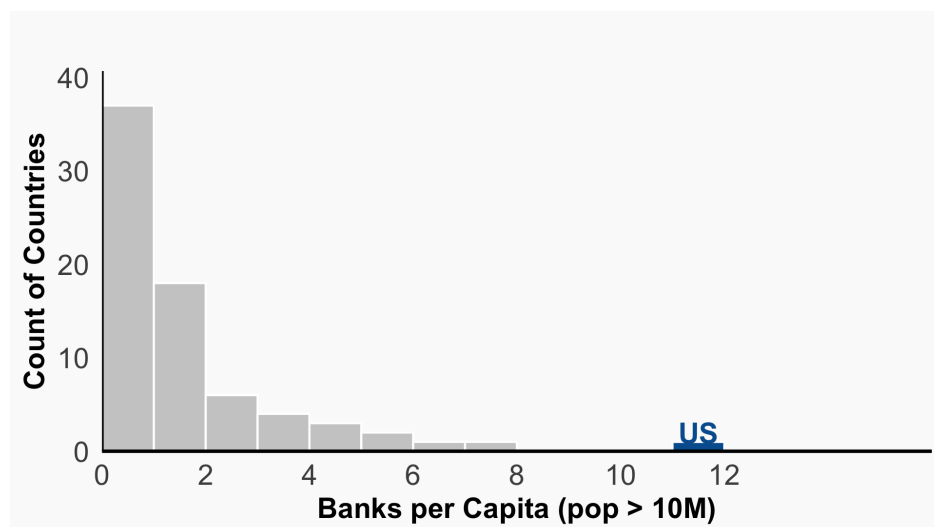
Figure A.1: This figure shows the number of commercial banks incorporated in the United States since 1930. (Source: FDIC)

Figure A.2: Concentration by industry, 2022



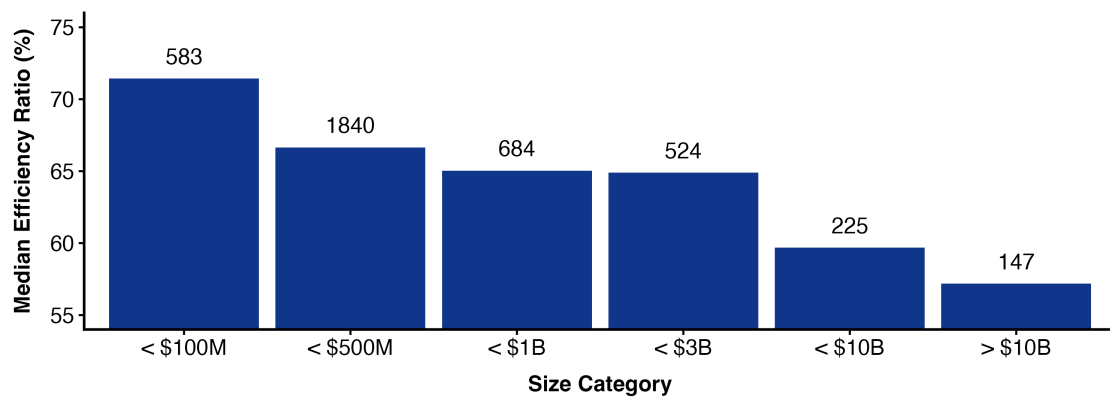
Note: This figure shows a Lorenz curve of U.S. firms classified as banks (blue), grocery stores (green), and pharmacies (orange) in 2022 Dun & Bradstreet data. The x-axis is the percentile of firms by number of employees reported to Dun & Bradstreet. The y-axis is the cumulative number of the total employees of the sector that are employed by the share of firms from the x-axis. While it is a challenge to compare industries that have returns to scale of labor, the graph shows that the smallest 75% of firms are cumulatively far smaller in banking than in grocery stores or pharmacies.

Figure A.3: Banks per capita



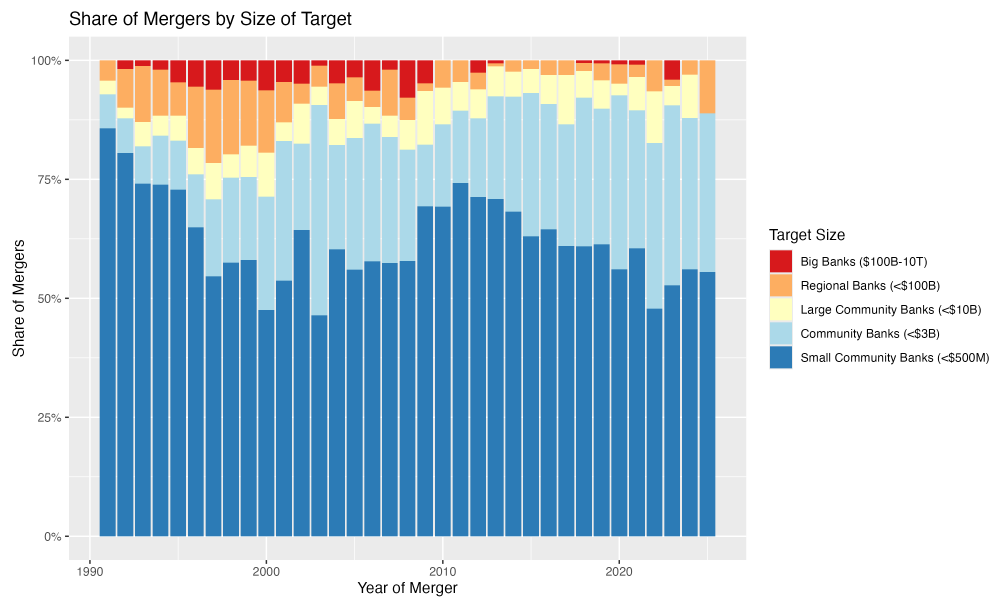
Note: This figure shows of number of banks per one million people where each observation is a country. The population are countries with populations exceeding 10 million people. The U.S. is a far outlier compared to countries with 11 banks per one million people (1 bank per 90,000 people) compared to less than eight for other countries.

Figure A.4: Median inefficiency ratio, by bank size, 2023



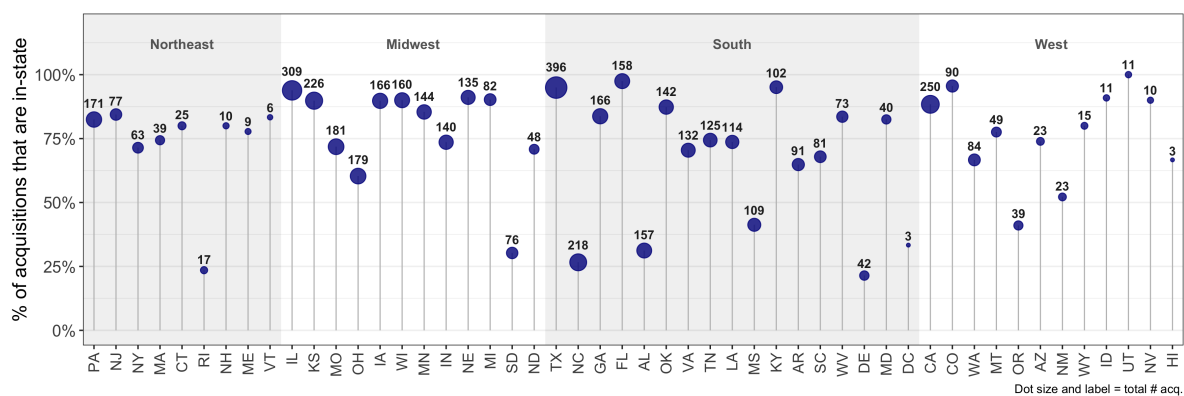
Note: This bar graph shows the median inefficiency ratio of banks in each size category. There is a clear pattern consistent with economies of scale that larger banks have a smaller percent of their net income attributed to non-interest expenses.

Figure A.5: Target size (2023 USD)



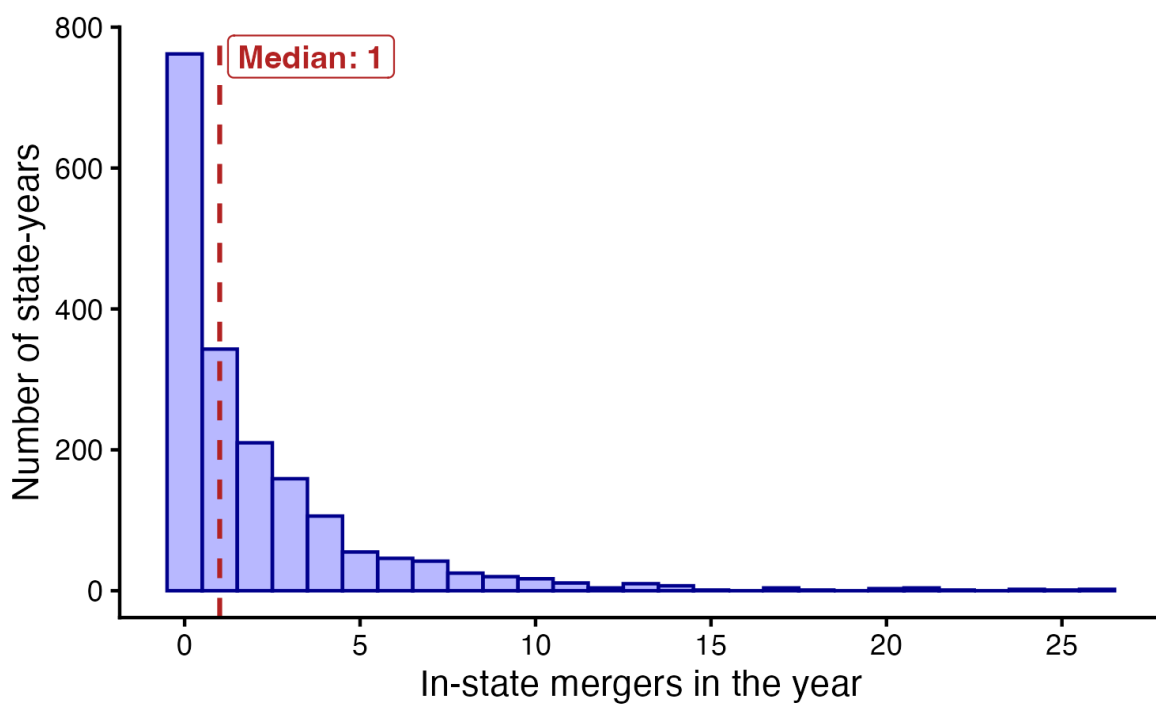
Note: This figure shows the share of transactions in the merger sample cut by the size of the target bank. The majority of target banks have less than \$500M in assets (in real terms , 2023 USD) in the quarter before they are acquired.

Figure A.6: In-state acquisitions



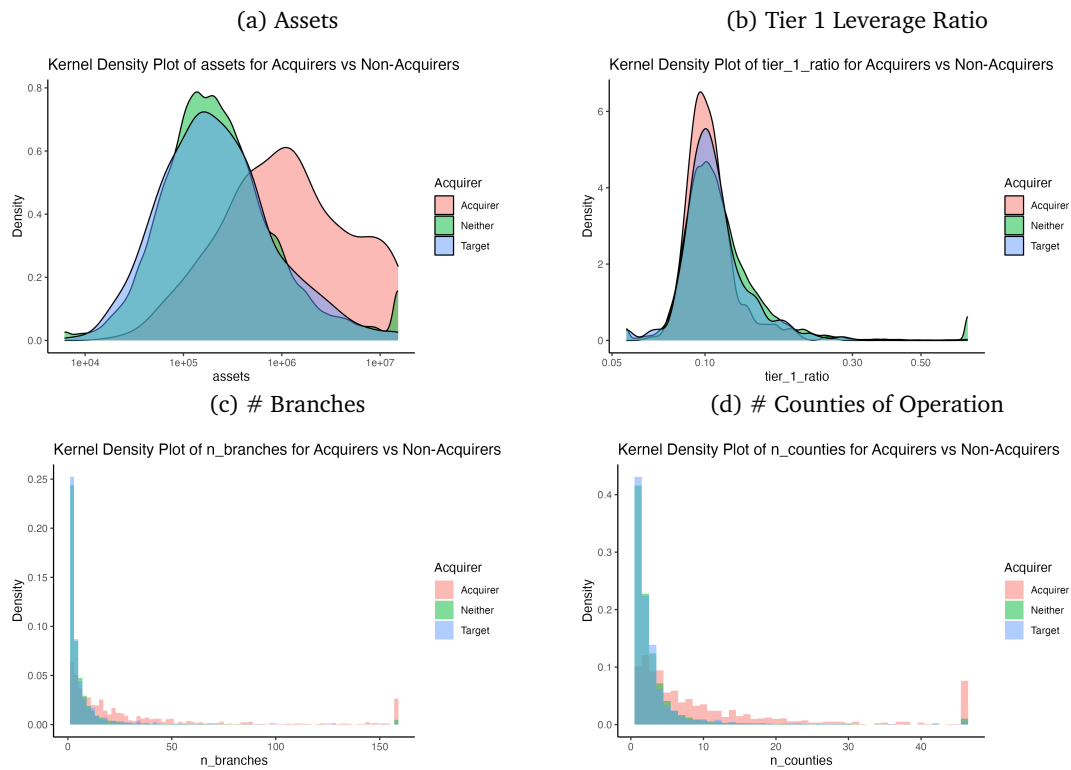
Note: This graph shows the percent of mergers occurring between 1995-2024 between banks headquartered in the same state. The size of the circle for a given state is the total number of mergers occurring between 1995-2024.

Figure A.7: Merger frequency by market



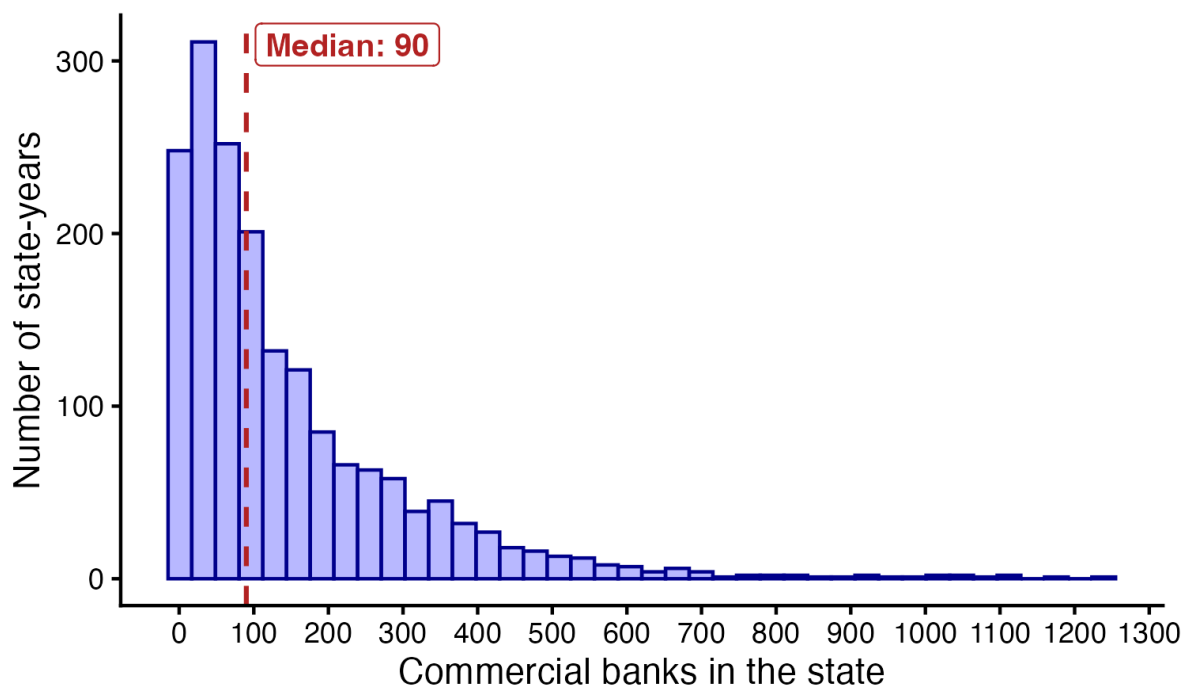
Note: This histogram shows the distribution number of commercial bank mergers occurring between 1995-2024 in a given year between banks headquartered in the same state, where each observation is a state-year cell. Mergers between banks headquartered in different states are excluded

Figure A.8: Acquirer and target size and balance sheets



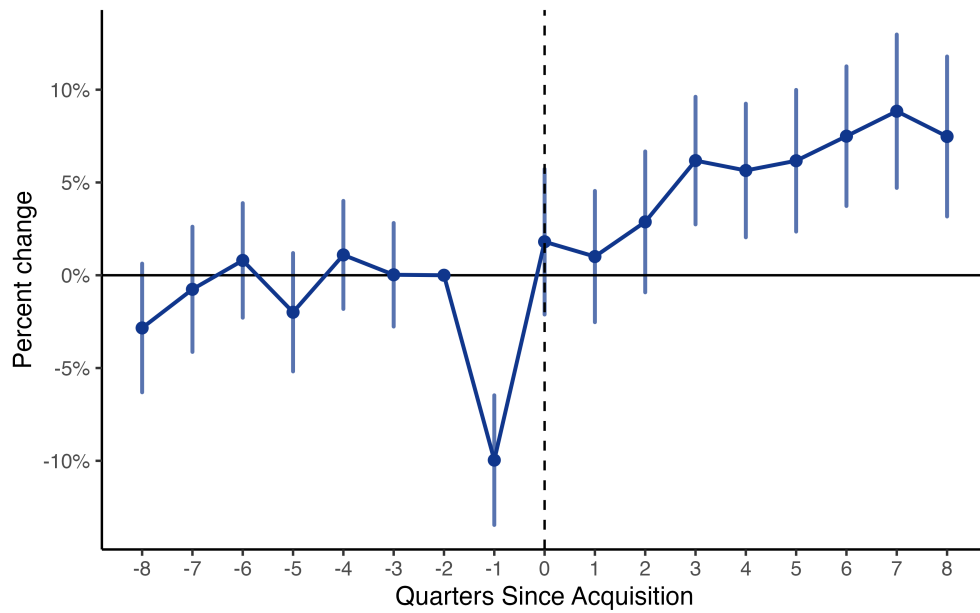
Note: The figures show kernel density plots of outcomes for acquirers, targets, and banks who were neither targets nor acquirers in the entire panel. The figures make clear that buyers are larger in assets, branches, and number of counties operating and have slightly lower leverage ratios.

Figure A.9: Market size



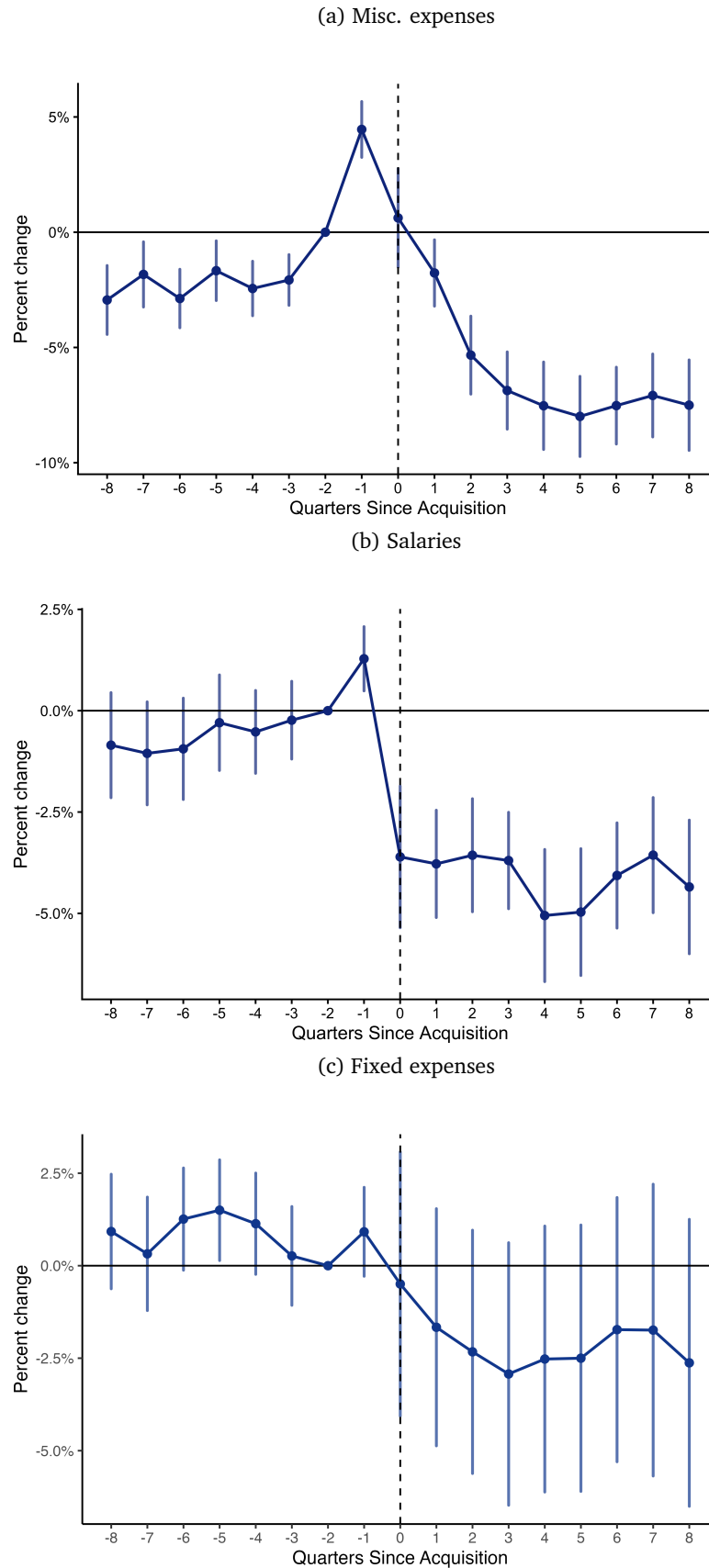
Note: This histogram shows the distribution number of commercial banks in each market, where each observation is a state-year cell between 1990 and 2024.

Figure A.10: Net income



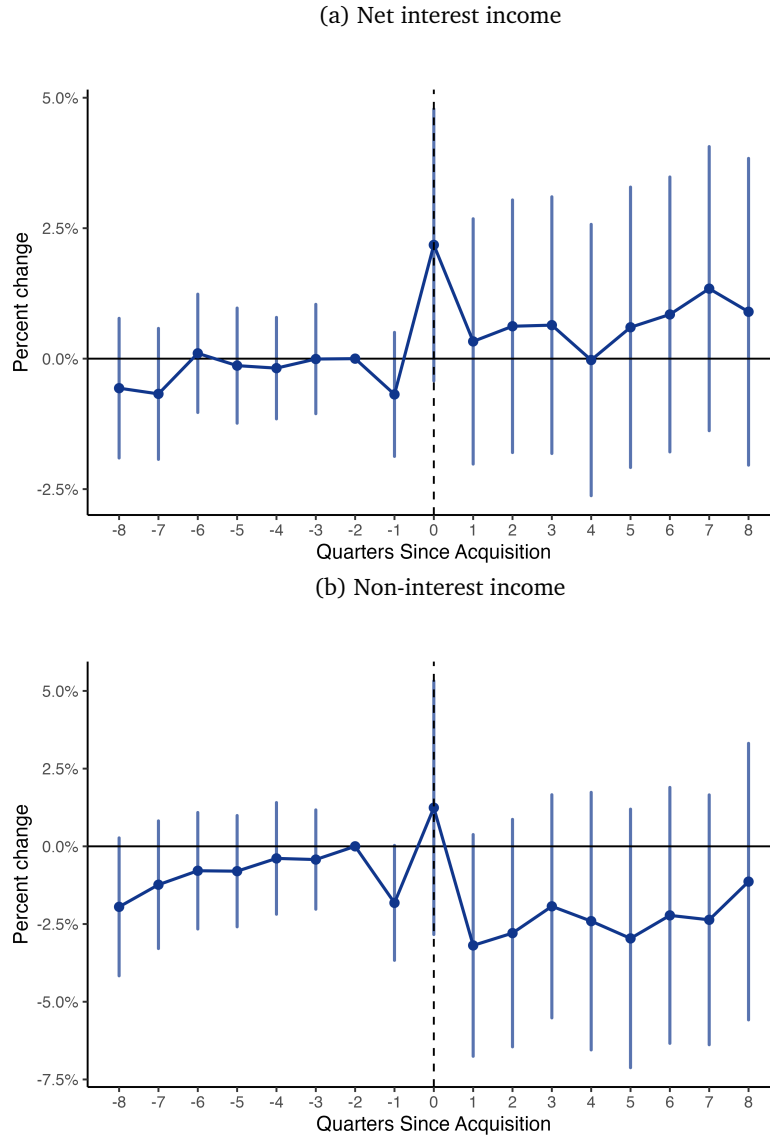
Note: This figure shows the estimates of β_s from Equation (1) on the constructed population of bank mergers compared with a stacked population of banks that were never acquirers nor targets between 1990-2024. The estimates imply that net income of the *pro-forma* (combined) bank increased by 6.6% two years after the merger.

Figure A.11: Difference-in-difference evidence of cost synergies



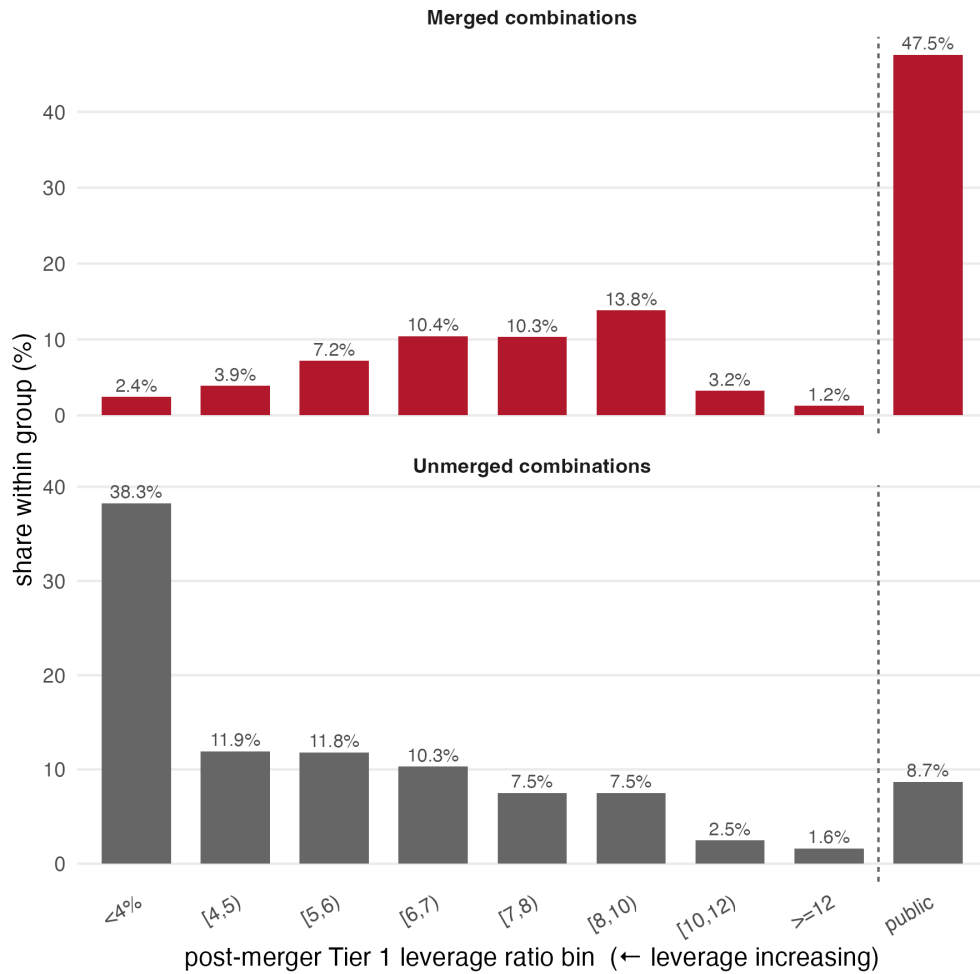
Note: This figure shows the estimates of β_s from Equation (1) on the constructed population of bank mergers compared with a stacked population of banks that were never acquirers nor targets between 1990-2024.

Figure A.12: Difference-in-difference evidence of revenue synergies



Note: This figure shows the estimates of β_s from Equation (1) on the constructed population of bank mergers compared with a stacked population of banks that were never acquirers nor targets between 1990-2024. The outcome variables are net interest income (Panel (a)) and non interest income (Panel (b)). The estimates rule out, at the 95% confidence level, an increase that exceed 1.7% in net interest income and 1.3% in non-interest income.

Figure A.13: Merger counts by Tier 1 Leverage Ratio



Note: This graph shows a histogram of number of combinations in each post-merger Tier 1 Leverage Ratio bin. The relative counts of combinations in each bin that merge and do not merge create Figure 8.

B Additional tables

Table B.1: Glossary

Quantity	Call Reports item
Assets	RCFD 2170
Risk-weighted assets	
w_{it-8}	
Net income	RIAD 4340
Good will impairment losses	RIAD C216
Amortization expense	RIAD C232
Non-interest expenses	RIAD 4093
Fixed expenses	RIAD XXXX
Misc. expenses	RIAD 4092
Net interest income	RIAD 4073
Non-interest income	RIAD B497
Economic profits ("Profit")	RIAD 4340 + RIAD C216 + RIAD 232
Economic costs	RIAD 4093 - RIAD C216 - RIAD 232
Inefficiency Ratio	(RIAD 4093)/(RIAD 7073 + RIAD B947)
Unmarked HTM/AFS assets	(AFS Fair Value + HTM Fair Value) -(AFS Cost + HTM Cost)

Table B.2: Commercial bank merger sample: Descriptive statistics by cohort

1990-1994				1995-1999			
	Acq.	Tar.	Unin.		Acq.	Tar.	Unin.
Inefficiency Ratio	66.4	78.2	70.5	Inefficiency Ratio	61.2	65.8	66.1
ROA: Net income / Assets	0.28	0.18	0.22	ROA: Net income / Assets	0.32	0.26	4.39
ROA (adj): Economic profits / Assets	0.29	0.19	0.23	ROA (adj): Economic profits / Assets	0.34	0.27	4.40
ROE: Net income / Equity	3.39	0.96	1.76	ROE: Net income / Equity	3.68	2.82	2.80
ROE (adj): Economic profits / Equity	3.53	1.02	1.81	ROE (adj): Economic profits / Equity	3.89	2.93	2.88
Tier 1 Leverage Ratio	8.0	8.7	9.6	Tier 1 Leverage Ratio	8.5	9.4	11.4
Loans / Assets	57.1	53.6	53.3	Loans / Assets	61.6	59.3	57.6
Deposits / Assets	84.7	87.9	86.8	Deposits / Assets	80.3	85.2	84.1
2000-2004				2005 – 2007-11			
	Acq.	Tar.	Unin.		Acq.	Tar.	Unin.
Inefficiency Ratio	62.3	75.3	69.2	Inefficiency Ratio	60.2	70.1	71.0
ROA: Net income / Assets	0.31	0.20	0.24	ROA: Net income / Assets	0.32	0.27	0.23
ROA (adj): Economic profits / Assets	0.33	0.21	0.25	ROA (adj): Economic profits / Assets	0.33	0.27	0.23
ROE: Net income / Equity	3.41	1.72	2.76	ROE: Net income / Equity	3.26	2.40	4.72
ROE (adj): Economic profits / Equity	3.57	1.80	2.81	ROE (adj): Economic profits / Equity	3.35	2.44	4.76
Tier 1 Leverage Ratio	8.7	10.0	11.6	Tier 1 Leverage Ratio	9.0	11.0	13.1
Loans / Assets	66.1	63.2	60.8	Loans / Assets	68.5	66.2	63.4
Deposits / Assets	77.3	82.0	82.0	Deposits / Assets	76.2	81.4	80.4
Recession (2007-12 – 2009-06)				2009-07 – 2014			
	Acq.	Tar.	Unin.		Acq.	Tar.	Unin.
Inefficiency Ratio	68.6	95.9	119.9	Inefficiency Ratio	66.0	92.1	83.1
ROA: Net income / Assets	0.27	-0.00	0.09	ROA: Net income / Assets	0.24	0.04	0.12
ROA (adj): Economic profits / Assets	0.28	0.01	0.11	ROA (adj): Economic profits / Assets	0.25	0.05	0.13
ROE: Net income / Equity	2.77	-0.02	1.88	ROE: Net income / Equity	2.06	-0.01	0.43
ROE (adj): Economic profits / Equity	2.88	0.02	2.06	ROE (adj): Economic profits / Equity	2.14	0.11	0.50
Tier 1 Leverage Ratio	8.9	9.9	12.2	Tier 1 Leverage Ratio	10.2	10.6	11.6
Loans / Assets	69.4	74.7	64.6	Loans / Assets	64.7	58.7	60.5
Deposits / Assets	75.9	81.9	79.7	Deposits / Assets	82.1	84.8	82.6
2015-2019				2020-2024			
	Acq.	Tar.	Unin.		Acq.	Tar.	Unin.
Inefficiency Ratio	64.6	74.6	71.5	Inefficiency Ratio	61.7	71.9	68.6
ROA: Net income / Assets	0.27	0.18	0.30	ROA: Net income / Assets	0.30	0.24	0.33
ROA (adj): Economic profits / Assets	0.28	0.18	0.30	ROA (adj): Economic profits / Assets	0.30	0.24	0.33
ROE: Net income / Equity	2.46	1.22	2.09	ROE: Net income / Equity	2.90	1.83	2.59
ROE (adj): Economic profits / Equity	2.53	1.26	2.12	ROE (adj): Economic profits / Equity	2.97	1.87	2.63
Tier 1 Leverage Ratio	10.4	10.8	12.6	Tier 1 Leverage Ratio	10.2	11.1	12.6
Loans / Assets	68.9	64.3	64.0	Loans / Assets	65.4	58.4	60.8
Deposits / Assets	81.8	84.2	82.2	Deposits / Assets	84.4	84.6	83.3

Note: This table shows the means of profit and balance sheet variables for commercial bank acquirers, commercial bank targets, and banks uninvolved in the merger market. Means for the acquirer and target are evaluated for each bank four quarters prior to their merger date. Means for the uninvolved banks are taken for the sample period.

Table B.3: County economic characteristics: merger-exposed vs. non-exposed county-years

Variable	Merger county-yrs	Non-merger county-yrs	Diff.
Unemployment rate (%)	5.81	6.11	-0.30***
Poverty rate (%)	14.05	15.84	-1.79***
Median HH income (\$)	55,463	49,431	6,032***
Log population	11.52	10.04	1.48***
Share age 65+	0.163	0.181	-0.018***
Share bachelor's+ (%)	24.51	18.57	5.94***
Nonmetro (RUCC ≥ 4)	0.353	0.663	-0.310***
N county-years	9,698	54,722	

* $p < .05$, ** $p < .01$, *** $p < .001$.

Note: An observation is a county $\times$ year. A county-year (c, t) is merger-exposed if a bank acquired in year t (2005–2024) operated a branch in county c (footprint at the merger's SOD year).

Table B.4: Merger propensity versus unrealized gains, pre- and post-2001

	Unrealized Gains (pp)		1(Seller) (bps)			
	First stage		OLS		2SLS	
	Pre-2001 (1)	Post-2001 (2)	Pre-2001 (3)	Post-2001 (4)	Pre-2001 (5)	Post-2001 (6)
$w_{i,t-8} \times \Delta y_t^{5-yr}$	-0.645*** (0.021)	-2.347*** (0.047)				
Unrealized Gains (pp)			-18.654+ (9.667)	-5.399 (3.983)	-65.319 (70.739)	20.994** (8.055)
First-stage F					951.8	2518.1
Size $\times$ quarter-year FE	X	X	X	X	X	X
Observations	260,872	508,092	260,872	508,092	260,872	508,092
R ²	0.394	0.601	0.004	0.005	0.004	0.005

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table shows the estimates from Equation (10) and (11) for the 1994-2001 and 2002-2024 subpopulations separately. As expected, only the 2002-2024 population yields a statistically significant second stage coefficient τ on unrealized gains. The differential effect of unrealized gains 2002-2024 is consistent with unrealized gains shifting capital and thereby increasing merger propensity.

C Robustness for Section 3

This section contains robustness checks for the baseline difference-in-differences estimates presented in Section 3. We show robustness to merger timing date, and violations of parallel trends coming from the control banks used for to estimate the causal effect on pro-forma banks or simultaneous changes to macroeconomic shocks in places mergers were targeted.

C.1 Merger timing

C.1.1 Bank Holding Company merge date

Figure C.1 shows the result of performing analysis at the BHC level.

C.1.2 Placebo test on county economic variables

Profits from mergers could also seem large because they are simultaneous with windfalls to the economic conditions of places where the mergers took place. We account for this by performing event study analysis at the county level. Using the date a county c experienced its first bank acquisition post 1990 in year $g(c)$, we estimate the regression for years t

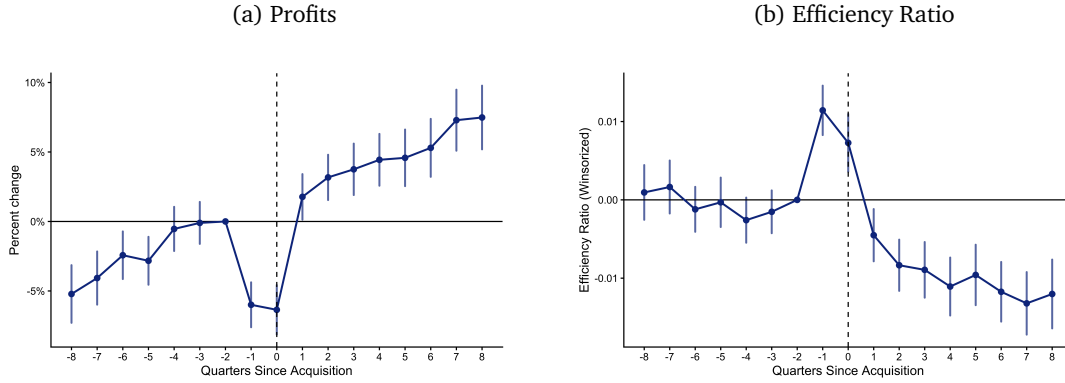
$$Y_{ct} = \alpha_c + \tau_t + \sum_{s=-5}^5 \beta_s \mathbf{1}(t - g(c) = s) \quad (57)$$

The specification uses variation in merger timing between counties experiencing a merger versus those that had not yet been treated. We use Sun and Abraham (2021) to address concerns about this staggered design. With random merger timing, the coefficient β_s is the causal effect on county c of the merger. Figure C.2 shows the results for the ten economic variables described in the main text. Effects are largely insignificant at the 95% confidence level five years pre- and post-merger date.

C.2 Violations of parallel trends

While difference-in-differences assumption of parallel trends is fundamentally untestable, we provide information of three forms consistent with the assumption being reasonable. First, the merger date coincides with a change in outcomes that seem specifically merger related. Second, matching to observably similar banks delivers similar quantitative estimates. Lastly, the bank merger date does not coincide with a change in trend of macroeconomic variables and consumer welfare related to bank demand but outside of the expected effects of the merger.

Figure C.1: Difference-in-differences around merger date of Bank Holding Company



Note: This figure shows the estimates of β_s from Equation (1) using the merger sample. Results are nearly identical to the full stacked difference-in-differences analysis. Mergers occur at the bank holding company level and are compared to control bank holding companies.

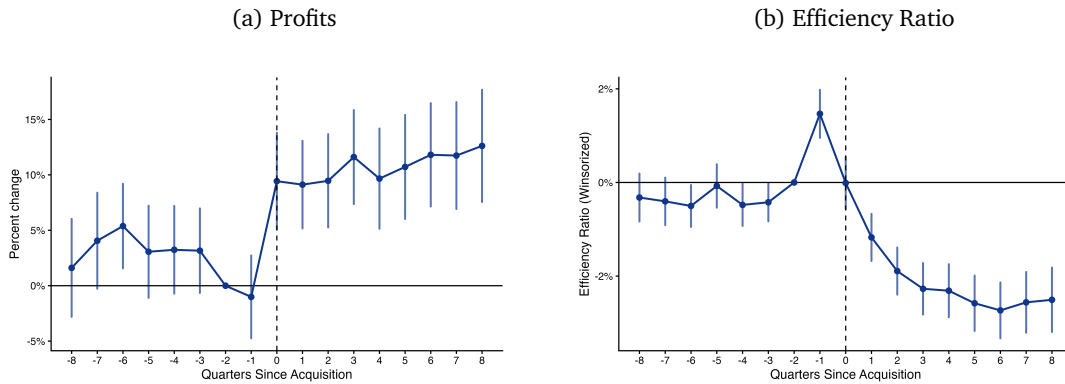
C.2.1 Narrative evidence

The merger date coincides with a change in outcomes that seem specifically merger related. Specifically, while one salient confounding story violating parallel trends could be that mergers coincide with a change in the latent demand for deposits or loans, the merger affects cost items at the merger date that are fundamentally merger-specific, like consulting and legal. Figure C.3 shows the estimates of Equation (1)

C.2.2 Control banks

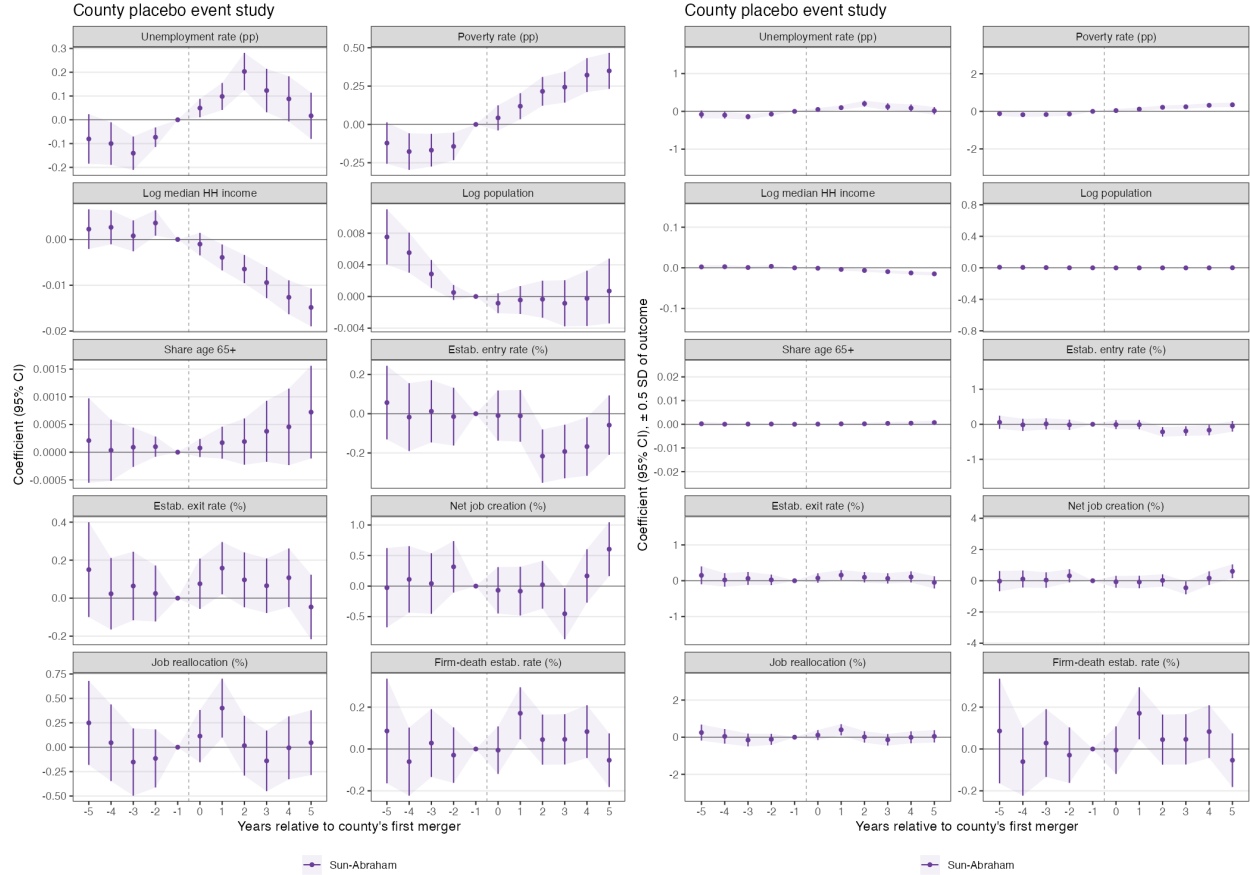
Our results are robust to using a matched sample. With the matched sample, we run Equation (1) with time, rather than time-entity fixed effects

Figure C.4: Matched difference-in-differences



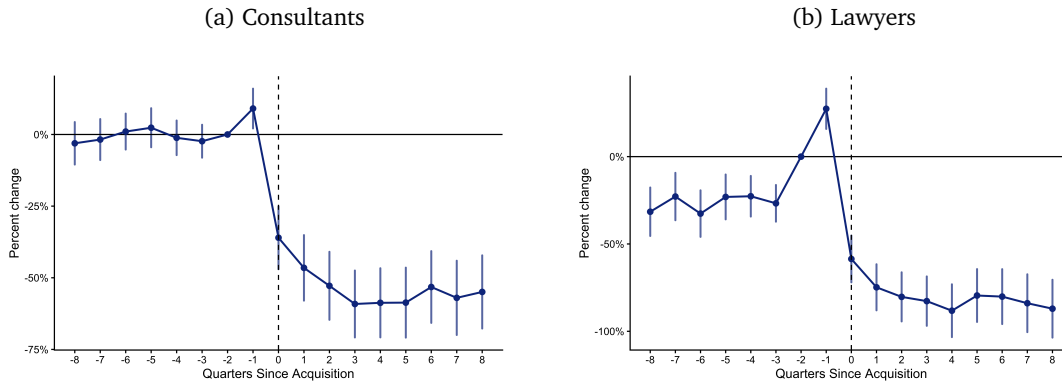
Note: This figure shows the estimates of β_s from Equation (1) without cohort-level effects using the merger sample compared to a matched control sample. Results are nearly identical to the full stacked difference-in-differences analysis.

Figure C.2: Placebo: The effect of first bank acquisition on counties



Note: The panels show the result of estimation Equation (57) on counties around the earliest year post-1990 that the county had at least one branch of an acquired bank. The left panel shows the results with 95% confidence intervals. The right panel rescales the left panel so that the y-axis for every graph has the length of the standard deviation of the variable in sample. When rescaling, it is clear that statistically significant effects, while observable especially for household median income, are economically very small. The results are consistent with one aspect of parallel trends, that absent acquisition, a bank would have been performing at least as well as before, hence biasing estimates of profit increases up in difference-in-differences. If anything, merger target banks are located in counties that are about to degrade on economic indicators, including decreasing population and increasing poverty and unemployment rates.

Figure C.3: Professional services



Note: This figure shows the estimates of β_s from Equation (1) for professional services. The spike in professional services at $t = -1$ and consistent lack of pre-trend is consistent with the merger date being correctly identified as the RSSD subsumption date. Furthermore, the reduction in these estimates mirror the non-interest-expense reductions and strengthen the case that cost reductions after mergers in the difference-in-difference design were indeed due to the mergers themselves.

D Proofs and additional theoretical results for Section 4

D.1 Toy model with testable empirical predictions

In order to discipline our reduced form results, we formalize the logic of this constraint mathematically. A bank i wishes to evaluate a target j with a price of π_j . A merged bank of i and j , indexed by $\mathcal{M}(i, j)$ has an idiosyncratic value of future profit synergies of $\Delta\pi_{\mathcal{M}(i, j)} \equiv \pi_{ij} - \pi_i - \pi_j$. The bank will finance their investment in the cheapest possible manner. In the event of a cash-financed merger, the firm can choose the lesser of two evils: keeping a permanently lower equity capital ratio and incurring a shadow cost c^{Reg} , or facing a marginal cost c^{Ext} to raise external equity to recapitalize. To perform a stock merger, the firm incurs a marginal cost c^{Ext} representing the inducements or premium required to entice the seller to accept buyer stock as opposed to cash that, as discussed, is roughly equivalent to the ability to solicit equity from external shareholders.²⁸ The buyer's choice financing choice becomes:

$$\min \left\{ \underbrace{\min \left\{ \underbrace{c^{Reg}}_{\text{Without recapitalizing}}, \underbrace{c^{Ext}}_{\text{Issuing equity to recapitalize}} \right\}}_{\text{Cash merger}}, \underbrace{c^{Ext}}_{\text{Stock merger}} \right\} \quad (58)$$

Or, simply, the firm chooses its cheapest financing, net of the method of payment and net of its *ex post* choice of whether to recapitalize. In other words, the buyer bank i 's valuation of bank j is:

$$V_{ij} \equiv \Delta\pi_{\mathcal{M}(i, j)} - \underbrace{\min \left\{ \underbrace{c^{Reg}}_{\text{Not issuing equity}}, \underbrace{c^{Ext}}_{\text{Issuing equity}} \right\}}_{\text{regulatory capital wedge (RCW)}}. \quad (59)$$

In industries without capital regulation, $c^{Reg} \approx 0$, and the value of a potential merger is simply the value of its potential profit synergies, $V_{it} \approx \Delta\pi_{ij}$. In contrast, the bank proceeds with the merger if $V_i > 0$ (so merger $M_{ij} = \mathbf{1}(V_i > 0)$), taking the shadow cost of capital regulation into account. We refer to this wedge in the merger incentives as the *regulatory capital wedge* (RCW).

The setup delivers empirical predictions we bring to data. For simplicity of exposition, the following predictions assume that c_i^{Reg} , π_i , and c_i^{Ext} are mutually statistically independent and $\Delta\pi \sim \mathcal{N}(1, 1)$. The generalization of each of the three predictions, with proofs, is in Appendix D.

²⁸We leave out the possibility of purchasing another bank in debt, which would further increase leverage. We implicitly assume that any cost of increasing leverage (i.e., increasing the probability of incurring bankruptcy cost by decreasing distance to default) is weakly smaller than c^{Reg} .

Prediction 1. If $c_1 > c_2$, then

$$\Pr(\text{Not issuing equity} \mid M_{ij} = 1, c_i^{\text{Ext}} = c_1) \geq \Pr(\text{Not issuing equity} \mid M_{ij} = 1, c_i^{\text{Ext}} = c_2). \quad (60)$$

Prediction 1 relies on variation in c_i^{Ext} independent of variation in regulatory cost and merger synergies π_{ij} . It states that, given that the buyer has chosen to acquire the target, the use of cash is increasing in the cost of external equity issuance faced by the buyer.

Prediction 2.

$$\frac{d\Pr(M_{ij} = 1 \mid c^{\text{Reg}} = r, c^{\text{Ext}} = \bar{c})}{dr} < 0 \quad \text{When } r < c_i^{\text{Ext}} \quad (61)$$

$$\frac{d\Pr(M_{ij} = 1 \mid c^{\text{Reg}} = r, c^{\text{Ext}} = \bar{c})}{dr} = 0 \quad \text{When } r > c_i^{\text{Ext}} \quad (62)$$

Prediction 2 states that the inverse relationship between regulatory cost and merger propensity is conditioned on the cost of issuing external equity. When the cost of equity issuance is high, the RCW is active. The propensity of a combination consummating is decreasing in the regulatory cost that the combination would enforce. The prediction implicitly holds fixed c_i^{Ext} and the value of the merger $\Delta\pi_i$ while varying c_i^{Reg} .

Prediction 3.

$$\frac{d\mathbb{E}[\Delta\pi_{ij} \mid c^{\text{Reg}} = r, c^{\text{Ext}} = \bar{c}, M_{ij} = 1]}{dr} \geq 0 \quad (63)$$

Prediction 3 is another corollary of the setup; if c_i^{Reg} is a constraint (RCW is active), the mergers that do consummate are censored to be of higher value. The choice to acquire is a discrete choice, and so, with the assumptions established, V_{ij} is nonincreasing in c_i^{Reg} .

Write G for the CDF of c^{Reg} , and φ, Φ for the standard normal density and CDF, so that $\Pr(\Delta\pi > x) = \Phi(1 - x)$. The proofs then follow:

Proof, Prediction 1. It suffices to show that $P(c) \equiv \Pr(\text{Not issuing equity} \mid M_{ij} = 1, c_i^{\text{Ext}} = c)$ is nondecreasing in c . Given $c^{\text{Ext}} = c$, not issuing is the event $\{c^{\text{Reg}} \leq c\}$. By independence,

$$N(c) \equiv \Pr(M = 1, c^{\text{Reg}} \leq c \mid c^{\text{Ext}} = c) = \int_{[0, c]} \Phi(1 - r) dG(r), \quad T(c) \equiv \Phi(1 - c)(1 - G(c)), \quad (64)$$

with $T(c) = \Pr(M = 1, c^{\text{Reg}} > c \mid c^{\text{Ext}} = c)$. N is nondecreasing and T is nonincreasing (a product of nonnegative nonincreasing functions), so $P(c) = N(c)/(N(c) + T(c)) = 1/(1 + T(c)/N(c))$ is nondecreasing wherever $N(c) > 0$, and $P(c) = 0$ where $N(c) = 0$. $\square$

Proof, Prediction 2. By independence, conditioning on $(c^{\text{Reg}} = r, c^{\text{Ext}} = \bar{c})$ leaves the distribution of $\Delta\pi$ unchanged, and by (67) the merger occurs iff $\Delta\pi > \min\{r, \bar{c}\}$. Hence, $p(r) = \Phi(1 -$

$\min\{r, \bar{c}\}$); taking derivatives in the two regimes of r yields

$$p'(r) = \begin{cases} -\varphi(1-r) < 0, & r < \bar{c} \quad (\text{wedge active}), \\ 0, & r > \bar{c} \quad (\text{wedge slack}). \end{cases} \quad (65)$$

□

Proof, Prediction 3. $m(r) \equiv \mathbb{E}[\Delta\pi \mid M = 1, c^{\text{Reg}} = r, c^{\text{Ext}} = \bar{c}] = 1 + \lambda(\min\{r, \bar{c}\} - 1)$, where $\lambda(z) \equiv \varphi(z)/(1 - \Phi(z))$, so

$$m'(r) = \begin{cases} \lambda(r-1) [\lambda(r-1) - (r-1)] > 0, & r < \bar{c} \quad (\text{wedge active}), \\ 0, & r > \bar{c} \quad (\text{wedge slack}). \end{cases} \quad (66)$$

□

D.2 General forms of Predictions 1–3

Definitions. Suppress buyer-seller subscripts. The random variables assigned to are $(\Delta\pi, c^{\text{Reg}}, c^{\text{Ext}})$: the merger synergy, the shadow cost of operating closer to the regulatory capital floor, and the cost of inducing shareholders to absorb equity. Define

$$w \equiv \min\{c^{\text{Reg}}, c^{\text{Ext}}\}, \quad M = \mathbf{1}\{\Delta\pi > w\}. \quad (67)$$

Let F_c denote the probability distribution of c^{Reg} among consummated mergers at $c^{\text{Ext}} = c$:

$$F_c(s) \equiv \Pr(c^{\text{Reg}} \leq s \mid M = 1, c^{\text{Ext}} = c), \quad s \geq 0, \quad (68)$$

“Not issuing equity” is the event $\{c^{\text{Reg}} \leq c^{\text{Ext}}\}$ and $M = 1$.

To interpret method of payment as meaningful, we need that among consummated mergers, the buyer’s facing a higher c^{Ext} have weakly lower regulatory cost c^{Reg} stochastically. To use differences in profit synergies as indicative of meaningful c^{Reg} , profit synergies must move independently from c^{Reg} . These two statements motivate Assumptions 1 and 2.

Assumption 1 (First order stochastic dominance). For $c_1 > c_2$ and every $s \geq 0$, $F_{c_1}(s) \geq F_{c_2}(s)$.

Remark 1. $(c^{\text{Reg}}, \Delta\pi) \perp c^{\text{Ext}}$ is sufficient to satisfy Assumption 1.

Assumption 2. $\forall \bar{c}, \Delta\pi \perp c^{\text{Reg}} \mid c^{\text{Ext}} = \bar{c}$

Remark 2. Mutual independence of $(c^{\text{Reg}}, \Delta\pi, c^{\text{Ext}})$ (the simplifying assumption in the main text) is sufficient to satisfy both Assumption 1 and Assumption 2.

Proposition 1. Under Assumption 1, if $c_1 > c_2$, then

$$P(\text{Not issuing equity} \mid M = 1, c^{\text{Ext}} = c_1) \geq Pr(\text{Not issuing equity} \mid M = 1, c^{\text{Ext}} = c_2)$$

Proof. First, notice

$$F_{c_1}(c_1) = P(\text{Not issuing equity} \mid M = 1, c^{\text{Ext}} = c_1)$$

and

$$F_{c_2}(c_2) = P(\text{Not issuing equity} \mid M = 1, c^{\text{Ext}} = c_2).$$

Then,

$$F_{c_1}(c_1) \underbrace{\geq}_{(1)} F_{c_1}(c_2) \underbrace{\geq}_{(2)} F_{c_2}(c_2)$$

(1) holds because F_{c_1} , being a CDF, is nondecreasing in its argument and $c_1 > c_2$.

(2) is Assumption 1 evaluated at $s = c_2$. □

Proposition 2. Fix $\bar{c}$. Let $p(r) \equiv \Pr(M = 1 \mid c^{\text{Reg}} = r, c^{\text{Ext}} = \bar{c})$. Under Assumption 2:

- (a) p is nonincreasing on $[0, \bar{c}]$ and constant on $[\bar{c}, \infty)$.
- (b) If in addition $\Delta\pi$ given $c^{\text{Ext}} = \bar{c}$ admits a density $f_{\bar{c}}$, then

$$p'(r) = \begin{cases} -f_{\bar{c}}(r) \leq 0, & r < \bar{c} \quad (\text{wedge active}), \\ 0, & r > \bar{c} \quad (\text{wedge slack}), \end{cases} \quad (69)$$

with $p'(r) < 0$ wherever $f_{\bar{c}}(r) > 0$.

Proof. Write $S_{\bar{c}}(x) \equiv \Pr(\Delta\pi > x \mid c^{\text{Ext}} = \bar{c})$; by Assumption 2, further conditioning on $c^{\text{Reg}} = r$ leaves this probability unchanged. Since $M = \mathbf{1}\{\Delta\pi > \min\{r, \bar{c}\}\}$ by (67), $p(r) = S_{\bar{c}}(\min\{r, \bar{c}\})$.

(a) $x \mapsto S_{\bar{c}}(x)$ is nonincreasing, and the threshold $\min\{r, \bar{c}\}$ is nondecreasing in r below $\bar{c}$ and constant above.

(b) Below $\bar{c}$ the threshold is r itself, so $p(r) = S_{\bar{c}}(r)$ and $S'_{\bar{c}} = -f_{\bar{c}}$; above $\bar{c}$ the threshold is fixed. □

Proposition 3. Suppose $\mathbb{E}[|\Delta\pi| \mid c^{\text{Ext}} = \bar{c}] < \infty$, and let $m(r) \equiv \mathbb{E}[\Delta\pi \mid M = 1, c^{\text{Reg}} = r, c^{\text{Ext}} = \bar{c}]$, defined wherever $p(r) > 0$. Under Assumption 2:

- (a) m is nondecreasing on $[0, \bar{c}]$ and constant on $[\bar{c}, \infty)$.
- (b) If in addition $\Delta\pi$ given $c^{\text{Ext}} = \bar{c}$ admits a density $f_{\bar{c}}$, then with $h(w) \equiv f_{\bar{c}}(w) / \Pr(\Delta\pi > w \mid c^{\text{Ext}} = \bar{c})$,

$$m'(r) = \begin{cases} h(r)(m(r) - r) \geq 0, & r < \bar{c} \quad (\text{wedge active}), \\ 0, & r > \bar{c} \quad (\text{wedge slack}), \end{cases} \quad (70)$$

with $m'(r) > 0$ wherever $f_{\bar{c}}(r) > 0$.

Proof. Notice that under Assumption 2, $m(r)$ can be rewritten as

$$m(r) = \mathbb{E}[\Delta\pi \mid \Delta\pi > \min\{r, \bar{c}\}, c^{\text{Ext}} = \bar{c}].$$

(a) The threshold $w = \min\{r, \bar{c}\}$ is nondecreasing in r and constant above $\bar{c}$, so it suffices that the truncated mean is nondecreasing in w .

Take $w_1 \geq w_2$ with $\Pr(\Delta\pi > w_1 \mid \bar{c}) > 0$ (all probabilities and expectations below condition on $c^{\text{Ext}} = \bar{c}$). For every t , $\Pr(\Delta\pi > t \mid \Delta\pi > w_1) \geq \Pr(\Delta\pi > t \mid \Delta\pi > w_2)$: for $t \leq w_1$ the left side is 1; for $t > w_1 \geq w_2$ the two sides are $\Pr(\Delta\pi > t)/\Pr(\Delta\pi > w_1)$ and $\Pr(\Delta\pi > t)/\Pr(\Delta\pi > w_2)$, and $\Pr(\Delta\pi > w_1) \leq \Pr(\Delta\pi > w_2)$.

So the censored distribution of $\Delta\pi$ above w_1 first order stochastically dominates the censored distribution of $\Delta\pi$ above w_2 . So, $\mathbb{E}[\Delta\pi \mid \Delta\pi > w_1] \geq \mathbb{E}[\Delta\pi \mid \Delta\pi > w_2]$.

(b) With $S_{\bar{c}}$ as in the proof of Proposition 2, write $m(w) = (\int_w^\infty x f_{\bar{c}}(x) dx)/S_{\bar{c}}(w)$ and differentiate:

$$m'(w) = \frac{-w f_{\bar{c}}(w) S_{\bar{c}}(w) + f_{\bar{c}}(w) \int_w^\infty x f_{\bar{c}}(x) dx}{S_{\bar{c}}(w)^2} = h(w) (m(w) - w); \quad (71)$$

below $\bar{c}$ the truncation point is $w = r$, above it is fixed. $f_{\bar{c}}(w) > 0$ implies $h(w) > 0$ and implies $S_{\bar{c}}(w) > 0$, and $m(w) > w$ by construction. So, $m'(r)$ is weakly positive. $\square$

Remark 3. Where the wedge is active, the slope of merger propensity is (minus) the density of marginal deals extinguished by regulatory cost, and the selection gradient is the exit rate of those marginal deals, $h(r)$, times the markup of the average selected synergy over the marginal one, $m(r) - r$.

E Proofs and microfoundations for Section 5

E.1 Proof for Theorem 1

Proof. Given a set of firms $\mathcal{J}'_{kt}$ posting a set of prices ρ^D, ρ^L , the allocation of consumers to firms in horizontal Bertrand-Nash competition is generically unique because consumer utilities functions specified by $\alpha, \rho^D, \xi^D, \rho^L, \xi^L$ have unique maxima (Equation (30)).

Given a set of firms, each firm has a single optimal choice of ρ^D_{jkt} and a single optimal choice of ρ^L_{jkt} . (Firm problem has unique maximum, Equation (29)).

The most novel part of the proof is showing a unique allocation in the merger market governed by Equation (28). The number of banks in a market k at time t $\mathcal{J}_{kt}$ is an integer, $\mathcal{J}_{kt} \in \mathbb{Z}$. To prove that the allocation is unique, represent Step 1 of the game using a directed graph. The adjacency matrix to that directed graph, P is a square $\mathcal{J}_{kt} \times \mathcal{J}_{kt}$ matrix that captures the outcomes of Step 1.

This graph is an upper triangular matrix where $P_{ji} = 1$ indicates that bank j chooses to sell to bank i and $P_{j\ell} = 0 \forall$ elements $\ell \neq i$. P represents a unique allocation given a set of surpluses because every bank will generically have a highest surplus option to sell to. Furthermore, the mapping of primitives Ω_{kt} to surpluses is unique because of the belief structure posited. Banks choose merger partners by building counterfactual profits holding the other players fixed without additional M&A, so no expectations over opponent play could create lumps in the objective that would risk the disappearance of a unique maximum (e.g., a coordination game that requires the payoff of one player to affect another).

The chain resolution stage simply rewrites the allocation in a directed graph that is now defined by the adjacency matrix $P^{\mathcal{J}_{kt}-1}$. Since raising a matrix to a power is well-defined if the matrix is square, then a well-defined P implies a well-defined $P^{\mathcal{J}_{kt}-1}$. $\square$

E.2 Microfoundations of the merger market allocation rule

For each bank $j \in \mathcal{J}_{kt}$, the eligible buyer pool is the set of weakly larger banks,

$$\mathcal{B}_{jt} := \{i \in \mathcal{J}_{kt} \setminus \{j\} : \text{book assets of } i \geq \text{book assets of } j\},$$

$V_{jt} + \Phi(Z_{ijt})$ denotes buyer i 's valuation of acquiring bank j as in (27), where V_{jt} is defined to add back in the standalone value of j , $V_{jt} \equiv \mathcal{C}^S + \pi_j(\Omega_{kt})$. The allocation rule used to compute the likelihood of realized merger choices is:

$$d_{jt} = \begin{cases} \arg \max_{i \in \mathcal{B}_{jt}} \Phi(Z_{ijt}), & \text{if } \max_{i \in \mathcal{B}_{jt}} \Phi(Z_{ijt}) \geq 0, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{A})$$

We present three mechanisms that microfound this allocation rule (A): a centralized seller-proposing mechanism, simultaneous second-price auctions, and simultaneous ascending auctions.

The mechanisms imply different transfers, but because the likelihood depends only on acquirer

identities and never on transaction prices, they are observationally equivalent without transaction prices.

Remark 4 (Role of the size restriction). *Limiting $\mathcal{B}_{jt}$ to weakly larger banks makes simultaneous execution coherent by ruling out the possibility of cycles of buyers and sellers. For instance, if seller A names buyer B while B itself names buyer C , the planner resolves the chain transitively: B is sold to C , and A is absorbed by C through B . Such chains ascend the size order and the resolution is well-defined. The size restriction thus substitutes for the sequencing device that renders allocation well-defined in a sequential model. Monotonicity is strict only under a strict size order; with ties, a tie-breaking rule is required to rule out two equal-size banks naming each other.*

We impose the following belief assumptions.

Assumption 3 (Zero conjectural variation). *Each agent's beliefs assign probability one to the event that no transaction other than its own occurs in period t .*

Assumption 4 (Myopia). *Each agent treats period- t outcomes as once-and-for-all: the value of retaining bank j is V_{jt} , the value to buyer i of acquiring bank j at price p is $\Phi(Z_{ijt}) + V_{jt} - p$, and there is no option value of transacting in future periods.*

Assumption 5 (Narrow bracketing). *Buyer i 's valuation of target j equals $\Phi(Z_{ijt})$ irrespective of the set of other banks i owns or wins.*

E.2.1 Centralized seller-proposing mechanism

Game.

1. Each seller $j \in \mathcal{J}_{kt}$ simultaneously reports an assignment $a_j \in \mathcal{B}_{jt} \cup \{j\}$, where $a_j = j$ denotes retention.
2. The planner executes every report: if $a_j = i \neq j$, bank j is sold to bank i ; buyers cannot decline.
3. Transfers: a seller assigned to buyer i receives $p_{ij} = \Phi(Z_{ijt}) + V_{jt}$.

Proposition 4. *Under Assumptions 3–4, it is weakly dominant for every seller j to report $a_j = \arg \max_{i \in \mathcal{B}_{jt}} \Phi(Z_{ijt})$ if $\max_{i \in \mathcal{B}_{jt}} \Phi(Z_{ijt}) \geq V_{jt}$ and $a_j = j$ otherwise. The induced allocation is (A).*

Proof. Under Assumption 3, seller j believes its report affects only its own assignment and transfer. Under Assumption 4, staying independent yields 0 and reporting $a_j = i$ yields the transfer $p_{ij} = \Phi(Z_{ijt})$. The seller's payoff is hence $\Phi(Z_{ijt})$ or zero, independent of other reports, so the stated report maximizes j 's payoff against every profile of other reports and is weakly dominant. The mechanism executes reports directly, so the allocation is (A). $\square$

Remark 5 (Resolving chains). Under Assumption 3, sellers assign probability zero to such events when reporting. If the bank-level transaction records ($d_A = B$, $d_B = C$) could be incorporated into the likelihood, estimation would precisely reflect the model.

Remark 6 (Transfers). Full surplus extraction ($p_{ij} = \Phi(Z_{ijt})$) is sufficient but not necessary. Any transfer rule strictly increasing in the named buyer's valuation and satisfying $p_{ij} \geq V_{jt} \iff \Phi(Z_{ijt}) \geq 0$ leaves the seller's induced ranking over $\mathcal{B}_{jt} \cup \{j\}$, and hence Proposition 4, unchanged. One example of a transfer rule could be Nash Bargaining over the incremental surplus created, where price p minus standalone value V_j satisfies $p_{ij} - V_j \equiv \arg \max_q (\Phi_{it} - q)^{\lambda^B} q^{\lambda^S}$ and λ^B/λ^S is constant for every potential merger pair. In this case, $p_{ij} = V_{jt} + \frac{\lambda^S}{\lambda^B + \lambda^S} \Phi_{ij}$.

E.2.2 Simultaneous second-price auctions

Game.

1. All banks in $\mathcal{J}_{kt}$ are put up for sealed-bid second-price auction simultaneously. The eligible bidder pool for bank j is $\mathcal{B}_{jt} \cup \{j\}$; equivalently, each bank bids on every bank with weakly smaller book assets, itself included.
2. Bank i submits a bid $b_{ij} \geq 0$ on each eligible target j ; the self-bid b_{jj} serves as j 's reservation price.
3. In each auction, the bank is awarded to the highest bidder at the second-highest bid; if the self-bid is highest, the bank is retained. All winning bids are binding and executed.

Proposition 5. Under Assumptions 3–5, the strategy profile $b_{ij} = \Phi(Z_{ijt}) + V_{jt}$ for all $i \neq j$ and $b_{jj} = V_{jt}$ is weakly dominant. The induced allocation is (A).

Proof. By Assumption 5, bidder i 's value of winning auction j at price p is $\Phi(Z_{ijt}) - p$, independent of the outcomes of all other auctions; by Assumptions 3–4, no other transaction enters i 's perceived problem. Each bidder's problem therefore separates across auctions into isolated private-value second-price auctions, in which bidding one's valuation is weakly dominant (Vickrey, 1961); the analogous argument gives $b_{jj} = V_{jt}$ for the seller. Each auction thus awards j to $\arg \max_{i \in \mathcal{B}_{jt}} \Phi(Z_{ijt})$ if and only if the maximand exceeds V_{jt} , which is (A): competitive bidding computes the maximum that the seller computes directly in the centralized mechanism. $\square$

Remark 7. Because winning bids are binding, simultaneity can produce chains: if B wins A while C wins B , then C acquires B and, through B , executes B 's winning bid on A . Ex post, chain execution is in tension with Assumption 5, since the price paid for A reflects B 's valuation rather than C 's. Under Assumption 3, however, bidders assign probability zero to such chains at the bidding stage, so equilibrium bids and hence auction outcomes are unaffected.

E.2.3 Simultaneous ascending auctions

Game. The game is identical to the simultaneous second-price game except for the auction format: each bank is put up for ascending-price auction, all auctions run concurrently, each seller j sets reservation price V_{jt} , and in each auction the last remaining bidder wins at the price at which its final rival exits.

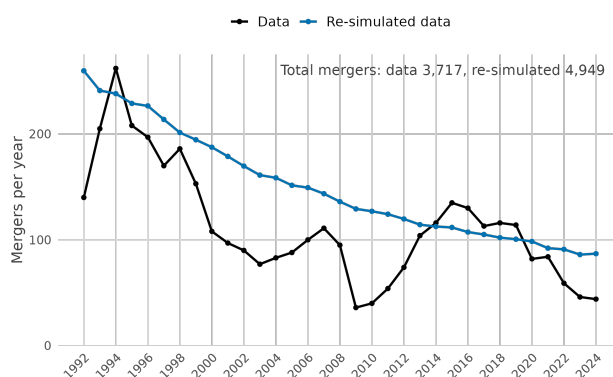
Proposition 6. *Under Assumptions 3–5, it is weakly dominant for each bidder i to remain in the auction for bank j until the price reaches $\Phi(Z_{ijt}) + V_{jt}$. The induced allocation is (A).*

Proof. With private values, the ascending auction is strategically equivalent to the sealed-bid second-price auction: exiting at one's valuation is weakly dominant and induces the same allocation and price (Vickrey, 1961; Milgrom and Weber, 1982). Assumptions 3–5 imply that each bidder's valuation in every auction equal to $\Phi(Z_{ijt})$, independent of concurrent auctions, exactly as in the proof of Proposition 5. The allocation in each auction is therefore (A). $\square$

F Model fit

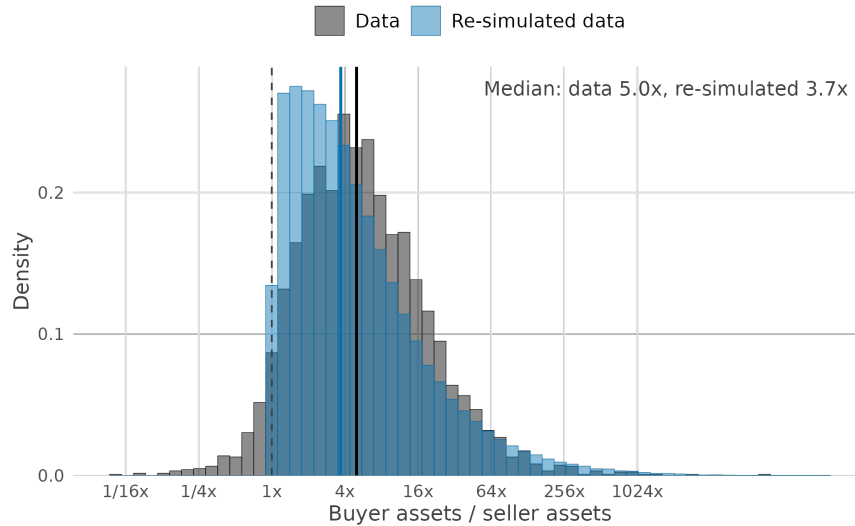
This section shows how data re-simulated from the model parameters compare to empirical data. The model is estimated using within-state merger choices and the outcomes that those mergers have on the profitability of banks in the market. As a result, the total number of mergers in the real data with the model's estimate, without additional assumptions, will not precisely match the number of mergers implied by simulations. Banks could have been purchased by out of state acquirers, failed, or exited, and the period has also been marked by modest bank entry, all of which are assumed not to exist in the re-simulation. We do not include the other phenomena in the computation of counterfactuals to create a transparent comparison between baseline and counterfactual policy. However, future sensitivity analysis could incorporate these channels.

Figure F.1: Count of mergers used for estimation versus implied by model parameters



Note: This graph shows the number of in-state mergers used for estimation annually from 1992 to 2024 in black. The blue line plots the average number of mergers occurring in 45 re-simulations around the model's estimated parameters.

Figure F.2: Relative buyer and seller size



Note: This figure shows the mergers in the real data used for model estimation versus re-simulated mergers using the estimated parameter values. The re-simulated values produce mergers that are smaller (3.7x vs. 5x) than in the data.

Figure F.3: Model fit by bank size

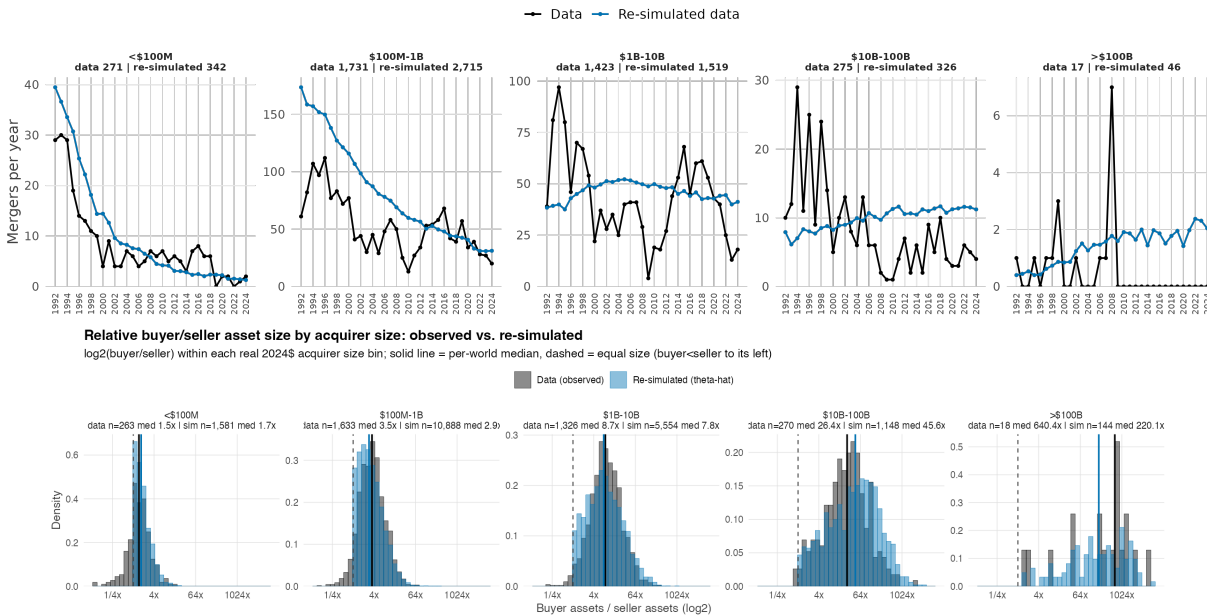


Figure F.4: Note: This figure shows the mergers in the real data used for model estimation versus re-simulated mergers using the estimated parameter values. The re-simulated values produce similar trends in mergers over time than in the data. The mergers in re-simulated data are roughly the same size as in the data. The difference between the simulation and data is in over-predicting the size of targets for regional banks (10-100B) and under-predicting the size of targets for the largest banks (100B+).